

Forall-x: Calgary, Model-Generated Solutions

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This document presents proofs for all of the exercises and examples in the Forall-x: Calgary book, covering Truth Functional Logic and First Order Logic. The Modal Logic proofs are not covered.

All of these proofs were generated by the Qwen3.5-35B-A3B-UD model quantised to Q8_K_XL format with inference running on an Nvidia A100 or H100 GPU.

The proofs were generated and checked to be correct using a proof checking tool and have been transcribed directly into this document. Consequently although all of the proofs are correct and give the expected result, they may not be the most efficient or attractive solutions to the questions.

In some cases – for example where a question asks for an equivalence to be proven – the solutions may be split into multiple parts.

Each proof is presented using Fitch notation and is followed by the specific commands needed to enter the proof into Daeducer, the proof tool they were created with.

Proofs

Chapter 17, Example p.134

Proof presented using Fitch notation:

1		$((P \wedge Q) \vee (P \wedge R))$	PR
2			
3			
4			
5			
6		P	

2		$(P \wedge Q)$	AS
3		P	$\wedge E, 2$
4		$(P \wedge R)$	AS
5		P	$\wedge E, 4$
6		P	$\vee E, 1, 2-3, 4-5$

Daeducer commands needed to generate the proof:

```
1 premise ((P ^ Q) v (P ^ R))
2 assumption (P ^ Q)
3 and_elim_left 2
4 discharge
5 assumption (P ^ R)
6 and_elim_left 5
7 discharge
8 or_elim 1, 2, 3, 5, 6
9 qed
```

Chapter 17, Example p.134

Proof presented using Fitch notation:

1	$(A \wedge (B \vee C))$	PR
2	A	$\wedge E$, 1
3	$(B \vee C)$	$\wedge E$, 1
4	B	AS
5	$(A \wedge B)$	$\wedge I$, 2, 4
6	$((A \wedge B) \vee (A \wedge C))$	$\vee E$, 5
7	C	AS
8	$(A \wedge C)$	$\wedge I$, 2, 7
9	$((A \wedge B) \vee (A \wedge C))$	$\vee E$, 8
10	$((A \wedge B) \vee (A \wedge C))$	$\vee E$, 3, 4-6, 7-9

Daeducer commands needed to generate the proof:

```

1 premise (A ^ (B v C))
2 and_elim_left 1
3 and_elim_right 1
4 assumption B
5 and_intro 2, 4
6 or_intro_left 5, (A ^ C)
7 discharge
8 assumption C
9 and_intro 2, 8
10 or_intro_right 9, (A ^ B)
11 discharge
12 or_elim 3, 4, 6, 8, 10
13 qed

```

Chapter 17, Example p.137

Proof presented using Fitch notation:

1	D	PR
2	$\neg D$	AS
3	$\perp$	$\neg E$, 2, 1
4	$\neg\neg D$	$\neg I$, 2-3

Daeducer commands needed to generate the proof:

```

1 premise D
2 assumption !D
3 not_elim 2, 1
4 discharge
5 not_intro 2, 3
6 qed

```

Chapter 17, Example p.138

Proof presented using Fitch notation:

1	$\neg\neg D$	PR
2	$\neg D$	AS
3	$\perp$	$\neg E$, 1, 2
4	D	IP, 2–3

Daeducer commands needed to generate the proof:

```
1 premise !!D
2 assumption !D
3 not_elim 1, 2
4 discharge
5 indirect 2, 3
6 qed
```

Chapter 17, Exercise A.1

Proof presented using Fitch notation:

1	$(A \wedge (B \wedge C))$	PR
2	$((B \vee C) \rightarrow D)$	PR
3	$(B \wedge C)$	$\wedge E$, 1
4	B	$\wedge E$, 3
5	$(B \vee C)$	$\vee E$, 4
6	D	$\rightarrow E$, 2, 5

Daeducer commands needed to generate the proof:

```
1 premise (A ^ (B ^ C))
2 premise ((B v C) -> D)
3 and_elim_right 1
4 and_elim_left 3
5 or_intro_left 4, C
6 imp_elim 2, 5
7 qed
```

Chapter 17, Exercise B.1

Proof presented using Fitch notation:

1		$(P \wedge S)$	PR
2		$(S \rightarrow R)$	PR
3		S	$\wedge E, 1$
4		R	$\rightarrow E, 2, 3$
5		$(R \vee E)$	$\vee E, 4$

Daeducer commands needed to generate the proof:

```
1 premise (P ^ S)
2 premise (S -> R)
3 and_elim_right 1
4 imp_elim 2, 3
5 or_intro_left 4, E
6 qed
```

Chapter 17, Exercise B.2

Proof presented using Fitch notation:

1		$(A \rightarrow D)$	PR	
2			$(A \wedge B)$	AS
3			A	$\wedge E, 2$
4			D	$\rightarrow E, 1, 3$
5			$(D \vee E)$	$\vee E, 4$
6		$((A \wedge B) \rightarrow (D \vee E))$	$\rightarrow I, 2, 5$	

Daeducer commands needed to generate the proof:

```
1 premise (A -> D)
2 assumption (A ^ B)
3 and_elim_left 2
4 imp_elim 1, 3
5 or_intro_left 4, E
6 discharge
7 imp_intro 2, 5
8 qed
```

Chapter 17, Exercise B.3

Proof presented using Fitch notation:

1	$(\neg L \rightarrow (J \vee L))$	PR
2	$\neg L$	PR
3	$(J \vee L)$	$\rightarrow$ E, 1, 2
4	J	AS
5	J	R, 4
6	L	AS
7	$\perp$	$\neg$ E, 2, 6
8	J	X, 7
9	J	$\vee$ E, 3, 4–5, 6–8

Daeducer commands needed to generate the proof:

```

1 premise (!L -> (J v L))
2 premise !L
3 imp_elim 1, 2
4 assumption J
5 reiteration 4
6 discharge
7 assumption L
8 not_elim 2, 7
9 explosion 8, J
10 discharge
11 or_elim 3, 4, 5, 7, 9
12 qed

```

Chapter 17, Exercise C.1

Proof presented using Fitch notation:

1	$(J \rightarrow \neg J)$	PR
2	J	AS
3	$\neg J$	$\rightarrow$ E, 1, 2
4	$\perp$	$\neg$ E, 3, 2
5	$\neg J$	$\neg$ I, 2–4

Daeducer commands needed to generate the proof:

```

1 premise (J -> !J)
2 assumption J
3 imp_elim 1, 2
4 not_elim 3, 2
5 discharge
6 not_intro 2, 4
7 qed

```

Chapter 17, Exercise C.2

Proof presented using Fitch notation:

1		$(Q \rightarrow (Q \wedge \neg Q))$	PR
2		Q	AS
3		$(Q \wedge \neg Q)$	$\rightarrow$ E, 1, 2
4		$\neg Q$	$\wedge$ E, 3
5		$\perp$	$\neg$ E, 4, 2
6		$\neg Q$	$\neg$ I, 2–5

Daeducer commands needed to generate the proof:

```

1 premise (Q -> (Q ^ !Q))
2 assumption Q
3 imp_elim 1, 2
4 and_elim_right 3
5 not_elim 4, 2
6 discharge
7 not_intro 2, 5
8 qed

```

Chapter 17, Exercise C.3

Proof presented using Fitch notation:

1		$(A \rightarrow (B \rightarrow C))$	PR
2		$(A \wedge B)$	AS
3		A	$\wedge$ E, 2
4		B	$\wedge$ E, 2
5		$(B \rightarrow C)$	$\rightarrow$ E, 1, 3
6		C	$\rightarrow$ E, 5, 4
7		$((A \wedge B) \rightarrow C)$	$\rightarrow$ I, 2, 6

Daeducer commands needed to generate the proof:

```

1 premise (A -> (B -> C))
2 assumption (A ^ B)
3 and_elim_left 2
4 and_elim_right 2
5 imp_elim 1, 3
6 imp_elim 5, 4
7 discharge
8 imp_intro 2, 6
9 qed

```

Chapter 17, Exercise C.4

Proof presented using Fitch notation:

1	$(K \wedge L)$	PR
2	K	$\wedge E, 1$
3	L	$\wedge E, 1$
4	K	AS
5	L	R, 3
6	$(K \rightarrow L)$	$\rightarrow I, 4, 5$
7	L	AS
8	K	R, 2
9	$(L \rightarrow K)$	$\rightarrow I, 7, 8$
10	$((K \rightarrow L) \wedge (L \rightarrow K))$	$\wedge I, 6, 9$

Daeducer commands needed to generate the proof:

```

1 premise (K ^ L)
2 and_elim_left 1
3 and_elim_right 1
4 assumption K
5 reiteration 3
6 discharge
7 imp_intro 4, 5
8 assumption L
9 reiteration 2
10 discharge
11 imp_intro 8, 9
12 and_intro 7, 11
13 qed

```

Chapter 17, Exercise C.5

Proof presented using Fitch notation:

1	$((C \wedge D) \vee E)$	PR
2	$(C \wedge D)$	AS
3	D	$\wedge E, 2$
4	$(E \vee D)$	$\vee E, 3$
5	E	AS
6	$(E \vee D)$	$\vee E, 5$
7	$(E \vee D)$	$\vee E, 1, 2-4, 5-6$

Daeducer commands needed to generate the proof:

```

1 premise ((C ^ D) v E)
2 assumption (C ^ D)
3 and_elim_right 2
4 or_intro_right 3, E
5 discharge
6 assumption E

```

```

7  or_intro_left 6, D
8  discharge
9  or_elim 1, 2, 4, 6, 7
10 qed

```

Chapter 17, Exercise C.6

Proof presented using Fitch notation:

1		$((A \rightarrow B) \wedge (B \rightarrow A))$	PR
2		$((B \rightarrow C) \wedge (C \rightarrow B))$	PR
3		$(A \rightarrow B)$	$\wedge E, 1$
4		$(B \rightarrow C)$	$\wedge E, 2$
5		A	AS
6		B	$\rightarrow E, 3, 5$
7		C	$\rightarrow E, 4, 6$
8		$(A \rightarrow C)$	$\rightarrow I, 5, 7$
9		$(B \rightarrow A)$	$\wedge E, 1$
10		$(C \rightarrow B)$	$\wedge E, 2$
11		C	AS
12		B	$\rightarrow E, 10, 11$
13		A	$\rightarrow E, 9, 12$
14		$(C \rightarrow A)$	$\rightarrow I, 11, 13$
15		$((A \rightarrow C) \wedge (C \rightarrow A))$	$\wedge I, 8, 14$

Daeducer commands needed to generate the proof:

```

1  premise ((A -> B) ^ (B -> A))
2  premise ((B -> C) ^ (C -> B))
3  and_elim_left 1
4  and_elim_left 2
5  assumption A
6  imp_elim 3, 5
7  imp_elim 4, 6
8  discharge
9  imp_intro 5, 7
10 and_elim_right 1
11 and_elim_right 2
12 assumption C
13 imp_elim 11, 12
14 imp_elim 10, 13
15 discharge
16 imp_intro 12, 14
17 and_intro 9, 16
18 qed

```


Chapter 17, Exercise C.7

Proof presented using Fitch notation:

1		$(\neg F \rightarrow G)$	PR
2		$(F \rightarrow H)$	PR
3			
3		$\neg(G \vee H)$	AS
4			
4		F	AS
5		H	$\rightarrow$ E, 2, 4
6		$(G \vee H)$	$\vee$ E, 5
7		$\perp$	$\neg$ E, 3, 6
8		$\neg F$	$\neg$ I, 4–7
9		G	$\rightarrow$ E, 1, 8
10		$(G \vee H)$	$\vee$ E, 9
11		$\perp$	$\neg$ E, 3, 10
12		$(G \vee H)$	IP, 3–11

Daeducer commands needed to generate the proof:

```

1  premise (!F -> G)
2  premise (F -> H)
3  assumption !(G v H)
4  assumption F
5  imp_elim 2, 4
6  or_intro_right 5, G
7  not_elim 3, 6
8  discharge
9  not_intro 4, 7
10 imp_elim 1, 9
11 or_intro_left 10, H
12 not_elim 3, 11
13 discharge
14 indirect 3, 12
15 qed

```

Chapter 17, Exercise C.8

Proof presented using Fitch notation:

1		$((Z \wedge K) \vee (K \wedge M))$	PR
2		$(K \rightarrow D)$	PR
3		$(Z \wedge K)$	AS
4		K	$\wedge E$, 3
5		D	$\rightarrow E$, 2, 4
6		$(K \wedge M)$	AS
7		K	$\wedge E$, 6
8		D	$\rightarrow E$, 2, 7
9		D	$\vee E$, 1, 3–5, 6–8

Daeducer commands needed to generate the proof:

```

1 premise ((Z ^ K) v (K ^ M))
2 premise (K -> D)
3 assumption (Z ^ K)
4 and_elim_right 3
5 imp_elim 2, 4
6 discharge
7 assumption (K ^ M)
8 and_elim_left 7
9 imp_elim 2, 8
10 discharge
11 or_elim 1, 3, 5, 7, 9
12 qed

```

Chapter 17, Exercise C.9

Proof presented using Fitch notation:

1		$(P \wedge (Q \vee R))$	PR
2		$(P \rightarrow \neg R)$	PR
3		P	$\wedge E$, 1
4		$(Q \vee R)$	$\wedge E$, 1
5		$\neg R$	$\rightarrow E$, 2, 3
6		Q	AS
7		$(Q \vee E)$	$\vee E$, 6
8		R	AS
9		$\perp$	$\neg E$, 5, 8
10		$(Q \vee E)$	X, 9
11		$(Q \vee E)$	$\vee E$, 4, 6–7, 8–10

Daeducer commands needed to generate the proof:

```

1 premise (P ^ (Q v R))

```

```

2  premise (P -> !R)
3  and_elim_left 1
4  and_elim_right 1
5  imp_elim 2, 3
6  assumption Q
7  or_intro_left 6, E
8  discharge
9  assumption R
10 not_elim 5, 9
11 explosion 10, (Q ∨ E)
12 discharge
13 or_elim 4, 6, 7, 9, 11
14 qed

```

Chapter 17, Exercise C.10

Proof presented using Fitch notation:

1	$((S \rightarrow T) \wedge (T \rightarrow S))$	PR
2	$(S \rightarrow T)$	$\wedge E, 1$
3	$(T \rightarrow S)$	$\wedge E, 1$
4	S	AS
5	$(T \vee S)$	$\vee E, 4$
6	$(S \rightarrow (T \vee S))$	$\rightarrow I, 4, 5$
7	$(T \vee S)$	AS
8	T	AS
9	S	$\rightarrow E, 3, 8$
10	S	AS
11	S	R, 10
12	S	$\vee E, 7, 8-9, 10-11$
13	$((T \vee S) \rightarrow S)$	$\rightarrow I, 7, 12$
14	$((S \rightarrow (T \vee S)) \wedge ((T \vee S) \rightarrow S))$	$\wedge I, 6, 13$

Daeducer commands needed to generate the proof:

```

1  premise ((S -> T) ^ (T -> S))
2  and_elim_left 1
3  and_elim_right 1
4  assumption S
5  or_intro_right 4, T
6  discharge
7  imp_intro 4, 5
8  assumption (T ∨ S)
9  assumption T
10 imp_elim 3, 9
11 discharge
12 assumption S
13 reiteration 12
14 discharge
15 or_elim 8, 9, 10, 12, 13
16 discharge
17 imp_intro 8, 15
18 and_intro 7, 17
19 qed

```

Chapter 17, Exercise C.11

Proof presented using Fitch notation:

1	$\neg(P \rightarrow Q)$	PR
2	Q	AS
3	P	AS
4	Q	R, 2
5	$(P \rightarrow Q)$	$\rightarrow$ I, 3, 4
6	$\perp$	$\neg$ E, 1, 5
7	$\neg Q$	$\neg$ I, 2-6

Daeducer commands needed to generate the proof:

```

1 premise !(P -> Q)
2 assumption Q
3 assumption P
4 reiteration 2
5 discharge
6 imp_intro 3, 4
7 not_elim 1, 6
8 discharge
9 not_intro 2, 7
10 qed

```

Chapter 17, Exercise C.12

Proof presented using Fitch notation:

1	$\neg(P \rightarrow Q)$	PR
2	$\neg P$	AS
3	P	AS
4	$\perp$	$\neg$ E, 2, 3
5	Q	X, 4
6	$(P \rightarrow Q)$	$\rightarrow$ I, 3, 5
7	$\perp$	$\neg$ E, 1, 6
8	P	IP, 2-7

Daeducer commands needed to generate the proof:

```

1 premise !(P -> Q)
2 assumption !P
3 assumption P
4 not_elim 2, 3
5 explosion 4, Q
6 discharge
7 imp_intro 3, 5
8 not_elim 1, 7
9 discharge
10 indirect 2, 8
11 qed

```

Chapter 18, Example p.149

Proof presented using Fitch notation:

1	$((A \wedge B) \vee (A \wedge C))$	PR
2	$(A \wedge B)$	AS
3	A	$\wedge E$, 2
4	B	$\wedge E$, 2
5	$(B \vee C)$	$\vee E$, 4
6	$(A \wedge (B \vee C))$	$\wedge I$, 3, 5
7	$(A \wedge C)$	AS
8	A	$\wedge E$, 7
9	C	$\wedge E$, 7
10	$(B \vee C)$	$\vee E$, 9
11	$(A \wedge (B \vee C))$	$\wedge I$, 8, 10
12	$(A \wedge (B \vee C))$	$\vee E$, 1, 2–6, 7–11

Daeducer commands needed to generate the proof:

```

1 premise ((A ^ B) v (A ^ C))
2 assumption (A ^ B)
3 and_elim_left 2
4 and_elim_right 2
5 or_intro_left 4, C
6 and_intro 3, 5
7 discharge
8 assumption (A ^ C)
9 and_elim_left 8
10 and_elim_right 8
11 or_intro_right 10, B
12 and_intro 9, 11
13 discharge
14 or_elim 1, 2, 6, 8, 12
15 qed

```

Chapter 18, Example p.153

Proof presented using Fitch notation:

1			$(A \rightarrow B)$	AS
2				$\neg B$
3				
4				A
5				B
6				$\perp$
7			$\neg A$	$\neg E, 2, 4$
8		$(\neg B \rightarrow \neg A)$	$\rightarrow I, 2, 6$	
9	$((A \rightarrow B) \rightarrow (\neg B \rightarrow \neg A))$	$\rightarrow I, 1, 7$		

Daeducer commands needed to generate the proof:

```

1  assumption (A -> B)
2  assumption !B
3  assumption A
4  imp_elim 1, 3
5  not_elim 2, 4
6  discharge
7  not_intro 3, 5
8  discharge
9  imp_intro 2, 7
10 discharge
11 imp_intro 1, 9
12 qed

```

Chapter 18, Example p.156

Proof presented using Fitch notation:

1		$(A \vee B)$	PR
2		$\neg A$	PR
3			A
4			$\perp$
5			B
6			B
7			B
8		B	$\vee E, 1, 3-5, 6-7$

Daeducer commands needed to generate the proof:

```

1  premise (A v B)
2  premise !A
3  assumption A
4  not_elim 2, 3
5  explosion 4, B
6  discharge
7  assumption B
8  reiteration 7
9  discharge

```

```

10  or_elim 1, 3, 5, 7, 8
11  qed

```

Chapter 18, Example p.158

Proof presented using Fitch notation:

1		$\neg(A \vee \neg A)$	AS
2		A	AS
3		$(A \vee \neg A)$	$\vee E$, 2
4		$\perp$	$\neg E$, 1, 3
5		$\neg A$	$\neg I$, 2–4
6		$\neg A$	AS
7		$(A \vee \neg A)$	$\vee E$, 6
8		$\perp$	$\neg E$, 1, 7
9		$\neg\neg A$	$\neg I$, 6–8
10		$\perp$	$\neg E$, 9, 5
11		$(A \vee \neg A)$	IP, 1–10

Daeducer commands needed to generate the proof:

```

1  assumption !(A v !A)
2  assumption A
3  or_intro_left 2, !A
4  not_elim 1, 3
5  discharge
6  not_intro 2, 4
7  assumption !A
8  or_intro_right 7, A
9  not_elim 1, 8
10 discharge
11 not_intro 7, 9
12 not_elim 11, 6
13 discharge
14 indirect 1, 12
15 qed

```

Chapter 18, Exercise A.1

Proof presented using Fitch notation:

1	$(A \rightarrow B)$	PR
2	$(A \rightarrow C)$	PR
3	A	AS
4	B	$\rightarrow$ E, 1, 3
5	C	$\rightarrow$ E, 2, 3
6	$(B \wedge C)$	$\wedge$ I, 4, 5
7	$(A \rightarrow (B \wedge C))$	$\rightarrow$ I, 3, 6

Daeducer commands needed to generate the proof:

```

1 premise (A -> B)
2 premise (A -> C)
3 assumption A
4 imp_elim 1, 3
5 imp_elim 2, 3
6 and_intro 4, 5
7 discharge
8 imp_intro 3, 6
9 qed

```

Chapter 18, Exercise A.2

Proof presented using Fitch notation:

1	$((A \wedge B) \rightarrow C)$	PR
2	A	AS
3	B	AS
4	$(A \wedge B)$	$\wedge$ I, 2, 3
5	C	$\rightarrow$ E, 1, 4
6	$(B \rightarrow C)$	$\rightarrow$ I, 3, 5
7	$(A \rightarrow (B \rightarrow C))$	$\rightarrow$ I, 2, 6

Daeducer commands needed to generate the proof:

```

1 premise ((A ^ B) -> C)
2 assumption A
3 assumption B
4 and_intro 2, 3
5 imp_elim 1, 4
6 discharge
7 imp_intro 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```


Chapter 18, Exercise A.3

Proof presented using Fitch notation:

1	$(A \rightarrow (B \rightarrow C))$	PR
2	$(A \rightarrow B)$	AS
3	A	AS
4	$(B \rightarrow C)$	$\rightarrow$ E, 1, 3
5	B	$\rightarrow$ E, 2, 3
6	C	$\rightarrow$ E, 4, 5
7	$(A \rightarrow C)$	$\rightarrow$ I, 3, 6
8	$((A \rightarrow B) \rightarrow (A \rightarrow C))$	$\rightarrow$ I, 2, 7

Daeducer commands needed to generate the proof:

```

1 premise (A -> (B -> C))
2 assumption (A -> B)
3 assumption A
4 imp_elim 1, 3
5 imp_elim 2, 3
6 imp_elim 4, 5
7 discharge
8 imp_intro 3, 6
9 discharge
10 imp_intro 2, 8
11 qed

```

Chapter 18, Exercise A.4

Proof presented using Fitch notation:

1	$(A \vee (B \wedge C))$	PR
2	A	AS
3	$(A \vee B)$	$\vee$ E, 2
4	$(A \vee C)$	$\vee$ E, 2
5	$((A \vee B) \wedge (A \vee C))$	$\wedge$ I, 3, 4
6	$(B \wedge C)$	AS
7	B	$\wedge$ E, 6
8	$(A \vee B)$	$\vee$ E, 7
9	C	$\wedge$ E, 6
10	$(A \vee C)$	$\vee$ E, 9
11	$((A \vee B) \wedge (A \vee C))$	$\wedge$ I, 8, 10
12	$((A \vee B) \wedge (A \vee C))$	$\vee$ E, 1, 2-5, 6-11

Daeducer commands needed to generate the proof:

```

1 premise (A ∨ (B ∧ C))

```

```

2  assumption A
3  or_intro_left 2, B
4  or_intro_left 2, C
5  and_intro 3, 4
6  discharge
7  assumption (B  $\wedge$  C)
8  and_elim_left 7
9  or_intro_right 8, A
10 and_elim_right 7
11 or_intro_right 10, A
12 and_intro 9, 11
13 discharge
14 or_elim 1, 2, 5, 7, 12
15 qed

```

Chapter 18, Exercise A.5

Proof presented using Fitch notation:

1		$((A \wedge B) \vee (A \wedge C))$	PR
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

Daeducer commands needed to generate the proof:

```

1  premise ((A  $\wedge$  B)  $\vee$  (A  $\wedge$  C))
2  assumption (A  $\wedge$  B)
3  and_elim_left 2
4  and_elim_right 2
5  or_intro_left 4, C
6  and_intro 3, 5
7  discharge
8  assumption (A  $\wedge$  C)
9  and_elim_left 8
10 and_elim_right 8
11 or_intro_right 10, B
12 and_intro 9, 11
13 discharge
14 or_elim 1, 2, 6, 8, 12
15 qed

```

Chapter 18, Exercise A.6

Proof presented using Fitch notation:

1		$(A \vee B)$	PR
2		$(A \rightarrow C)$	PR
3		$(B \rightarrow D)$	PR
4			
5			
6			
7			
8			
9			
10			

$$\begin{array}{llll}
 1 & | & (A \vee B) & \text{PR} \\
 2 & | & (A \rightarrow C) & \text{PR} \\
 3 & | & (B \rightarrow D) & \text{PR} \\
 4 & | & | & \\
 5 & | & | & \\
 6 & | & | & \\
 7 & | & | & \\
 8 & | & | & \\
 9 & | & | & \\
 10 & | & &
 \end{array}$$

Daeducer commands needed to generate the proof:

```

1 premise (A ∨ B)
2 premise (A -> C)
3 premise (B -> D)
4 assumption A
5 imp_elim 2, 4
6 or_intro_left 5, D
7 discharge
8 assumption B
9 imp_elim 3, 8
10 or_intro_right 9, C
11 discharge
12 or_elim 1, 4, 6, 8, 10
13 qed

```

Chapter 18, Exercise A.7

Proof presented using Fitch notation:

1		$(\neg A \vee \neg B)$	PR
2			
3			
4			
5			
6			
7			
8			
9			
10			

$$\begin{array}{llll}
 1 & | & (\neg A \vee \neg B) & \text{PR} \\
 2 & | & | & \\
 3 & | & | & \\
 4 & | & | & \\
 5 & | & | & \\
 6 & | & | & \\
 7 & | & | & \\
 8 & | & | & \\
 9 & | & | & \\
 10 & | & &
 \end{array}$$

Daeducer commands needed to generate the proof:

```

1  premise (!A ∨ !B)
2  assumption (A ∧ B)
3  assumption !A
4  and_elim_left 2
5  not_elim 3, 4
6  discharge
7  assumption !B
8  and_elim_right 2
9  not_elim 7, 8
10 discharge
11 or_elim 1, 3, 5, 7, 9
12 discharge
13 not_intro 2, 11
14 qed

```

Chapter 18, Exercise A.8

Proof presented using Fitch notation:

1		$(A \wedge \neg B)$	PR
2			
3			
4			
5			
6			
7			

Daeducer commands needed to generate the proof:

```

1  premise (A ∧ ¬B)
2  assumption (A → B)
3  and_elim_left 1
4  and_elim_right 1
5  imp_elim 2, 3
6  not_elim 4, 5
7  discharge
8  not_intro 2, 6
9  qed

```

Chapter 18, Exercise C.1

Proof presented using Fitch notation:

1			$\neg A$	AS
2				A
3				$\perp$
4			$(A \rightarrow \perp)$	$\rightarrow I, 2, 3$
5		$(\neg A \rightarrow (A \rightarrow \perp))$		$\rightarrow I, 1, 4$

Daeducer commands needed to generate the proof:

```

1  assumption !A
2  assumption A
3  not_elim 1, 2
4  discharge
5  imp_intro 2, 3
6  discharge
7  imp_intro 1, 5
8  qed

```

Chapter 18, Exercise C.2

Proof presented using Fitch notation:

1			$(A \wedge \neg A)$	AS
2			A	$\wedge E, 1$
3			$\neg A$	$\wedge E, 1$
4			$\perp$	$\neg E, 3, 2$
5		$\neg(A \wedge \neg A)$		$\neg I, 1-4$

Daeducer commands needed to generate the proof:

```

1  assumption (A ~ !A)
2  and_elim_left 1
3  and_elim_right 1
4  not_elim 3, 2
5  discharge
6  not_intro 1, 4
7  qed

```

Chapter 18, Exercise C.3

Proof presented using Fitch notation:

1		$((A \rightarrow C) \wedge (B \rightarrow C))$	AS
2		$(A \vee B)$	AS
3		$(A \rightarrow C)$	$\wedge E, 1$
4		$(B \rightarrow C)$	$\wedge E, 1$
5		A	AS
6		C	$\rightarrow E, 3, 5$
7		B	AS
8		C	$\rightarrow E, 4, 7$
9		C	$\vee E, 2, 5-6, 7-8$
10		$((A \vee B) \rightarrow C)$	$\rightarrow I, 2, 9$
11		$((A \rightarrow C) \wedge (B \rightarrow C)) \rightarrow ((A \vee B) \rightarrow C)$	$\rightarrow I, 1, 10$

Daeducer commands needed to generate the proof:

```

1  assumption ((A -> C) ^ (B -> C))
2  assumption (A v B)
3  and_elim_left 1
4  and_elim_right 1
5  assumption A
6  imp_elim 3, 5
7  discharge
8  assumption B
9  imp_elim 4, 8
10 discharge
11 or_elim 2, 5, 6, 8, 9
12 discharge
13 imp_intro 2, 11
14 discharge
15 imp_intro 1, 13
16 qed

```

Chapter 18, Exercise C.4

Proof presented using Fitch notation:

1		$\neg(A \rightarrow B)$	AS
2		$\neg A$	AS
3		A	AS
4		$\perp$	$\neg E$, 2, 3
5		B	X, 4
6		$(A \rightarrow B)$	$\rightarrow I$, 3, 5
7		$\perp$	$\neg E$, 1, 6
8		A	IP, 2–7
9		B	AS
10		A	AS
11		B	R, 9
12		$(A \rightarrow B)$	$\rightarrow I$, 10, 11
13		$\perp$	$\neg E$, 1, 12
14		$\neg B$	$\neg I$, 9–13
15		$(A \wedge \neg B)$	$\wedge I$, 8, 14
16		$(\neg(A \rightarrow B) \rightarrow (A \wedge \neg B))$	$\rightarrow I$, 1, 15

Daeducer commands needed to generate the proof:

```

1  assumption !(A -> B)
2  assumption !A
3  assumption A
4  not_elim 2, 3
5  explosion 4, B
6  discharge
7  imp_intro 3, 5
8  not_elim 1, 7
9  discharge
10 indirect 2, 8
11 assumption B
12 assumption A
13 reiteration 11
14 discharge
15 imp_intro 12, 13
16 not_elim 1, 15
17 discharge
18 not_intro 11, 16
19 and_intro 10, 18
20 discharge
21 imp_intro 1, 19
22 qed

```

Chapter 18, Exercise C.5

Proof presented using Fitch notation:

1			$(\neg A \vee B)$	AS
2				
				A
				AS
3				
				$\neg A$
				AS
4				
				$\perp$
				$\neg E, 3, 2$
5				
				B
				X, 4
6				
				B
				AS
7				
				B
				R, 6
8				B
				$\vee E, 1, 3-5, 6-7$
9			$(A \rightarrow B)$	$\rightarrow I, 2, 8$
10		$((\neg A \vee B) \rightarrow (A \rightarrow B))$		$\rightarrow I, 1, 9$

Daeducer commands needed to generate the proof:

```

1  assumption (!A ∨ B)
2  assumption A
3  assumption !A
4  not_elim 3, 2
5  explosion 4, B
6  discharge
7  assumption B
8  reiteration 7
9  discharge
10 or_elim 1, 3, 5, 7, 8
11 discharge
12 imp_intro 2, 10
13 discharge
14 imp_intro 1, 12
15 qed

```


Chapter 18, Exercise D.1

Proof presented using Fitch notation:

1			$\neg\neg A$	AS
2				$\neg A$
3				$\perp$
4			A	IP, 2-3
5		$(\neg\neg A \rightarrow A)$		$\rightarrow$ I, 1, 4

Daeducer commands needed to generate the proof:

```

1  assumption !!A
2  assumption !A
3  not_elim 1, 2
4  discharge
5  indirect 2, 3
6  discharge
7  imp_intro 1, 5
8  qed

```

Chapter 18, Exercise D.2

Proof presented using Fitch notation:

1		$(\neg A \rightarrow \neg B)$	PR
2			AS
3			AS
4			$\rightarrow$ E, 1, 3
5			$\neg$ E, 4, 2
6		A	IP, 3-5
7		$(B \rightarrow A)$	$\rightarrow$ I, 2, 6

Daeducer commands needed to generate the proof:

```

1  premise (!A -> !B)
2  assumption B
3  assumption !A
4  imp_elim 1, 3
5  not_elim 4, 2
6  discharge
7  indirect 3, 5
8  discharge
9  imp_intro 2, 7
10 qed

```

Chapter 18, Exercise D.3

Proof presented using Fitch notation:

1	$(A \rightarrow B)$	PR
2	$\neg(\neg A \vee B)$	AS
3	A	AS
4	B	$\rightarrow$ E, 1, 3
5	$(\neg A \vee B)$	$\vee$ E, 4
6	$\perp$	$\neg$ E, 2, 5
7	$\neg A$	$\neg$ I, 3–6
8	$(\neg A \vee B)$	$\vee$ E, 7
9	$\perp$	$\neg$ E, 2, 8
10	$(\neg A \vee B)$	IP, 2–9

Daeducer commands needed to generate the proof:

```

1  premise (A -> B)
2  assumption !(A v B)
3  assumption A
4  imp_elim 1, 3
5  or_intro_right 4, !A
6  not_elim 2, 5
7  discharge
8  not_intro 3, 6
9  or_intro_left 8, B
10 not_elim 2, 9
11 discharge
12 indirect 2, 10
13 qed

```

Chapter 18, Exercise D.4

Proof presented using Fitch notation:

1		$\neg(A \wedge B)$	AS
2		$\neg(\neg A \vee \neg B)$	AS
3		$\neg A$	AS
4		$(\neg A \vee \neg B)$	$\vee E, 3$
5		$\perp$	$\neg E, 2, 4$
6		A	IP, 3–5
7		$\neg B$	AS
8		$(\neg A \vee \neg B)$	$\vee E, 7$
9		$\perp$	$\neg E, 2, 8$
10		B	IP, 7–9
11		$(A \wedge B)$	$\wedge I, 6, 10$
12		$\perp$	$\neg E, 1, 11$
13		$(\neg A \vee \neg B)$	IP, 2–12
14		$(\neg(A \wedge B) \rightarrow (\neg A \vee \neg B))$	$\rightarrow I, 1, 13$

Daeducer commands needed to generate the proof:

```

1  assumption !(A ~ B)
2  assumption !(~A ~ !B)
3  assumption !A
4  or_intro_left 3, !B
5  not_elim 2, 4
6  discharge
7  indirect 3, 5
8  assumption !B
9  or_intro_right 8, !A
10 not_elim 2, 9
11 discharge
12 indirect 8, 10
13 and_intro 7, 12
14 not_elim 1, 13
15 discharge
16 indirect 2, 14
17 discharge
18 imp_intro 1, 16
19 qed

```

Chapter 18, Exercise D.5

Proof presented using Fitch notation:

1	$(A \rightarrow (B \vee C))$	PR
2	$\neg((A \rightarrow B) \vee (A \rightarrow C))$	AS
3	$(A \rightarrow B)$	AS
4	$((A \rightarrow B) \vee (A \rightarrow C))$	$\vee$ E, 3
5	$\perp$	$\neg$ E, 2, 4
6	$\neg(A \rightarrow B)$	$\neg$ I, 3–5
7	$(A \rightarrow C)$	AS
8	$((A \rightarrow B) \vee (A \rightarrow C))$	$\vee$ E, 7
9	$\perp$	$\neg$ E, 2, 8
10	$\neg(A \rightarrow C)$	$\neg$ I, 7–9
11	$\neg A$	AS
12	A	AS
13	$\perp$	$\neg$ E, 11, 12
14	B	X, 13
15	$(A \rightarrow B)$	$\rightarrow$ I, 12, 14
16	$\perp$	$\neg$ E, 6, 15
17	A	IP, 11–16
18	$(B \vee C)$	$\rightarrow$ E, 1, 17
19	B	AS
20	A	AS
21	B	R, 19
22	$(A \rightarrow B)$	$\rightarrow$ I, 20, 21
23	$\perp$	$\neg$ E, 6, 22
24	$\neg B$	$\neg$ I, 19–23
25	C	AS
26	A	AS
27	C	R, 25
28	$(A \rightarrow C)$	$\rightarrow$ I, 26, 27
29	$\perp$	$\neg$ E, 10, 28
30	$\neg C$	$\neg$ I, 25–29
31	$\perp$	$\vee$ E, 18, 19–23, 25–29
32	$((A \rightarrow B) \vee (A \rightarrow C))$	IP, 2–31

Daeducer commands needed to generate the proof:

```
1  premise (A -> (B v C))
2  assumption !((A -> B) v (A -> C))
3  assumption (A -> B)
4  or_intro_left 3, (A -> C)
5  not_elim 2, 4
6  discharge
7  not_intro 3, 5
8  assumption (A -> C)
9  or_intro_right 8, (A -> B)
10 not_elim 2, 9
11 discharge
12 not_intro 8, 10
13 assumption !A
14 assumption A
15 not_elim 13, 14
16 explosion 15, B
17 discharge
18 imp_intro 14, 16
19 not_elim 7, 18
20 discharge
21 indirect 13, 19
22 imp_elim 1, 21
23 assumption B
24 assumption A
25 reiteration 23
26 discharge
27 imp_intro 24, 25
28 not_elim 7, 27
29 discharge
30 not_intro 23, 28
31 assumption C
32 assumption A
33 reiteration 31
34 discharge
35 imp_intro 32, 33
36 not_elim 12, 35
37 discharge
38 not_intro 31, 36
39 or_elim 22, 23, 28, 31, 36
40 discharge
41 indirect 2, 39
42 qed
```

Chapter 18, Exercise D.6

Proof presented using Fitch notation:

1		$\neg((A \rightarrow B) \vee (B \rightarrow A))$	AS
2		$(A \rightarrow B)$	AS
3		$((A \rightarrow B) \vee (B \rightarrow A))$	$\vee E$, 2
4		$\perp$	$\neg E$, 1, 3
5		$\neg(A \rightarrow B)$	$\neg I$, 2–4
6		$(B \rightarrow A)$	AS
7		$((A \rightarrow B) \vee (B \rightarrow A))$	$\vee E$, 6
8		$\perp$	$\neg E$, 1, 7
9		$\neg(B \rightarrow A)$	$\neg I$, 6–8
10		$\neg A$	AS
11		A	AS
12		$\neg A$	R, 10
13		$\perp$	$\neg E$, 12, 11
14		B	X, 13
15		$(A \rightarrow B)$	$\rightarrow I$, 11, 14
16		$\perp$	$\neg E$, 5, 15
17		A	IP, 10–16
18		B	AS
19		A	AS
20		B	R, 18
21		$(A \rightarrow B)$	$\rightarrow I$, 19, 20
22		$\perp$	$\neg E$, 5, 21
23		$\neg B$	$\neg I$, 18–22
24		$\neg B$	AS
25		B	AS
26		$\neg B$	R, 24
27		$\perp$	$\neg E$, 26, 25
28		A	X, 27
29		$(B \rightarrow A)$	$\rightarrow I$, 25, 28
30		$\perp$	$\neg E$, 9, 29
31		B	IP, 24–30
32		A	AS

33				B	AS
34				A	R, 32
35				$(B \rightarrow A)$	$\rightarrow$ I, 33, 34
36				$\perp$	$\neg$ E, 9, 35
37			$\neg A$		$\neg$ I, 32–36
38			$\perp$		$\neg$ E, 37, 17
39		$((A \rightarrow B) \vee (B \rightarrow A))$			IP, 1–38

Daeducer commands needed to generate the proof:

```

1  assumption !((A -> B) ∨ (B -> A))
2  assumption (A -> B)
3  or_intro_left 2, (B -> A)
4  not_elim 1, 3
5  discharge
6  not_intro 2, 4
7  assumption (B -> A)
8  or_intro_right 7, (A -> B)
9  not_elim 1, 8
10 discharge
11 not_intro 7, 9
12 assumption !A
13 assumption A
14 reiteration 12
15 not_elim 14, 13
16 explosion 15, B
17 discharge
18 imp_intro 13, 16
19 not_elim 6, 18
20 discharge
21 indirect 12, 19
22 assumption B
23 assumption A
24 reiteration 22
25 discharge
26 imp_intro 23, 24
27 not_elim 6, 26
28 discharge
29 not_intro 22, 27
30 assumption !B
31 assumption B
32 reiteration 30
33 not_elim 32, 31
34 explosion 33, A
35 discharge
36 imp_intro 31, 34
37 not_elim 11, 36
38 discharge
39 indirect 30, 37
40 assumption A
41 assumption B
42 reiteration 40
43 discharge
44 imp_intro 41, 42
45 not_elim 11, 44
46 discharge
47 not_intro 40, 45
48 not_elim 47, 21
49 discharge
50 indirect 1, 48
51 qed

```

Chapter 18, Exercise D.7

Proof presented using Fitch notation:

1			$((A \rightarrow B) \rightarrow A)$	AS
2				
3				
4				
5				
6				
7				
8				
9				
10				

1			$((A \rightarrow B) \rightarrow A)$	AS
2				
3				
4				
5				
6				
7				
8				
9				
10				

Daeducer commands needed to generate the proof:

```

1  assumption ((A -> B) -> A)
2  assumption !A
3  assumption A
4  not_elim 2, 3
5  explosion 4, B
6  discharge
7  imp_intro 3, 5
8  imp_elim 1, 7
9  not_elim 2, 8
10 discharge
11 indirect 2, 9
12 discharge
13 imp_intro 1, 11
14 qed

```


Chapter 19, Example p.163

Proof presented using Fitch notation:

1		$(A \vee B)$	PR
2		$\neg A$	PR
3			A AS
4			$\perp$ $\neg E, 2, 3$
5			B X, 4
6			B AS
7			B R, 6
8		B	$\vee E, 1, 3-5, 6-7$

Daeducer commands needed to generate the proof:

```

1 premise (A ∨ B)
2 premise !A
3 assumption A
4 not_elim 2, 3
5 explosion 4, B
6 discharge
7 assumption B
8 reiteration 7
9 discharge
10 or_elim 1, 3, 5, 7, 8
11 qed

```

Chapter 19, Example p.163

Proof presented using Fitch notation:

1		$(A \vee B)$	PR
2		$\neg B$	PR
3			A AS
4			A R, 3
5			B AS
6			$\perp$ $\neg E, 2, 5$
7			A X, 6
8		A	$\vee E, 1, 3-4, 5-7$

Daeducer commands needed to generate the proof:

```

1 premise (A ∨ B)
2 premise !B
3 assumption A
4 reiteration 3
5 discharge
6 assumption B
7 not_elim 2, 6
8 explosion 7, A
9 discharge
10 or_elim 1, 3, 4, 6, 8

```

Chapter 19, Example p.164

Proof presented using Fitch notation:

1		$(A \rightarrow B)$	PR
2		$\neg B$	PR
3			
4			
5			
6			

3		A	AS
4		B	$\rightarrow E, 1, 3$
5		$\perp$	$\neg E, 2, 4$
6		$\neg A$	$\neg I, 3-5$

Daeducer commands needed to generate the proof:

```

1 premise (A -> B)
2 premise !B
3 assumption A
4 imp_elim 1, 3
5 not_elim 2, 4
6 discharge
7 not_intro 3, 5
8 qed

```

Chapter 19, Example p.164

Proof presented using Fitch notation:

1		$\neg\neg A$	PR
2			
3			
4			

2		$\neg A$	AS
3		$\perp$	$\neg E, 1, 2$
4		A	IP, 2-3

Daeducer commands needed to generate the proof:

```

1 premise !!A
2 assumption !A
3 not_elim 1, 2
4 discharge
5 indirect 2, 3
6 qed

```

Chapter 19, Example p.165

Proof presented using Fitch notation:

1		$(A \rightarrow B)$	PR
2		$(\neg A \rightarrow B)$	PR
3		$\neg B$	AS
4		A	AS
5		B	$\rightarrow$ E, 1, 4
6		$\perp$	$\neg$ E, 3, 5
7		$\neg A$	$\neg$ I, 4–6
8		$\neg A$	AS
9		B	$\rightarrow$ E, 2, 8
10		$\perp$	$\neg$ E, 3, 9
11		$\neg\neg A$	$\neg$ I, 8–10
12		$\perp$	$\neg$ E, 11, 7
13		B	IP, 3–12

Daeducer commands needed to generate the proof:

```

1 premise (A -> B)
2 premise (!A -> B)
3 assumption !B
4 assumption A
5 imp_elim 1, 4
6 not_elim 3, 5
7 discharge
8 not_intro 4, 6
9 assumption !A
10 imp_elim 2, 9
11 not_elim 3, 10
12 discharge
13 not_intro 9, 11
14 not_elim 13, 8
15 discharge
16 indirect 3, 14
17 qed

```

Chapter 19, Example p.166

Proof presented using Fitch notation:

1	P	PR
2	$\neg(D \vee \neg D)$	AS
3	D	AS
4	$(D \vee \neg D)$	$\vee E$, 3
5	$\perp$	$\neg E$, 2, 4
6	$\neg D$	$\neg I$, 3–5
7	$(D \vee \neg D)$	$\vee E$, 6
8	$\perp$	$\neg E$, 2, 7
9	$(D \vee \neg D)$	IP, 2–8
10	D	AS
11	$(P \wedge D)$	$\wedge I$, 1, 10
12	$((P \wedge D) \vee (P \wedge \neg D))$	$\vee E$, 11
13	$\neg D$	AS
14	$(P \wedge \neg D)$	$\wedge I$, 1, 13
15	$((P \wedge D) \vee (P \wedge \neg D))$	$\vee E$, 14
16	$((P \wedge D) \vee (P \wedge \neg D))$	$\vee E$, 9, 10–12, 13–15

Daeducer commands needed to generate the proof:

```

1  premise P
2  assumption !(D ∨ !D)
3  assumption D
4  or_intro_left 3, !D
5  not_elim 2, 4
6  discharge
7  not_intro 3, 5
8  or_intro_right 7, D
9  not_elim 2, 8
10 discharge
11 indirect 2, 9
12 assumption D
13 and_intro 1, 12
14 or_intro_left 13, (P ∧ !D)
15 discharge
16 assumption !D
17 and_intro 1, 16
18 or_intro_right 17, (P ∧ D)
19 discharge
20 or_elim 11, 12, 14, 16, 18
21 qed

```

Chapter 19, Exame p.166

Proof presented using Fitch notation:

1	$(A \rightarrow \neg A)$	PR
2	A	AS
3	$\neg A$	$\rightarrow E, 1, 2$
4	$\perp$	$\neg E, 3, 2$
5	$\neg A$	$\neg I, 2-4$

Daeducer commands needed to generate the proof:

```

1 premise (A -> !A)
2 assumption A
3 imp_elim 1, 2
4 not_elim 3, 2
5 discharge
6 not_intro 2, 4
7 qed

```

Chapter 19, Example p.167

Proof presented using Fitch notation:

1	$\neg(A \wedge B)$	PR
2	$\neg(\neg A \vee \neg B)$	AS
3	$\neg A$	AS
4	$(\neg A \vee \neg B)$	$\vee E, 3$
5	$\perp$	$\neg E, 2, 4$
6	$\neg\neg A$	$\neg I, 3-5$
7	$\neg B$	AS
8	$(\neg A \vee \neg B)$	$\vee E, 7$
9	$\perp$	$\neg E, 2, 8$
10	$\neg\neg B$	$\neg I, 7-9$
11	$\neg A$	AS
12	$\perp$	$\neg E, 6, 11$
13	A	IP, 11-12
14	$\neg B$	AS
15	$\perp$	$\neg E, 10, 14$
16	B	IP, 14-15
17	$(A \wedge B)$	$\wedge I, 13, 16$
18	$\perp$	$\neg E, 1, 17$
19	$(\neg A \vee \neg B)$	IP, 2-18

Daeducer commands needed to generate the proof:

```

1  premise !(A ^ B)
2  assumption !(A v !B)
3  assumption !A
4  or_intro_left 3, !B
5  not_elim 2, 4
6  discharge
7  not_intro 3, 5
8  assumption !B
9  or_intro_right 8, !A
10 not_elim 2, 9
11 discharge
12 not_intro 8, 10
13 assumption !A
14 not_elim 7, 13
15 discharge
16 indirect 13, 14
17 assumption !B
18 not_elim 12, 17
19 discharge
20 indirect 17, 18
21 and_intro 16, 20
22 not_elim 1, 21
23 discharge
24 indirect 2, 22
25 qed

```

Chapter 19, Example p.167

Proof presented using Fitch notation:

1		$(\neg A \vee \neg B)$	PR
2			
3			
4			
5			
6			
7			
8			
9			
10			

Daeducer commands needed to generate the proof:

```

1  premise (!A v !B)
2  assumption (A ^ B)
3  and_elim_left 2
4  and_elim_right 2
5  assumption !A
6  not_elim 5, 3
7  discharge
8  assumption !B
9  not_elim 8, 4
10 discharge
11 or_elim 1, 5, 6, 8, 9
12 discharge
13 not_intro 2, 11
14 qed

```

Chapter 19, Example p.167

Proof presented using Fitch notation:

1	$\neg(A \vee B)$	PR
2	A	AS
3	$(A \vee B)$	$\vee E$, 2
4	$\perp$	$\neg E$, 1, 3
5	$\neg A$	$\neg I$, 2–4
6	B	AS
7	$(A \vee B)$	$\vee E$, 6
8	$\perp$	$\neg E$, 1, 7
9	$\neg B$	$\neg I$, 6–8
10	$(\neg A \wedge \neg B)$	$\wedge I$, 5, 9

Daeducer commands needed to generate the proof:

```

1 premise !(A v B)
2 assumption A
3 or_intro_left 2, B
4 not_elim 1, 3
5 discharge
6 not_intro 2, 4
7 assumption B
8 or_intro_right 7, A
9 not_elim 1, 8
10 discharge
11 not_intro 7, 9
12 and_intro 6, 11
13 qed

```

Chapter 19, Example p.167

Proof presented using Fitch notation:

1	$(\neg A \wedge \neg B)$	PR
2	$\neg A$	$\wedge E$, 1
3	$\neg B$	$\wedge E$, 1
4	$(A \vee B)$	AS
5	A	AS
6	$\perp$	$\neg E$, 2, 5
7	B	AS
8	$\perp$	$\neg E$, 3, 7
9	$\perp$	$\vee E$, 4, 5–6, 7–8
10	$\neg(A \vee B)$	$\neg I$, 4–9

Daeducer commands needed to generate the proof:

```

1  premise (!A ~ !B)
2  and_elim_left 1
3  and_elim_right 1
4  assumption (A ~ B)
5  assumption A
6  not_elim 2, 5
7  discharge
8  assumption B
9  not_elim 3, 8
10 discharge
11 or_elim 4, 5, 6, 8, 9
12 discharge
13 not_intro 4, 11
14 qed

```

Chapter 19, Exercise A.1

Proof presented using Fitch notation:

1		$(W \rightarrow \neg B)$	PR
2		$(A \wedge W)$	PR
3		$(B \vee (J \wedge K))$	PR
4		W	$\wedge E$, 2
5		$\neg B$	$\rightarrow E$, 1, 4
6		B	AS
7		$\perp$	$\neg E$, 5, 6
8		K	X, 7
9		$(J \wedge K)$	AS
10		K	$\wedge E$, 9
11		K	$\vee E$, 3, 6–8, 9–10

Daeducer commands needed to generate the proof:

```

1  premise (W -> !B)
2  premise (A ~ W)
3  premise (B ~ (J ~ K))
4  and_elim_right 2
5  imp_elim 1, 4
6  assumption B
7  not_elim 5, 6
8  explosion 7, K
9  discharge
10 assumption (J ~ K)
11 and_elim_right 10
12 discharge
13 or_elim 3, 6, 8, 10, 11
14 qed

```


Chapter 19, Exercise A.2

Proof presented using Fitch notation:

1		$((L \rightarrow \neg O) \wedge (\neg O \rightarrow L))$	PR
2		$(L \vee \neg O)$	PR
3		$(L \rightarrow \neg O)$	$\wedge E, 1$
4		$(\neg O \rightarrow L)$	$\wedge E, 1$
5		L	AS
6		L	R, 5
7		$\neg O$	AS
8		L	$\rightarrow E, 4, 7$
9		L	$\vee E, 2, 5-6, 7-8$

Daeducer commands needed to generate the proof:

```
1 premise ((L -> !O) ^ (!O -> L))
2 premise (L v !O)
3 and_elim_left 1
4 and_elim_right 1
5 assumption L
6 reiteration 5
7 discharge
8 assumption !O
9 imp_elim 4, 8
10 discharge
11 or_elim 2, 5, 6, 8, 9
12 qed
```

Chapter 19, Exercise A.3

Proof presented using Fitch notation:

1		$(Z \rightarrow (C \wedge \neg N))$	PR
2		$(\neg Z \rightarrow (N \wedge \neg C))$	PR
3		$\neg(N \vee C)$	AS
4		N	AS
5		$(N \vee C)$	$\vee E$, 4
6		$\perp$	$\neg E$, 3, 5
7		$\neg N$	$\neg I$, 4–6
8		C	AS
9		$(N \vee C)$	$\vee E$, 8
10		$\perp$	$\neg E$, 3, 9
11		$\neg C$	$\neg I$, 8–10
12		Z	AS
13		$(C \wedge \neg N)$	$\rightarrow E$, 1, 12
14		C	$\wedge E$, 13
15		$\perp$	$\neg E$, 11, 14
16		$\neg Z$	$\neg I$, 12–15
17		$(N \wedge \neg C)$	$\rightarrow E$, 2, 16
18		N	$\wedge E$, 17
19		$\perp$	$\neg E$, 7, 18
20		$(N \vee C)$	IP, 3–19

Daeducer commands needed to generate the proof:

```

1 premise (Z -> (C ^ !N))
2 premise (!Z -> (N ^ !C))
3 assumption !(N v C)
4 assumption N
5 or_intro_left 4, C
6 not_elim 3, 5
7 discharge
8 not_intro 4, 6
9 assumption C
10 or_intro_right 9, N
11 not_elim 3, 10
12 discharge
13 not_intro 9, 11
14 assumption Z
15 imp_elim 1, 14
16 and_elim_left 15
17 not_elim 13, 16
18 discharge
19 not_intro 14, 17
20 imp_elim 2, 19
21 and_elim_left 20
22 not_elim 8, 21
23 discharge
24 indirect 3, 22

```

Chapter 19, Exercise B.1

Proof presented using Fitch notation:

1		$(E \vee F)$	PR
2		$(F \vee G)$	PR
3		$\neg F$	PR
4			
5			
6			
7			
8			
9		E	$\vee E$, 1, 4–5, 6–8
10			
11			
12			
13			
14			
15		G	$\vee E$, 2, 10–12, 13–14
16		$(E \wedge G)$	$\wedge I$, 9, 15

Daeducer commands needed to generate the proof:

```

1  premise (E v F)
2  premise (F v G)
3  premise !F
4  assumption E
5  reiteration 4
6  discharge
7  assumption F
8  not_elim 3, 7
9  explosion 8, E
10 discharge
11 or_elim 1, 4, 5, 7, 9
12 assumption F
13 not_elim 3, 12
14 explosion 13, G
15 discharge
16 assumption G
17 reiteration 16
18 discharge
19 or_elim 2, 12, 14, 16, 17
20 and_intro 11, 19
21 qed

```

Chapter 19, Exercise B.2

Proof presented using Fitch notation:

1		$(M \vee (N \rightarrow M))$	PR
2		$\neg M$	AS
3		M	AS
4		$\perp$	$\neg$ E, 2, 3
5		$\neg N$	X, 4
6		$(N \rightarrow M)$	AS
7		N	AS
8		M	$\rightarrow$ E, 6, 7
9		$\perp$	$\neg$ E, 2, 8
10		$\neg N$	$\neg$ I, 7–9
11		$\neg N$	$\vee$ E, 1, 3–5, 6–10
12		$(\neg M \rightarrow \neg N)$	$\rightarrow$ I, 2, 11

Daeducer commands needed to generate the proof:

```

1 premise (M v (N -> M))
2 assumption !M
3 assumption M
4 not_elim 2, 3
5 explosion 4, !N
6 discharge
7 assumption (N -> M)
8 assumption N
9 imp_elim 7, 8
10 not_elim 2, 9
11 discharge
12 not_intro 8, 10
13 discharge
14 or_elim 1, 3, 5, 7, 12
15 discharge
16 imp_intro 2, 14
17 qed

```

Chapter 19, Exercise B.3

Proof presented using Fitch notation:

1		$((M \vee N) \wedge (O \vee P))$	PR
2		$(N \rightarrow P)$	PR
3		$\neg P$	PR
4		$(M \vee N)$	$\wedge E, 1$
5		$(O \vee P)$	$\wedge E, 1$
6		N	AS
7		P	$\rightarrow E, 2, 6$
8		$\perp$	$\neg E, 3, 7$
9		$\neg N$	$\neg I, 6-8$
10		M	AS
11		M	R, 10
12		N	AS
13		$\perp$	$\neg E, 9, 12$
14		M	X, 13
15		M	$\vee E, 4, 10-11, 12-14$
16		O	AS
17		O	R, 16
18		P	AS
19		$\perp$	$\neg E, 3, 18$
20		O	X, 19
21		O	$\vee E, 5, 16-17, 18-20$
22		$(M \wedge O)$	$\wedge I, 15, 21$

Daeducer commands needed to generate the proof:

```

1 premise ((M ∨ N) ∧ (O ∨ P))
2 premise (N -> P)
3 premise !P
4 and_elim_left 1
5 and_elim_right 1
6 assumption N
7 imp_elim 2, 6
8 not_elim 3, 7
9 discharge
10 not_intro 6, 8
11 assumption M
12 reiteration 11
13 discharge
14 assumption N
15 not_elim 10, 14
16 explosion 15, M
17 discharge
18 or_elim 4, 11, 12, 14, 16
19 assumption O
20 reiteration 19

```

```

21 discharge
22 assumption P
23 not_elim 3, 22
24 explosion 23, 0
25 discharge
26 or_elim 5, 19, 20, 22, 24
27 and_intro 18, 26
28 qed

```

Chapter 19, Exercise B.4

Proof presented using Fitch notation:

1		$((X \wedge Y) \vee (X \wedge Z))$	PR
2		$\neg(X \wedge D)$	PR
3		$(D \vee M)$	PR
4		$(X \wedge Y)$	AS
5		X	$\wedge E$, 4
6		$(X \wedge Z)$	AS
7		X	$\wedge E$, 6
8		X	$\vee E$, 1, 4–5, 6–7
9		D	AS
10		$(X \wedge D)$	$\wedge I$, 8, 9
11		$\perp$	$\neg E$, 2, 10
12		$\neg D$	$\neg I$, 9–11
13		D	AS
14		$\perp$	$\neg E$, 12, 13
15		M	X, 14
16		M	AS
17		M	R, 16
18		M	$\vee E$, 3, 13–15, 16–17

Daeducer commands needed to generate the proof:

```

1 premise  $((X \wedge Y) \vee (X \wedge Z))$ 
2 premise  $\neg(X \wedge D)$ 
3 premise  $(D \vee M)$ 
4 assumption  $(X \wedge Y)$ 
5 and_elim_left 4
6 discharge
7 assumption  $(X \wedge Z)$ 
8 and_elim_left 7
9 discharge
10 or_elim 1, 4, 5, 7, 8
11 assumption D
12 and_intro 10, 11
13 not_elim 2, 12
14 discharge
15 not_intro 11, 13
16 assumption D
17 not_elim 15, 16

```

```

18 explosion 17, M
19 discharge
20 assumption M
21 reiteration 20
22 discharge
23 or_elim 3, 16, 18, 20, 21
24 qed

```

Chapter 20, Example p.171

Proof presented using Fitch notation:

1			$(A \wedge \neg A)$	AS
2			A	$\wedge E, 1$
3			$\neg A$	$\wedge E, 1$
4			$\perp$	$\neg E, 3, 2$
5			$\neg(A \wedge \neg A)$	$\neg I, 1-4$

Daeducer commands needed to generate the proof:

```

1 assumption (A ^ !A)
2 and_elim_left 1
3 and_elim_right 1
4 not_elim 3, 2
5 discharge
6 not_intro 1, 4
7 qed

```

Chapter 20, Exercise A.1

Proof presented using Fitch notation:

1			O	AS
2			O	R, 1
3			$(O \rightarrow O)$	$\rightarrow I, 1, 2$

Daeducer commands needed to generate the proof:

```

1 assumption 0
2 reiteration 1
3 discharge
4 imp_intro 1, 2
5 qed

```

Chapter 20, Exercise A.2

Proof presented using Fitch notation:

1			$\neg(N \vee \neg N)$	AS
2				
3				
4				
5				
6				
7				
8				
1			$\neg(N \vee \neg N)$	AS
2				AS
3				$(N \vee \neg N)$ $\vee E, 2$
4				$\perp$ $\neg E, 1, 3$
5				$\neg N$ $\neg I, 2-4$
6				$(N \vee \neg N)$ $\vee E, 5$
7				$\perp$ $\neg E, 1, 6$
8				$(N \vee \neg N)$ $IP, 1-7$

Daeducer commands needed to generate the proof:

```

1  assumption !(N v !N)
2  assumption N
3  or_intro_left 2, !N
4  not_elim 1, 3
5  discharge
6  not_intro 2, 4
7  or_intro_right 6, N
8  not_elim 1, 7
9  discharge
10 indirect 1, 8
11 qed

```

Chapter 20, Exercise A.3

Proof presented using Fitch notation:

1			J	AS
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
1			J	AS
2				$(J \vee (L \wedge \neg L))$ $\vee E, 1$
3				$(J \rightarrow (J \vee (L \wedge \neg L)))$ $\rightarrow I, 1, 2$
4				$(J \vee (L \wedge \neg L))$ AS
5				J AS
6				J R, 5
7				$(L \wedge \neg L)$ AS
8				L $\wedge E, 7$
9				$\neg L$ $\wedge E, 7$
10				$\perp$ $\neg E, 8, 9$
11				J X, 10
12				J $\vee E, 4, 5-6, 7-11$
13				$((J \vee (L \wedge \neg L)) \rightarrow J)$ $\rightarrow I, 4, 12$
14				$((J \rightarrow (J \vee (L \wedge \neg L))) \wedge ((J \vee (L \wedge \neg L)) \rightarrow J))$ $\wedge I, 3, 13$

Daeducer commands needed to generate the proof:

```

1  assumption J
2  or_intro_left 1, (L ~ !L)
3  discharge
4  imp_intro 1, 2
5  assumption (J v (L ~ !L))
6  assumption J
7  reiteration 6
8  discharge
9  assumption (L ~ !L)
10 and_elim_left 9
11 and_elim_right 9
12 not_elim 11, 10
13 explosion 12, J
14 discharge
15 or_elim 5, 6, 7, 9, 13
16 discharge
17 imp_intro 5, 15
18 and_intro 4, 17
19 qed

```

Chapter 20, Exercise A.4

Proof presented using Fitch notation:

1			$((A \rightarrow B) \rightarrow A)$	AS
2				
3				
4				
5				
6				
7				
8				
9				
10				

$$\begin{array}{ll}
 1 & | \quad | \quad | \quad ((A \rightarrow B) \rightarrow A) \quad \text{AS} \\
 2 & | \quad | \quad | \quad | \quad \neg A \quad \text{AS} \\
 3 & | \quad | \quad | \quad | \quad | \quad A \quad \text{AS} \\
 4 & | \quad | \quad | \quad | \quad | \quad \perp \quad \neg\text{E}, 2, 3 \\
 5 & | \quad | \quad | \quad | \quad | \quad B \quad \text{X}, 4 \\
 6 & | \quad | \quad | \quad | \quad (A \rightarrow B) \quad \rightarrow\text{I}, 3, 5 \\
 7 & | \quad | \quad | \quad | \quad A \quad \rightarrow\text{E}, 1, 6 \\
 8 & | \quad | \quad | \quad | \quad \perp \quad \neg\text{E}, 2, 7 \\
 9 & | \quad | \quad | \quad A \quad \text{IP}, 2-8 \\
 10 & | \quad | \quad ((A \rightarrow B) \rightarrow A) \rightarrow A \quad \rightarrow\text{I}, 1, 9
 \end{array}$$

Daeducer commands needed to generate the proof:

```

1  assumption ((A -> B) -> A)
2  assumption !A
3  assumption A
4  not_elim 2, 3
5  explosion 4, B
6  discharge
7  imp_intro 3, 5
8  imp_elim 1, 7
9  not_elim 2, 8
10 discharge
11 indirect 2, 9
12 discharge
13 imp_intro 1, 11
14 qed

```

Chapter 20, Exercise B.1

Proof presented using Fitch notation:

1		$(C \rightarrow (E \wedge G))$	PR
2		$(\neg C \rightarrow G)$	PR
3		$\neg G$	AS
4		C	AS
5		$(E \wedge G)$	$\rightarrow$ E, 1, 4
6		G	$\wedge$ E, 5
7		$\perp$	$\neg$ E, 3, 6
8		$\neg C$	$\neg$ I, 4–7
9		G	$\rightarrow$ E, 2, 8
10		$\perp$	$\neg$ E, 3, 9
11		G	IP, 3–10

Daeducer commands needed to generate the proof:

```

1  premise (C -> (E ^ G))
2  premise (!C -> G)
3  assumption !G
4  assumption C
5  imp_elim 1, 4
6  and_elim_right 5
7  not_elim 3, 6
8  discharge
9  not_intro 4, 7
10 imp_elim 2, 9
11 not_elim 3, 10
12 discharge
13 indirect 3, 11
14 qed

```

Chapter 20, Exercise B.2

Proof presented using Fitch notation:

1	$(M \wedge (\neg N \rightarrow \neg M))$	PR
2	M	$\wedge E, 1$
3	$(\neg N \rightarrow \neg M)$	$\wedge E, 1$
4	$\neg N$	AS
5	$\neg M$	$\rightarrow E, 3, 4$
6	$\perp$	$\neg E, 5, 2$
7	N	IP, 4–6
8	$(N \wedge M)$	$\wedge I, 7, 2$
9	$((N \wedge M) \vee \neg M)$	$\vee E, 8$

Daeducer commands needed to generate the proof:

```

1 premise (M ^ (!N -> !M))
2 and_elim_left 1
3 and_elim_right 1
4 assumption !N
5 imp_elim 3, 4
6 not_elim 5, 2
7 discharge
8 indirect 4, 6
9 and_intro 8, 2
10 or_intro_left 9, !M
11 qed

```

Chapter 20, Exercise B.3

Proof presented using Fitch notation:

1	$((Z \wedge K) \rightarrow (Y \wedge M)) \wedge ((Y \wedge M) \rightarrow (Z \wedge K))$	PR
2	$(D \wedge (D \rightarrow M))$	PR
3	D	$\wedge E, 2$
4	$(D \rightarrow M)$	$\wedge E, 2$
5	M	$\rightarrow E, 4, 3$
6	Y	AS
7	$((Z \wedge K) \rightarrow (Y \wedge M))$	$\wedge E, 1$
8	$((Y \wedge M) \rightarrow (Z \wedge K))$	$\wedge E, 1$
9	$(Y \wedge M)$	$\wedge I, 6, 5$
10	$(Z \wedge K)$	$\rightarrow E, 8, 9$
11	Z	$\wedge E, 10$
12	$(Y \rightarrow Z)$	$\rightarrow I, 6, 11$

Daeducer commands needed to generate the proof:

```

1 premise (((Z ^ K) -> (Y ^ M)) ^ ((Y ^ M) -> (Z ^ K)))
2 premise (D ^ (D -> M))
3 and_elim_left 2
4 and_elim_right 2
5 imp_elim 4, 3
6 assumption Y
7 and_elim_left 1
8 and_elim_right 1
9 and_intro 6, 5
10 imp_elim 8, 9
11 and_elim_left 10
12 discharge
13 imp_intro 6, 11
14 qed

```

Chapter 20, Exercise B.4

Proof presented using Fitch notation:

1	((W ∨ X) ∨ (Y ∨ Z))	PR
2	(X → Y)	PR
3	¬Z	PR
4	(W ∨ X)	AS
5	W	AS
6	(W ∨ Y)	∨E, 5
7	X	AS
8	Y	→E, 2, 7
9	(W ∨ Y)	∨E, 8
10	(W ∨ Y)	∨E, 4, 5–6, 7–9
11	(Y ∨ Z)	AS
12	Y	AS
13	(W ∨ Y)	∨E, 12
14	Z	AS
15	⊥	¬E, 3, 14
16	(W ∨ Y)	X, 15
17	(W ∨ Y)	∨E, 11, 12–13, 14–16
18	(W ∨ Y)	∨E, 1, 4–10, 11–17

Daeducer commands needed to generate the proof:

```

1 premise ((W ∨ X) ∨ (Y ∨ Z))
2 premise (X -> Y)
3 premise !Z
4 assumption (W ∨ X)
5 assumption W
6 or_intro_left 5, Y
7 discharge
8 assumption X
9 imp_elim 2, 8
10 or_intro_right 9, W
11 discharge

```

```

12 or_elim 4, 5, 6, 8, 10
13 discharge
14 assumption (Y ∨ Z)
15 assumption Y
16 or_intro_right 15, W
17 discharge
18 assumption Z
19 not_elim 3, 18
20 explosion 19, (W ∨ Y)
21 discharge
22 or_elim 14, 15, 16, 18, 20
23 discharge
24 or_elim 1, 4, 12, 14, 22
25 qed

```

Chapter 20, Exercise C.1.a

Proof presented using Fitch notation:

1		$((R \rightarrow E) \wedge (E \rightarrow R))$	PR
2		$(R \rightarrow E)$	$\wedge E, 1$
3		$(E \rightarrow R)$	$\wedge E, 1$
4		$((E \rightarrow R) \wedge (R \rightarrow E))$	$\wedge I, 3, 2$

Daeducer commands needed to generate the proof:

```

1 premise ((R -> E) ^ (E -> R))
2 and_elim_left 1
3 and_elim_right 1
4 and_intro 3, 2
5 qed

```

Chapter 20, Exercise C.1.b

Proof presented using Fitch notation:

1		$((E \rightarrow R) \wedge (R \rightarrow E))$	PR
2		$(E \rightarrow R)$	$\wedge E, 1$
3		$(R \rightarrow E)$	$\wedge E, 1$
4		$((R \rightarrow E) \wedge (E \rightarrow R))$	$\wedge I, 3, 2$

Daeducer commands needed to generate the proof:

```

1 premise ((E -> R) ^ (R -> E))
2 and_elim_left 1
3 and_elim_right 1
4 and_intro 3, 2
5 qed

```

Chapter 20, Exercise C.2.a

Proof presented using Fitch notation:

1	G	PR
2	$\neg G$	AS
3	$\perp$	$\neg$ E, 2, 1
4	$\neg\neg G$	$\neg$ I, 2-3
5	$\neg\neg\neg G$	AS
6	$\perp$	$\neg$ E, 5, 4
7	$\neg\neg\neg\neg G$	$\neg$ I, 5-6

Daeducer commands needed to generate the proof:

```

1 premise G
2 assumption !G
3 not_elim 2, 1
4 discharge
5 not_intro 2, 3
6 assumption !!!G
7 not_elim 6, 5
8 discharge
9 not_intro 6, 7
10 qed

```

Chapter 20, Exercise C.2.b

Proof presented using Fitch notation:

1	$\neg\neg\neg\neg G$	PR
2	$\neg\neg\neg G$	AS
3	$\perp$	$\neg$ E, 1, 2
4	$\neg\neg G$	IP, 2-3
5	$\neg G$	AS
6	$\perp$	$\neg$ E, 4, 5
7	G	IP, 5-6

Daeducer commands needed to generate the proof:

```

1 premise !!!!G
2 assumption !!!!G
3 not_elim 1, 2
4 discharge
5 indirect 2, 3
6 assumption !G
7 not_elim 5, 6
8 discharge
9 indirect 6, 7
10 qed

```


Chapter 20, Exercise C.4.a

Proof presented using Fitch notation:

1	$(U \rightarrow I)$	PR
2	$(U \wedge \neg I)$	AS
3	U	$\wedge E, 2$
4	$\neg I$	$\wedge E, 2$
5	I	$\rightarrow E, 1, 3$
6	$\perp$	$\neg E, 4, 5$
7	$\neg(U \wedge \neg I)$	$\neg I, 2-6$

Daeducer commands needed to generate the proof:

```

1 premise (U -> I)
2 assumption (U ^ !I)
3 and_elim_left 2
4 and_elim_right 2
5 imp_elim 1, 3
6 not_elim 4, 5
7 discharge
8 not_intro 2, 6
9 qed

```

Chapter 20, Exercise C.4.b

Proof presented using Fitch notation:

1	$\neg(U \wedge \neg I)$	PR
2	U	AS
3	$\neg I$	AS
4	$(U \wedge \neg I)$	$\wedge I, 2, 3$
5	$\perp$	$\neg E, 1, 4$
6	I	IP, 3-5
7	$(U \rightarrow I)$	$\rightarrow I, 2, 6$

Daeducer commands needed to generate the proof:

```

1 premise !(U ^ !I)
2 assumption U
3 assumption !I
4 and_intro 2, 3
5 not_elim 1, 4
6 discharge
7 indirect 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```


Chapter 20, Exercise C.5.a

Proof presented using Fitch notation:

1	$\neg(C \rightarrow D)$	PR
2	$\neg C$	AS
3	C	AS
4	$\perp$	$\neg$ E, 2, 3
5	D	X, 4
6	$(C \rightarrow D)$	$\rightarrow$ I, 3, 5
7	$\perp$	$\neg$ E, 1, 6
8	C	IP, 2–7
9	D	AS
10	C	AS
11	D	R, 9
12	$(C \rightarrow D)$	$\rightarrow$ I, 10, 11
13	$\perp$	$\neg$ E, 1, 12
14	$\neg D$	$\neg$ I, 9–13
15	$(C \wedge \neg D)$	$\wedge$ I, 8, 14

Daeducer commands needed to generate the proof:

```

1 premise !(C -> D)
2 assumption !C
3 assumption C
4 not_elim 2, 3
5 explosion 4, D
6 discharge
7 imp_intro 3, 5
8 not_elim 1, 7
9 discharge
10 indirect 2, 8
11 assumption D
12 assumption C
13 reiteration 11
14 discharge
15 imp_intro 12, 13
16 not_elim 1, 15
17 discharge
18 not_intro 11, 16
19 and_intro 10, 18
20 qed

```

Chapter 20, Exercise C.5.b

Proof presented using Fitch notation:

1	$(C \wedge \neg D)$	PR
2	C	$\wedge E, 1$
3	$\neg D$	$\wedge E, 1$
4	$(C \rightarrow D)$	AS
5	D	$\rightarrow E, 4, 2$
6	$\perp$	$\neg E, 3, 5$
7	$\neg(C \rightarrow D)$	$\neg I, 4-6$

Daeducer commands needed to generate the proof:

```
1 premise (C ^ !D)
2 and_elim_left 1
3 and_elim_right 1
4 assumption (C -> D)
5 imp_elim 4, 2
6 not_elim 3, 5
7 discharge
8 not_intro 4, 6
9 qed
```

Chapter 20, Exercise C.6.a

Proof presented using Fitch notation:

1	$((\neg G \rightarrow H) \wedge (H \rightarrow \neg G))$	PR
2	$((G \rightarrow H) \wedge (H \rightarrow G))$	AS
3	$(G \rightarrow H)$	$\wedge E, 2$
4	$(H \rightarrow G)$	$\wedge E, 2$
5	$(\neg G \rightarrow H)$	$\wedge E, 1$
6	$(H \rightarrow \neg G)$	$\wedge E, 1$
7	H	AS
8	$\neg G$	$\rightarrow E, 6, 7$
9	G	$\rightarrow E, 4, 7$
10	$\perp$	$\neg E, 8, 9$
11	$\neg H$	$\neg I, 7-10$
12	$\neg G$	AS
13	H	$\rightarrow E, 5, 12$
14	$\perp$	$\neg E, 11, 13$
15	G	IP, 12-14
16	H	$\rightarrow E, 3, 15$
17	$\perp$	$\neg E, 11, 16$
18	$\neg((G \rightarrow H) \wedge (H \rightarrow G))$	$\neg I, 2-17$

Daeducer commands needed to generate the proof:

```

1 premise ((!G -> H) ^ (H -> !G))
2 assumption ((G -> H) ^ (H -> G))
3 and_elim_left 2
4 and_elim_right 2
5 and_elim_left 1
6 and_elim_right 1
7 assumption H
8 imp_elim 6, 7
9 imp_elim 4, 7
10 not_elim 8, 9
11 discharge
12 not_intro 7, 10
13 assumption !G
14 imp_elim 5, 13
15 not_elim 12, 14
16 discharge
17 indirect 13, 15
18 imp_elim 3, 17
19 not_elim 12, 18
20 discharge
21 not_intro 2, 19
22 qed

```

Chapter 20, Exercise C.6.b

Proof presented using Fitch notation:

1	$\neg((G \rightarrow H) \wedge (H \rightarrow G))$	PR
2	$\neg G$	AS
3	$\neg H$	AS
4	G	AS
5	$\perp$	$\neg E$, 2, 4
6	H	X, 5
7	$(G \rightarrow H)$	$\rightarrow I$, 4, 6
8	H	AS
9	$\perp$	$\neg E$, 3, 8
10	G	X, 9
11	$(H \rightarrow G)$	$\rightarrow I$, 8, 10
12	$((G \rightarrow H) \wedge (H \rightarrow G))$	$\wedge I$, 7, 11
13	$\perp$	$\neg E$, 1, 12
14	H	IP, 3–13
15	$(\neg G \rightarrow H)$	$\rightarrow I$, 2, 14
16	H	AS
17	G	AS
18	G	AS
19	H	R, 16
20	$(G \rightarrow H)$	$\rightarrow I$, 18, 19
21	H	AS
22	G	R, 17
23	$(H \rightarrow G)$	$\rightarrow I$, 21, 22
24	$((G \rightarrow H) \wedge (H \rightarrow G))$	$\wedge I$, 20, 23
25	$\perp$	$\neg E$, 1, 24
26	$\neg G$	$\neg I$, 17–25
27	$(H \rightarrow \neg G)$	$\rightarrow I$, 16, 26
28	$((\neg G \rightarrow H) \wedge (H \rightarrow \neg G))$	$\wedge I$, 15, 27

Daeducer commands needed to generate the proof:

```

1 premise !((G -> H) ^ (H -> G))
2 assumption !G
3 assumption !H
4 assumption G
5 not_elim 2, 4
6 explosion 5, H
7 discharge
8 imp_intro 4, 6

```

```

9  assumption H
10 not_elim 3, 9
11 explosion 10, G
12 discharge
13 imp_intro 9, 11
14 and_intro 8, 13
15 not_elim 1, 14
16 discharge
17 indirect 3, 15
18 discharge
19 imp_intro 2, 17
20 assumption H
21 assumption G
22 assumption G
23 reiteration 20
24 discharge
25 imp_intro 22, 23
26 assumption H
27 reiteration 21
28 discharge
29 imp_intro 26, 27
30 and_intro 25, 29
31 not_elim 1, 30
32 discharge
33 not_intro 21, 31
34 discharge
35 imp_intro 20, 33
36 and_intro 19, 35
37 qed

```

Chapter 21, Example p.176

Proof presented using Fitch notation:

1		$(A \rightarrow B)$	PR
2		$\neg B$	PR
3			
4			
5			
6			
3			AS
4			$\rightarrow E, 1, 3$
5			$\neg E, 2, 4$
6			$\neg I, 3-5$

Daeducer commands needed to generate the proof:

```

1  premise (A -> B)
2  premise !B
3  assumption A
4  imp_elim 1, 3
5  not_elim 2, 4
6  discharge
7  not_intro 3, 5
8  qed

```

Chapter 21, Example p.177

Proof presented using Fitch notation:

1	$\neg\neg A$	PR
2	$\neg A$	AS
3	$\perp$	$\neg E$, 1, 2
4	A	IP, 2-3

Daeducer commands needed to generate the proof:

```

1 premise !!A
2 assumption !A
3 not_elim 1, 2
4 discharge
5 indirect 2, 3
6 qed

```

Chapter 21, Example p.178

Proof presented using Fitch notation:

1	$(A \rightarrow B)$	PR
2	$(\neg A \rightarrow B)$	PR
3	$\neg B$	AS
4	A	AS
5	B	$\rightarrow E$, 1, 4
6	$\perp$	$\neg E$, 3, 5
7	$\neg A$	$\neg I$, 4-6
8	B	$\rightarrow E$, 2, 7
9	$\perp$	$\neg E$, 3, 8
10	B	IP, 3-9

Daeducer commands needed to generate the proof:

```

1 premise (A -> B)
2 premise (!A -> B)
3 assumption !B
4 assumption A
5 imp_elim 1, 4
6 not_elim 3, 5
7 discharge
8 not_intro 4, 6
9 imp_elim 2, 8
10 not_elim 3, 9
11 discharge
12 indirect 3, 10
13 qed

```

Chapter 21, Example p.180

Proof presented using Fitch notation:

1	$\neg(A \wedge B)$	PR
2	$\neg(\neg A \vee \neg B)$	AS
3	$\neg A$	AS
4	$(\neg A \vee \neg B)$	$\vee E$, 3
5	$\perp$	$\neg E$, 2, 4
6	A	IP, 3–5
7	$\neg B$	AS
8	$(\neg A \vee \neg B)$	$\vee E$, 7
9	$\perp$	$\neg E$, 2, 8
10	B	IP, 7–9
11	$(A \wedge B)$	$\wedge I$, 6, 10
12	$\perp$	$\neg E$, 1, 11
13	$(\neg A \vee \neg B)$	IP, 2–12

Daeducer commands needed to generate the proof:

```

1  premise !(A ^ B)
2  assumption !(A v !B)
3  assumption !A
4  or_intro_left 3, !B
5  not_elim 2, 4
6  discharge
7  indirect 3, 5
8  assumption !B
9  or_intro_right 8, !A
10 not_elim 2, 9
11 discharge
12 indirect 8, 10
13 and_intro 7, 12
14 not_elim 1, 13
15 discharge
16 indirect 2, 14
17 qed

```

Chapter 21, Example p.180

Proof presented using Fitch notation:

1		$(\neg A \vee \neg B)$	PR
2			
2			$(A \wedge B)$
2			AS
3			A
3			$\wedge E, 2$
4			B
4			$\wedge E, 2$
5			
5			
5			$\neg A$
5			AS
6			
6			
6			$\perp$
6			$\neg E, 5, 3$
7			
7			
7			$\neg B$
7			AS
8			
8			
8			$\perp$
8			$\neg E, 7, 4$
9			$\perp$
9			$\vee E, 1, 5-6, 7-8$
10		$\neg(A \wedge B)$	$\neg I, 2-9$

Daeducer commands needed to generate the proof:

```

1 premise (!A v !B)
2 assumption (A ^ B)
3 and_elim_left 2
4 and_elim_right 2
5 assumption !A
6 not_elim 5, 3
7 discharge
8 assumption !B
9 not_elim 8, 4
10 discharge
11 or_elim 1, 5, 6, 8, 9
12 discharge
13 not_intro 2, 11
14 qed

```


Chapter 21, Exercise A.1

Proof presented using Fitch notation:

1	$\neg(A \vee B)$	PR
2	A	AS
3	$(A \vee B)$	$\vee E$, 2
4	$\perp$	$\neg E$, 1, 3
5	$\neg A$	$\neg I$, 2–4
6	B	AS
7	$(A \vee B)$	$\vee E$, 6
8	$\perp$	$\neg E$, 1, 7
9	$\neg B$	$\neg I$, 6–8
10	$(\neg A \wedge \neg B)$	$\wedge I$, 5, 9

Daeducer commands needed to generate the proof:

```

1 premise !(A ∨ B)
2 assumption A
3 or_intro_left 2, B
4 not_elim 1, 3
5 discharge
6 not_intro 2, 4
7 assumption B
8 or_intro_right 7, A
9 not_elim 1, 8
10 discharge
11 not_intro 7, 9
12 and_intro 6, 11
13 qed

```

Chapter 21, Exercise A.2

Proof presented using Fitch notation:

1	$(\neg A \wedge \neg B)$	PR
2	$\neg A$	$\wedge E$, 1
3	$\neg B$	$\wedge E$, 1
4	$(A \vee B)$	AS
5	A	AS
6	$\perp$	$\neg E$, 2, 5
7	B	AS
8	$\perp$	$\neg E$, 3, 7
9	$\perp$	$\vee E$, 4, 5–6, 7–8
10	$\neg(A \vee B)$	$\neg I$, 4–9

Daeducer commands needed to generate the proof:

```

1  premise (!A ~ !B)
2  and_elim_left 1
3  and_elim_right 1
4  assumption (A ~ B)
5  assumption A
6  not_elim 2, 5
7  discharge
8  assumption B
9  not_elim 3, 8
10 discharge
11 or_elim 4, 5, 6, 8, 9
12 discharge
13 not_intro 4, 11
14 qed

```

Chapter 22, Exercise A.1

Proof presented using Fitch notation:

1			A	AS
2				
3				
4				
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Chapter 22, Exercise A.2

Proof presented using Fitch notation:

1	A	PR
2	$\neg B$	PR
3	$(A \rightarrow (A \rightarrow B))$	AS
4	$(A \rightarrow B)$	$\rightarrow E, 3, 1$
5	B	$\rightarrow E, 4, 1$
6	$\perp$	$\neg E, 2, 5$
7	$\neg(A \rightarrow (A \rightarrow B))$	$\neg I, 3-6$

Daeducer commands needed to generate the proof:

```
1 premise A
2 premise !B
3 assumption (A -> (A -> B))
4 imp_elim 3, 1
5 imp_elim 4, 1
6 not_elim 2, 5
7 discharge
8 not_intro 3, 6
9 qed
```

Chapter 22, Exercise A.3

Proof presented using Fitch notation:

1	$\neg A$	PR
2	A	AS
3	$\neg A$	R, 1
4	$(A \rightarrow \neg A)$	$\rightarrow I, 2, 3$

Daeducer commands needed to generate the proof:

```
1 premise !A
2 assumption A
3 reiteration 1
4 discharge
5 imp_intro 2, 3
6 qed
```

Chapter 22, Exercise A.4

Proof presented using Fitch notation:

1	$((A \rightarrow \neg A) \wedge (\neg A \rightarrow A))$	PR
2	$(A \rightarrow \neg A)$	$\wedge E$, 1
3	$(\neg A \rightarrow A)$	$\wedge E$, 1
4	$\neg A$	AS
5	A	$\rightarrow E$, 3, 4
6	$\perp$	$\neg E$, 4, 5
7	A	IP, 4–6
8	$\neg A$	$\rightarrow E$, 2, 7
9	$\perp$	$\neg E$, 8, 7

Daeducer commands needed to generate the proof:

```

1 premise ((A -> !A) ^ (!A -> A))
2 and_elim_left 1
3 and_elim_right 1
4 assumption !A
5 imp_elim 3, 4
6 not_elim 4, 5
7 discharge
8 indirect 4, 6
9 imp_elim 2, 8
10 not_elim 9, 8
11 qed

```

Chapter 22, Exercise A.5.a

Proof presented using Fitch notation:

1	W	PR
2	$\neg J$	PR
3	$(W \rightarrow (J \vee J))$	AS
4	$(J \vee J)$	$\rightarrow E$, 3, 1
5	J	AS
6	$\perp$	$\neg E$, 2, 5
7	J	AS
8	$\perp$	$\neg E$, 2, 7
9	$\perp$	$\vee E$, 4, 5–6, 7–8
10	$\neg(W \rightarrow (J \vee J))$	$\neg I$, 3–9

Daeducer commands needed to generate the proof:

```

1 premise W
2 premise !J
3 assumption (W -> (J v J))

```

```

4  imp_elim 3, 1
5  assumption J
6  not_elim 2, 5
7  discharge
8  assumption J
9  not_elim 2, 8
10 discharge
11 or_elim 4, 5, 6, 8, 9
12 discharge
13 not_intro 3, 11
14 qed

```

Chapter 22, Exercise A.5.b

Proof presented using Fitch notation:

1	W	PR
2	J	PR
3	$\neg(W \rightarrow (J \vee J))$	AS
4	W	AS
5	$(J \vee J)$	$\vee E, 2$
6	$(W \rightarrow (J \vee J))$	$\rightarrow I, 4, 5$
7	$\perp$	$\neg E, 3, 6$
8	$\neg\neg(W \rightarrow (J \vee J))$	$\neg I, 3-7$

Daeducer commands needed to generate the proof:

```

1  premise W
2  premise J
3  assumption !(W -> (J v J))
4  assumption W
5  or_intro_left 2, J
6  discharge
7  imp_intro 4, 5
8  not_elim 3, 7
9  discharge
10 not_intro 3, 8
11 qed

```

Chapter 22, Exercise A.6

Proof presented using Fitch notation:

1			$\neg(\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z)))$	AS
2				
3				
4				
5				
6				
7				
8				

2			$(X \vee (Y \vee Z))$	AS
3				
4				
5				
6				
7				
8				

3			$(\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z)))$	$\vee E, 2$
4			$\perp$	$\neg E, 1, 3$
5			$\neg(X \vee (Y \vee Z))$	$\neg I, 2-4$
6			$(\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z)))$	$\vee E, 5$
7			$\perp$	$\neg E, 1, 6$
8			$(\neg(X \vee (Y \vee Z)) \vee (X \vee (Y \vee Z)))$	IP, 1-7

Daeducer commands needed to generate the proof:

```

1  assumption !(X v (Y v Z)) v (X v (Y v Z))
2  assumption (X v (Y v Z))
3  or_intro_right 2, !(X v (Y v Z))
4  not_elim 1, 3
5  discharge
6  not_intro 2, 4
7  or_intro_left 6, (X v (Y v Z))
8  not_elim 1, 7
9  discharge
10 indirect 1, 8
11 qed

```

Chapter 22, Exercise A.7

Proof presented using Fitch notation:

1		$(B \rightarrow \neg S)$	AS
2		$\neg\neg B$	AS
3		$\neg B$	AS
4		$\perp$	$\neg$ E, 2, 3
5		B	IP, 3–4
6		$\neg S$	$\rightarrow$ E, 1, 5
7		$(\neg\neg B \rightarrow \neg S)$	$\rightarrow$ I, 2, 6
8		$((B \rightarrow \neg S) \rightarrow (\neg\neg B \rightarrow \neg S))$	$\rightarrow$ I, 1, 7
9		$(\neg\neg B \rightarrow \neg S)$	AS
10		B	AS
11		$\neg B$	AS
12		$\perp$	$\neg$ E, 11, 10
13		$\neg\neg B$	$\neg$ I, 11–12
14		$\neg S$	$\rightarrow$ E, 9, 13
15		$(B \rightarrow \neg S)$	$\rightarrow$ I, 10, 14
16		$((\neg\neg B \rightarrow \neg S) \rightarrow (B \rightarrow \neg S))$	$\rightarrow$ I, 9, 15
17		$(((B \rightarrow \neg S) \rightarrow (\neg\neg B \rightarrow \neg S)) \wedge ((\neg\neg B \rightarrow \neg S) \rightarrow (B \rightarrow \neg S)))$	$\wedge$ I, 8, 16

Daeducer commands needed to generate the proof:

```

1  assumption (B -> !S)
2  assumption !!B
3  assumption !B
4  not_elim 2, 3
5  discharge
6  indirect 3, 4
7  imp_elim 1, 6
8  discharge
9  imp_intro 2, 7
10 discharge
11 imp_intro 1, 9
12 assumption (!!B -> !S)
13 assumption B
14 assumption !B
15 not_elim 14, 13
16 discharge
17 not_intro 14, 15
18 imp_elim 12, 17
19 discharge
20 imp_intro 13, 18
21 discharge
22 imp_intro 12, 20
23 and_intro 11, 22
24 qed

```

Chapter 22, Exercise A.8

Proof presented using Fitch notation:

1	X	PR
2	O	PR
3	$\hline ((X \wedge O) \wedge \neg(X \vee O))$	AS
4	$\neg(X \vee O)$	$\wedge E, 3$
5	$(X \vee O)$	$\vee E, 1$
6	$\perp$	$\neg E, 4, 5$
7	$\neg((X \wedge O) \wedge \neg(X \vee O))$	$\neg I, 3-6$

Daeducer commands needed to generate the proof:

```

1 premise X
2 premise O
3 assumption ((X ^ O) ^ !(X v O))
4 and_elim_right 3
5 or_intro_left 1, O
6 not_elim 4, 5
7 discharge
8 not_intro 3, 6
9 qed

```

Chapter 22, Exercise A.9

Proof presented using Fitch notation:

1	$\neg(A \vee B)$	PR
2	C	PR
3	$(C \rightarrow A)$	PR
4	$\hline A$	$\rightarrow E, 3, 2$
5	$(A \vee B)$	$\vee E, 4$
6	$\perp$	$\neg E, 1, 5$

Daeducer commands needed to generate the proof:

```

1 premise !(A v B)
2 premise C
3 premise (C -> A)
4 imp_elim 3, 2
5 or_intro_left 4, B
6 not_elim 1, 5
7 qed

```


Chapter 22, Exercise A.10

Proof presented using Fitch notation:

1	$\neg A$	PR
2	$\neg B$	PR
3	$(A \vee B)$	AS
4	A	AS
5	$\perp$	$\neg E$, 1, 4
6	B	AS
7	$\perp$	$\neg E$, 2, 6
8	$\perp$	$\vee E$, 3, 4–5, 6–7
9	$\neg(A \vee B)$	$\neg I$, 3–8
10	B	AS
11	$\perp$	$\neg E$, 2, 10
12	A	X, 11
13	$(B \rightarrow A)$	$\rightarrow I$, 10, 12
14	$(\neg B \wedge (B \rightarrow A))$	$\wedge I$, 2, 13
15	$(\neg(A \vee B) \wedge (\neg B \wedge (B \rightarrow A)))$	$\wedge I$, 9, 14

Daeducer commands needed to generate the proof:

```

1 premise !A
2 premise !B
3 assumption (A v B)
4 assumption A
5 not_elim 1, 4
6 discharge
7 assumption B
8 not_elim 2, 7
9 discharge
10 or_elim 3, 4, 5, 7, 8
11 discharge
12 not_intro 3, 10
13 assumption B
14 not_elim 2, 13
15 explosion 14, A
16 discharge
17 imp_intro 13, 15
18 and_intro 2, 17
19 and_intro 12, 18
20 qed

```

Chapter 22, Exercise A.11

Proof presented using Fitch notation:

1	$\neg(A \vee (B \vee C))$	PR
2	C	AS
3	$(B \vee C)$	$\vee E, 2$
4	$(A \vee (B \vee C))$	$\vee E, 3$
5	$\perp$	$\neg E, 1, 4$
6	$\neg C$	$\neg I, 2-5$

Daeducer commands needed to generate the proof:

```

1 premise !(A ∨ (B ∨ C))
2 assumption C
3 or_intro_right 2, B
4 or_intro_right 3, A
5 not_elim 1, 4
6 discharge
7 not_intro 2, 5
8 qed

```

Chapter 22, Exercise A.12

Proof presented using Fitch notation:

1	$\neg A$	PR
2	B	PR
3	C	PR
4	$\neg C$	AS
5	$\perp$	$\neg E, 4, 3$
6	$\neg\neg C$	$\neg I, 4-5$
7	$(A \wedge (B \vee C))$	AS
8	A	$\wedge E, 7$
9	$\perp$	$\neg E, 1, 8$
10	$\neg(A \wedge (B \vee C))$	$\neg I, 7-9$
11	$(\neg(A \wedge (B \vee C)) \wedge \neg\neg C)$	$\wedge I, 10, 6$

Daeducer commands needed to generate the proof:

```

1 premise !A
2 premise B
3 premise C
4 assumption !C
5 not_elim 4, 3
6 discharge
7 not_intro 4, 5
8 assumption (A ∧ (B ∨ C))
9 and_elim_left 8
10 not_elim 1, 9
11 discharge

```

```

12 not_intro 8, 10
13 and_intro 12, 7
14 qed

```

Chapter 22, Exercise B.1

Proof presented using Fitch notation:

1			A	AS
2				
3				
4				
5				
6				
7				
8				
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Chapter 22, Exercise B.3.a

Proof presented using Fitch notation:

1	Z	PR
2	$(Z \vee ((\neg Z \rightarrow Z) \wedge (Z \rightarrow \neg Z)))$	$\vee E, 1$

Daeducer commands needed to generate the proof:

```
1 premise Z
2 or_intro_left 1, ((!Z -> Z) ^ (Z -> !Z))
3 qed
```

Chapter 22, Exercise B.3b

Proof presented using Fitch notation:

1	$\neg Z$	PR
2	$(Z \vee ((\neg Z \rightarrow Z) \wedge (Z \rightarrow \neg Z)))$	AS
3	Z	AS
4	$\perp$	$\neg E, 1, 3$
5	$((\neg Z \rightarrow Z) \wedge (Z \rightarrow \neg Z))$	AS
6	$(\neg Z \rightarrow Z)$	$\wedge E, 5$
7	Z	$\rightarrow E, 6, 1$
8	$\perp$	$\neg E, 1, 7$
9	$\perp$	$\vee E, 2, 3-4, 5-8$
10	$\neg(Z \vee ((\neg Z \rightarrow Z) \wedge (Z \rightarrow \neg Z)))$	$\neg I, 2-9$

Daeducer commands needed to generate the proof:

```
1 premise !Z
2 assumption (Z v ((!Z -> Z) ^ (Z -> !Z)))
3 assumption Z
4 not_elim 1, 3
5 discharge
6 assumption ((!Z -> Z) ^ (Z -> !Z))
7 and_elim_left 6
8 imp_elim 7, 1
9 not_elim 1, 8
10 discharge
11 or_elim 2, 3, 4, 6, 9
12 discharge
13 not_intro 2, 11
14 qed
```

Chapter 22, Exercise B.4

Proof presented using Fitch notation:

1		N	AS
2		N	R, 1
3		$(N \rightarrow N)$	$\rightarrow$ I, 1, 2
4		L	AS
5		$(N \rightarrow N)$	AS
6		L	R, 4
7		$((N \rightarrow N) \rightarrow L)$	$\rightarrow$ I, 5, 6
8		$(L \rightarrow ((N \rightarrow N) \rightarrow L))$	$\rightarrow$ I, 4, 7
9		$((N \rightarrow N) \rightarrow L)$	AS
10		$(N \rightarrow N)$	R, 3
11		L	$\rightarrow$ E, 9, 10
12		$((N \rightarrow N) \rightarrow L) \rightarrow L$	$\rightarrow$ I, 9, 11
13		$((L \rightarrow ((N \rightarrow N) \rightarrow L)) \wedge (((N \rightarrow N) \rightarrow L) \rightarrow L))$	$\wedge$ I, 8, 12
14		$((L \rightarrow ((N \rightarrow N) \rightarrow L)) \wedge (((N \rightarrow N) \rightarrow L) \rightarrow L)) \vee H$	$\vee$ E, 13

Daeducer commands needed to generate the proof:

```

1  assumption N
2  reiteration 1
3  discharge
4  imp_intro 1, 2
5  assumption L
6  assumption (N -> N)
7  reiteration 5
8  discharge
9  imp_intro 6, 7
10 discharge
11 imp_intro 5, 9
12 assumption ((N -> N) -> L)
13 reiteration 4
14 imp_elim 12, 13
15 discharge
16 imp_intro 12, 14
17 and_intro 11, 16
18 or_intro_left 17, H
19 qed

```

Chapter 22, Exercise B.5

Proof presented using Fitch notation:

1	$((A \rightarrow A) \wedge (A \rightarrow A)) \wedge (B \wedge \neg B)$	PR
2	$(B \wedge \neg B)$	$\wedge E, 1$
3	B	$\wedge E, 2$
4	$\neg B$	$\wedge E, 2$
5	$\perp$	$\neg E, 3, 4$

Daeducer commands needed to generate the proof:

```
1 premise (((A -> A) ^ (A -> A)) ^ (B ^ !B))
2 and_elim_right 1
3 and_elim_left 2
4 and_elim_right 2
5 not_elim 4, 3
6 qed
```

Chapter 22, Exercise B.6

Proof presented using Fitch notation:

1	B	PR
2	C	PR
3	B	AS
4	$(C \vee B)$	$\vee E, 3$
5	$(B \rightarrow (C \vee B))$	$\rightarrow I, 3, 4$
6	$(C \vee B)$	AS
7	B	R, 1
8	$((C \vee B) \rightarrow B)$	$\rightarrow I, 6, 7$
9	$((B \rightarrow (C \vee B)) \wedge ((C \vee B) \rightarrow B))$	$\wedge I, 5, 8$

Daeducer commands needed to generate the proof:

```
1 premise B
2 premise C
3 assumption B
4 or_intro_right 3, C
5 discharge
6 imp_intro 3, 4
7 assumption (C v B)
8 reiteration 1
9 discharge
10 imp_intro 7, 8
11 and_intro 6, 10
12 qed
```

Chapter 22, Exercise B.7

Proof presented using Fitch notation:

1		X	AS
2		$((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X$	$\vee E$, 1
3		$(X \rightarrow ((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X)$	$\rightarrow I$, 1, 2
4		$((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X$	AS
5		$((\neg X \rightarrow X) \wedge (X \rightarrow \neg X))$	AS
6		$(\neg X \rightarrow X)$	$\wedge E$, 5
7		$\neg X$	AS
8		X	$\rightarrow E$, 6, 7
9		$\perp$	$\neg E$, 7, 8
10		X	IP, 7–9
11		X	AS
12		X	R, 11
13		X	$\vee E$, 4, 5–10, 11–12
14		$((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X \rightarrow X$	$\rightarrow I$, 4, 13
15		$(X \rightarrow ((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X) \wedge (((\neg X \rightarrow X) \wedge (X \rightarrow \neg X)) \vee X) \rightarrow X$	$\wedge I$, 3, 14

Daeducer commands needed to generate the proof:

```

1  assumption X
2  or_intro_right 1, ((!X -> X) ^ (X -> !X))
3  discharge
4  imp_intro 1, 2
5  assumption (((!X -> X) ^ (X -> !X)) v X)
6  assumption ((!X -> X) ^ (X -> !X))
7  and_elim_left 6
8  assumption !X
9  imp_elim 7, 8
10 not_elim 8, 9
11 discharge
12 indirect 8, 10
13 discharge
14 assumption X
15 reiteration 14
16 discharge
17 or_elim 5, 6, 12, 14, 15
18 discharge
19 imp_intro 5, 17
20 and_intro 4, 19
21 qed

```

Chapter 22, Exercise B.8

Proof presented using Fitch notation:

1	F	PR
2	$\neg K$	PR
3	$\neg R$	PR
4	$\hline (F \wedge (K \wedge R))$	AS
5	F	$\wedge E$, 4
6	$(K \wedge R)$	$\wedge E$, 4
7	K	$\wedge E$, 6
8	$\perp$	$\neg E$, 2, 7
9	$\neg(F \wedge (K \wedge R))$	$\neg I$, 4–8
10	$\hline F$	AS
11	$\hline K$	AS
12	$\perp$	$\neg E$, 2, 11
13	R	X, 12
14	$(K \rightarrow R)$	$\rightarrow I$, 11, 13
15	$\hline R$	AS
16	$\perp$	$\neg E$, 3, 15
17	K	X, 16
18	$(R \rightarrow K)$	$\rightarrow I$, 15, 17
19	$((K \rightarrow R) \wedge (R \rightarrow K))$	$\wedge I$, 14, 18
20	$(F \rightarrow ((K \rightarrow R) \wedge (R \rightarrow K)))$	$\rightarrow I$, 10, 19
21	$\hline ((K \rightarrow R) \wedge (R \rightarrow K))$	AS
22	F	R, 1
23	$((K \rightarrow R) \wedge (R \rightarrow K)) \rightarrow F$	$\rightarrow I$, 21, 22
24	$((F \rightarrow ((K \rightarrow R) \wedge (R \rightarrow K))) \wedge (((K \rightarrow R) \wedge (R \rightarrow K)) \rightarrow F))$	$\wedge I$, 20, 23
25	$(\neg(F \wedge (K \wedge R)) \wedge ((F \rightarrow ((K \rightarrow R) \wedge (R \rightarrow K))) \wedge (((K \rightarrow R) \wedge (R \rightarrow K)) \rightarrow F)))$	$\wedge I$, 9, 24

Daeducer commands needed to generate the proof:

```

1 premise F
2 premise !K
3 premise !R
4 assumption (F ^ (K ^ R))
5 and_elim_left 4
6 and_elim_right 4
7 and_elim_left 6
8 not_elim 2, 7
9 discharge
10 not_intro 4, 8
11 assumption F
12 assumption K
13 not_elim 2, 12
14 explosion 13, R

```



```

15 discharge
16 imp_intro 12, 14
17 assumption R
18 not_elim 3, 17
19 explosion 18, K
20 discharge
21 imp_intro 17, 19
22 and_intro 16, 21
23 discharge
24 imp_intro 11, 22
25 assumption ((K -> R) ^ (R -> K))
26 reiteration 1
27 discharge
28 imp_intro 25, 26
29 and_intro 24, 28
30 and_intro 10, 29
31 qed

```

Chapter 22, Exercise B.9

Proof presented using Fitch notation:

1	$\neg(W \rightarrow W)$	PR
2	$((W \rightarrow W) \wedge (W \rightarrow W)) \wedge W$	PR
3	$(E \vee (W \rightarrow \neg(E \wedge W)))$	PR
4	$((W \rightarrow W) \wedge (W \rightarrow W))$	$\wedge E, 2$
5	$(W \rightarrow W)$	$\wedge E, 4$
6	$\perp$	$\neg E, 1, 5$

Daeducer commands needed to generate the proof:

```

1 premise !(W -> W)
2 premise (((W -> W) ^ (W -> W)) ^ W)
3 premise (E v (W -> !(E ^ W)))
4 and_elim_left 2
5 and_elim_left 4
6 not_elim 1, 5
7 qed

```

Chapter 22, Exercise B.10

Proof presented using Fitch notation:

1	C	PR
2	$\neg R$	PR
3	$(\neg R \vee C)$	$\vee E$, 2
4	$(C \wedge R)$	AS
5	$\neg R$	R, 2
6	$((C \wedge R) \rightarrow \neg R)$	$\rightarrow I$, 4, 5
7	$\neg(R \vee C)$	AS
8	$(R \vee C)$	$\vee E$, 1
9	$\perp$	$\neg E$, 7, 8
10	R	X, 9
11	$(\neg(R \vee C) \rightarrow R)$	$\rightarrow I$, 7, 10
12	$((C \wedge R) \rightarrow \neg R) \wedge (\neg(R \vee C) \rightarrow R)$	$\wedge I$, 6, 11
13	$((\neg R \vee C) \wedge (((C \wedge R) \rightarrow \neg R) \wedge (\neg(R \vee C) \rightarrow R)))$	$\wedge I$, 3, 12

Daeducer commands needed to generate the proof:

```

1 premise C
2 premise !R
3 or_intro_left 2, C
4 assumption (C ~ R)
5 reiteration 2
6 discharge
7 imp_intro 4, 5
8 assumption !(R v C)
9 or_intro_right 1, R
10 not_elim 8, 9
11 explosion 10, R
12 discharge
13 imp_intro 8, 11
14 and_intro 7, 13
15 and_intro 3, 14
16 qed

```

Chapter 22, Exercise B.11

Proof presented using Fitch notation:

1	$\neg\neg((C \rightarrow \neg C) \wedge (\neg C \rightarrow C))$	PR
2	$((G \vee G) \vee G)$	PR
3	$((C \rightarrow \neg C) \wedge (\neg C \rightarrow C))$	AS
4	$(C \rightarrow \neg C)$	$\wedge$ E, 3
5	$(\neg C \rightarrow C)$	$\wedge$ E, 3
6	C	AS
7	$\neg C$	$\rightarrow$ E, 4, 6
8	$\perp$	$\neg$ E, 7, 6
9	$\neg C$	$\neg$ I, 6–8
10	$\neg C$	AS
11	C	$\rightarrow$ E, 5, 10
12	$\perp$	$\neg$ E, 10, 11
13	$\neg\neg C$	$\neg$ I, 10–12
14	$\perp$	$\neg$ E, 13, 9
15	$\neg((C \rightarrow \neg C) \wedge (\neg C \rightarrow C))$	$\neg$ I, 3–14
16	$\perp$	$\neg$ E, 1, 15
17	$((G \rightarrow C) \wedge G)$	X, 16

Daeducer commands needed to generate the proof:

```

1 premise !!(C -> !C) ^ (!C -> C)
2 premise ((G v G) v G)
3 assumption ((C -> !C) ^ (!C -> C))
4 and_elim_left 3
5 and_elim_right 3
6 assumption C
7 imp_elim 4, 6
8 not_elim 7, 6
9 discharge
10 not_intro 6, 8
11 assumption !C
12 imp_elim 5, 11
13 not_elim 11, 12
14 discharge
15 not_intro 11, 13
16 not_elim 15, 10
17 discharge
18 not_intro 3, 16
19 not_elim 1, 18
20 explosion 19, ((G -> C) ^ G)
21 qed

```

Chapter 22, Exercise B.12

Proof presented using Fitch notation:

1	C	PR
2	L	PR
3	$\neg L$	AS
4	$\perp$	$\neg$ E, 3, 2
5	$\neg\neg L$	$\neg$ I, 3-4
6	$\neg C$	AS
7	$\perp$	$\neg$ E, 6, 1
8	$\neg\neg C$	$\neg$ I, 6-7
9	$(C \rightarrow \neg L)$	AS
10	C	R, 1
11	$((C \rightarrow \neg L) \rightarrow C)$	$\rightarrow$ I, 9, 10
12	$((C \rightarrow \neg L) \rightarrow C) \wedge \neg\neg C$	$\wedge$ I, 11, 8
13	$(\neg\neg L \wedge (((C \rightarrow \neg L) \rightarrow C) \wedge \neg\neg C))$	$\wedge$ I, 5, 12

Daeducer commands needed to generate the proof:

```

1  premise C
2  premise L
3  assumption !L
4  not_elim 3, 2
5  discharge
6  not_intro 3, 4
7  assumption !C
8  not_elim 7, 1
9  discharge
10 not_intro 7, 8
11 assumption (C -> !L)
12 reiteration 1
13 discharge
14 imp_intro 11, 12
15 and_intro 14, 10
16 and_intro 6, 15
17 qed

```

Chapter 24, Exercise A.1

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow F(x))$	PR
2	$\forall x(H(x) \rightarrow G(x))$	PR
3	$H(y)$	AS
4	$(H(y) \rightarrow G(y))$	$\forall E, 2$
5	$G(y)$	$\rightarrow E, 4, 3$
6	$(G(y) \rightarrow F(y))$	$\forall E, 1$
7	$F(y)$	$\rightarrow E, 6, 5$
8	$(H(y) \rightarrow F(y))$	$\rightarrow I, 3, 7$
9	$\forall x(H(x) \rightarrow F(x))$	$\forall I, 8$

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> F(x))
2 premise forall x (H(x) -> G(x))
3 assumption H(y)
4 forall_elim 2, y
5 imp_elim 4, 3
6 forall_elim 1, y
7 imp_elim 6, 5
8 discharge
9 imp_intro 3, 7
10 forall_intro 9, y, x
11 qed

```

Chapter 24, Exercise A.2

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow \neg F(x))$	PR
2	$\forall x(H(x) \rightarrow G(x))$	PR
3	$H(a)$	AS
4	$(H(a) \rightarrow G(a))$	$\forall E, 2$
5	$G(a)$	$\rightarrow E, 4, 3$
6	$(G(a) \rightarrow \neg F(a))$	$\forall E, 1$
7	$\neg F(a)$	$\rightarrow E, 6, 5$
8	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I, 3, 7$
9	$\forall x(H(x) \rightarrow \neg F(x))$	$\forall I, 8$

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> !F(x))
2 premise forall x (H(x) -> G(x))
3 assumption H(a)
4 forall_elim 2, a
5 imp_elim 4, 3

```

```

6 forall_elim 1, a
7 imp_elim 6, 5
8 discharge
9 imp_intro 3, 7
10 forall_intro 9, a, x
11 qed

```

Chapter 24, Exercise A.3

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow \neg F(x))$	PR
2	$\exists x(H(x) \wedge G(x))$	PR
3	$(H(a) \wedge G(a))$	AS
4	$H(a)$	$\wedge E, 3$
5	$G(a)$	$\wedge E, 3$
6	$(G(a) \rightarrow \neg F(a))$	$\forall E, 1$
7	$\neg F(a)$	$\rightarrow E, 6, 5$
8	$(H(a) \wedge \neg F(a))$	$\wedge I, 4, 7$
9	$\exists x(H(x) \wedge \neg F(x))$	$\exists I, 8$
10	$\exists x(H(x) \wedge \neg F(x))$	$\exists E, 2, 3-9$

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> !F(x))
2 premise exists x (H(x) ^ G(x))
3 assumption (H(a) ^ G(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 imp_elim 6, 5
8 and_intro 4, 7
9 exists_intro 8, (H(x) ^ !F(x))
10 discharge
11 exists_elim 2, 3, 9
12 qed

```

Chapter 24, Exercise A.4

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow F(x))$	PR
2	$\exists x(H(x) \wedge G(x))$	PR
3	$(H(a) \wedge G(a))$	AS
4	$H(a)$	$\wedge E$, 3
5	$G(a)$	$\wedge E$, 3
6	$(G(a) \rightarrow F(a))$	$\forall E$, 1
7	$F(a)$	$\rightarrow E$, 6, 5
8	$(H(a) \wedge F(a))$	$\wedge I$, 4, 7
9	$\exists x(H(x) \wedge F(x))$	$\exists I$, 8
10	$\exists x(H(x) \wedge F(x))$	$\exists E$, 2, 3–9

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> F(x))
2 premise exists x (H(x) ^ G(x))
3 assumption (H(a) ^ G(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 imp_elim 6, 5
8 and_intro 4, 7
9 exists_intro 8, (H(x) ^ F(x))
10 discharge
11 exists_elim 2, 3, 9
12 qed

```

Chapter 24, Exercise A.5

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow G(x))$	PR
2	$\forall x(H(x) \rightarrow \neg G(x))$	PR
3	$(F(a) \rightarrow G(a))$	$\forall E$, 1
4	$(H(a) \rightarrow \neg G(a))$	$\forall E$, 2
5	$H(a)$	AS
6	$\neg G(a)$	$\rightarrow E$, 4, 5
7	$\neg F(a)$	MT, 3, 6
8	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I$, 5, 7
9	$\forall x(H(x) \rightarrow \neg F(x))$	$\forall I$, 8

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> G(x))
2 premise forall x (H(x) -> !G(x))

```

```

3 forall_elim 1, a
4 forall_elim 2, a
5 assumption H(a)
6 imp_elim 4, 5
7 modus_tollens 3, 6
8 discharge
9 imp_intro 5, 7
10 forall_intro 9, a, x
11 qed

```

Chapter 24, Exercise A.6

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow \neg G(x))$	PR
2	$\forall x(H(x) \rightarrow G(x))$	PR
3	$H(a)$	AS
4	$(H(a) \rightarrow G(a))$	$\forall E, 2$
5	$G(a)$	$\rightarrow E, 4, 3$
6	$(F(a) \rightarrow \neg G(a))$	$\forall E, 1$
7	$F(a)$	AS
8	$\neg G(a)$	$\rightarrow E, 6, 7$
9	$\perp$	$\neg E, 8, 5$
10	$\neg F(a)$	$\neg I, 7-9$
11	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I, 3, 10$
12	$\forall x(H(x) \rightarrow \neg F(x))$	$\forall I, 11$

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> !G(x))
2 premise forall x (H(x) -> G(x))
3 assumption H(a)
4 forall_elim 2, a
5 imp_elim 4, 3
6 forall_elim 1, a
7 assumption F(a)
8 imp_elim 6, 7
9 not_elim 8, 5
10 discharge
11 not_intro 7, 9
12 discharge
13 imp_intro 3, 11
14 forall_intro 13, a, x
15 qed

```


Chapter 24, Exercise A.7

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow G(x))$	PR
2	$\exists x(H(x) \wedge \neg G(x))$	PR
3	$(H(a) \wedge \neg G(a))$	AS
4	$H(a)$	$\wedge E$, 3
5	$\neg G(a)$	$\wedge E$, 3
6	$(F(a) \rightarrow G(a))$	$\forall E$, 1
7	$\neg F(a)$	MT, 6, 5
8	$(H(a) \wedge \neg F(a))$	$\wedge I$, 4, 7
9	$\exists x(H(x) \wedge \neg F(x))$	$\exists I$, 8
10	$\exists x(H(x) \wedge \neg F(x))$	$\exists E$, 2, 3–9

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> G(x))
2 premise exists x (H(x) ^ !G(x))
3 assumption (H(a) ^ !G(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 modus_tollens 6, 5
8 and_intro 4, 7
9 exists_intro 8, (H(x) ^ !F(x))
10 discharge
11 exists_elim 2, 3, 9
12 qed

```

Chapter 24, Exercise A.8

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow \neg G(x))$	PR
2	$\exists x(H(x) \wedge G(x))$	PR
3	$(H(a) \wedge G(a))$	AS
4	$H(a)$	$\wedge E, 3$
5	$G(a)$	$\wedge E, 3$
6	$(F(a) \rightarrow \neg G(a))$	$\forall E, 1$
7	$F(a)$	AS
8	$\neg G(a)$	$\rightarrow E, 6, 7$
9	$\perp$	$\neg E, 8, 5$
10	$\neg F(a)$	$\neg I, 7-9$
11	$(H(a) \wedge \neg F(a))$	$\wedge I, 4, 10$
12	$\exists x(H(x) \wedge \neg F(x))$	$\exists I, 11$
13	$\exists x(H(x) \wedge \neg F(x))$	$\exists E, 2, 3-12$

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> !G(x))
2 premise exists x (H(x) ^ G(x))
3 assumption (H(a) ^ G(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 assumption F(a)
8 imp_elim 6, 7
9 not_elim 8, 5
10 discharge
11 not_intro 7, 9
12 and_intro 4, 11
13 exists_intro 12, (H(x) ^ !F(x))
14 discharge
15 exists_elim 2, 3, 13
16 qed

```

Chapter 24, Exercise A.9

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow F(x))$	PR
2	$\exists x(G(x) \wedge H(x))$	PR
3	$(G(a) \wedge H(a))$	AS
4	$G(a)$	$\wedge E, 3$
5	$H(a)$	$\wedge E, 3$
6	$(G(a) \rightarrow F(a))$	$\forall E, 1$
7	$F(a)$	$\rightarrow E, 6, 4$
8	$(H(a) \wedge F(a))$	$\wedge I, 5, 7$
9	$\exists x(H(x) \wedge F(x))$	$\exists I, 8$
10	$\exists x(H(x) \wedge F(x))$	$\exists E, 2, 3-9$

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> F(x))
2 premise exists x (G(x) ^ H(x))
3 assumption (G(a) ^ H(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 imp_elim 6, 4
8 and_intro 5, 7
9 exists_intro 8, (H(x) ^ F(x))
10 discharge
11 exists_elim 2, 3, 9
12 qed

```

Chapter 24, Exercise A.10

Proof presented using Fitch notation:

1	$\exists x(G(x) \wedge F(x))$	PR
2	$\forall x(G(x) \rightarrow H(x))$	PR
3	$(G(a) \wedge F(a))$	AS
4	$G(a)$	$\wedge E, 3$
5	$F(a)$	$\wedge E, 3$
6	$(G(a) \rightarrow H(a))$	$\forall E, 2$
7	$H(a)$	$\rightarrow E, 6, 4$
8	$(H(a) \wedge F(a))$	$\wedge I, 7, 5$
9	$\exists x(H(x) \wedge F(x))$	$\exists I, 8$
10	$\exists x(H(x) \wedge F(x))$	$\exists E, 1, 3-9$

Daeducer commands needed to generate the proof:

```

1 premise exists x (G(x) ^ F(x))

```

```

2 premise forall x (G(x) -> H(x))
3 assumption (G(a) ^ F(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 2, a
7 imp_elim 6, 4
8 and_intro 7, 5
9 exists_intro 8, (H(x) ^ F(x))
10 discharge
11 exists_elim 1, 3, 9
12 qed

```

Chapter 24, Exercise A.11

Proof presented using Fitch notation:

1	$\forall x(G(x) \rightarrow \neg F(x))$	PR
2	$\exists x(G(x) \wedge H(x))$	PR
3	$(G(a) \wedge H(a))$	AS
4	$G(a)$	$\wedge E, 3$
5	$H(a)$	$\wedge E, 3$
6	$(G(a) \rightarrow \neg F(a))$	$\forall E, 1$
7	$\neg F(a)$	$\rightarrow E, 6, 4$
8	$(H(a) \wedge \neg F(a))$	$\wedge I, 5, 7$
9	$\exists x(H(x) \wedge \neg F(x))$	$\exists I, 8$
10	$\exists x(H(x) \wedge \neg F(x))$	$\exists E, 2, 3-9$

Daeducer commands needed to generate the proof:

```

1 premise forall x (G(x) -> !F(x))
2 premise exists x (G(x) ^ H(x))
3 assumption (G(a) ^ H(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 imp_elim 6, 4
8 and_intro 5, 7
9 exists_intro 8, (H(x) ^ !F(x))
10 discharge
11 exists_elim 2, 3, 9
12 qed

```

Chapter 24, Exercise A.12

Proof presented using Fitch notation:

1	$\exists x(G(x) \wedge \neg F(x))$	PR
2	$\forall x(G(x) \rightarrow H(x))$	PR
3	$(G(a) \wedge \neg F(a))$	AS
4	$G(a)$	$\wedge E, 3$
5	$\neg F(a)$	$\wedge E, 3$
6	$(G(a) \rightarrow H(a))$	$\forall E, 2$
7	$H(a)$	$\rightarrow E, 6, 4$
8	$(H(a) \wedge \neg F(a))$	$\wedge I, 7, 5$
9	$\exists x(H(x) \wedge \neg F(x))$	$\exists I, 8$
10	$\exists x(H(x) \wedge \neg F(x))$	$\exists E, 1, 3-9$

Daeducer commands needed to generate the proof:

```

1 premise exists x (G(x) ^ !F(x))
2 premise forall x (G(x) -> H(x))
3 assumption (G(a) ^ !F(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 2, a
7 imp_elim 6, 4
8 and_intro 7, 5
9 exists_intro 8, (H(x) ^ !F(x))
10 discharge
11 exists_elim 1, 3, 9
12 qed

```

Chapter 24, Exercise A.13

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow G(x))$	PR
2	$\forall x(G(x) \rightarrow \neg H(x))$	PR
3	$(F(x) \rightarrow G(x))$	$\forall E, 1$
4	$(G(x) \rightarrow \neg H(x))$	$\forall E, 2$
5	$H(x)$	AS
6	$F(x)$	AS
7	$G(x)$	$\rightarrow E, 3, 6$
8	$\neg H(x)$	$\rightarrow E, 4, 7$
9	$\perp$	$\neg E, 8, 5$
10	$\neg F(x)$	$\neg I, 6-9$
11	$(H(x) \rightarrow \neg F(x))$	$\rightarrow I, 5, 10$
12	$\forall x(H(x) \rightarrow \neg F(x))$	$\forall I, 11$

Daeducer commands needed to generate the proof:

```

1  premise forall x (F(x) -> G(x))
2  premise forall x (G(x) -> !H(x))
3  forall_elim 1, x
4  forall_elim 2, x
5  assumption H(x)
6  assumption F(x)
7  imp_elim 3, 6
8  imp_elim 4, 7
9  not_elim 8, 5
10 discharge
11 not_intro 6, 9
12 discharge
13 imp_intro 5, 11
14 forall_intro 13, x, x
15 qed

```

Chapter 24, Exercise A.14

Proof presented using Fitch notation:

1	$\exists x(F(x) \wedge G(x))$	PR
2	$\forall x(G(x) \rightarrow H(x))$	PR
3	$(F(a) \wedge G(a))$	AS
4	$F(a)$	$\wedge E$, 3
5	$G(a)$	$\wedge E$, 3
6	$(G(a) \rightarrow H(a))$	$\forall E$, 2
7	$H(a)$	$\rightarrow E$, 6, 5
8	$(H(a) \wedge F(a))$	$\wedge I$, 7, 4
9	$\exists x(H(x) \wedge F(x))$	$\exists I$, 8
10	$\exists x(H(x) \wedge F(x))$	$\exists E$, 1, 3–9

Daeducer commands needed to generate the proof:

```

1  premise exists x (F(x) ^ G(x))
2  premise forall x (G(x) -> H(x))
3  assumption (F(a) ^ G(a))
4  and_elim_left 3
5  and_elim_right 3
6  forall_elim 2, a
7  imp_elim 6, 5
8  and_intro 7, 4
9  exists_intro 8, (H(x) ^ F(x))
10 discharge
11 exists_elim 1, 3, 9
12 qed

```

Chapter 24, Exercise A.15

Proof presented using Fitch notation:

1	$\forall x(F(x) \rightarrow \neg G(x))$	PR
2	$\exists x(G(x) \wedge H(x))$	PR
3	$(G(a) \wedge H(a))$	AS
4	$G(a)$	$\wedge E, 3$
5	$H(a)$	$\wedge E, 3$
6	$(F(a) \rightarrow \neg G(a))$	$\forall E, 1$
7	$F(a)$	AS
8	$\neg G(a)$	$\rightarrow E, 6, 7$
9	$\perp$	$\neg E, 8, 4$
10	$\neg F(a)$	$\neg I, 7-9$
11	$(H(a) \wedge \neg F(a))$	$\wedge I, 5, 10$
12	$\exists x(H(x) \wedge \neg F(x))$	$\exists I, 11$
13	$\exists x(H(x) \wedge \neg F(x))$	$\exists E, 2, 3-12$

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> !G(x))
2 premise exists x (G(x) ^ H(x))
3 assumption (G(a) ^ H(a))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, a
7 assumption F(a)
8 imp_elim 6, 7
9 not_elim 8, 4
10 discharge
11 not_intro 7, 9
12 and_intro 5, 11
13 exists_intro 12, (H(x) ^ !F(x))
14 discharge
15 exists_elim 2, 3, 13
16 qed

```

Chapter 36, Exercise B.1

Proof presented using Fitch notation:

1	$\forall x \exists y (R(x, y) \vee R(y, x))$	PR
2	$\forall x \neg R(m, x)$	PR
3	$\exists y (R(m, y) \vee R(y, m))$	$\forall E, 1$
4	$(R(m, a) \vee R(a, m))$	AS
5	$\neg R(m, a)$	$\forall E, 2$
6	$R(a, m)$	DS, 4, 5
7	$\exists x R(x, m)$	$\exists I, 6$
8	$\exists x R(x, m)$	$\exists E, 3, 4-7$

Daeducer commands needed to generate the proof:

```

1 premise forall x exists y (R(x, y) v R(y, x))
2 premise forall x !R(m, x)
3 forall_elim 1, m
4 assumption (R(m, a) v R(a, m))
5 forall_elim 2, a
6 or_syllogism_left 4, 5
7 exists_intro 6, R(x, m)
8 discharge
9 exists_elim 3, 4, 7
10 qed

```

Chapter 36, Exercise B.2

Proof presented using Fitch notation:

1	$\forall x (\exists y L(x, y) \rightarrow \forall z L(z, x))$	PR
2	$L(a, b)$	PR
3	$(\exists y L(a, y) \rightarrow \forall z L(z, a))$	$\forall E, 1$
4	$\exists y L(a, y)$	$\exists I, 2$
5	$\forall z L(z, a)$	$\rightarrow E, 3, 4$
6	$L(x, a)$	$\forall E, 5$
7	$\exists y L(x, y)$	$\exists I, 6$
8	$(\exists y L(x, y) \rightarrow \forall z L(z, x))$	$\forall E, 1$
9	$\forall z L(z, x)$	$\rightarrow E, 8, 7$
10	$L(x, x)$	$\forall E, 9$
11	$\forall x L(x, x)$	$\forall I, 10$

Daeducer commands needed to generate the proof:

```

1 premise forall x (exists y L(x, y) -> forall z L(z, x))
2 premise L(a, b)
3 forall_elim 1, a
4 exists_intro 2, L(a, y)

```



```

5  imp_elim 3, 4
6  forall_elim 5, x
7  exists_intro 6, L(x, y)
8  forall_elim 1, x
9  imp_elim 8, 7
10 forall_elim 9, x
11 forall_intro 10, x, x
12 qed

```

Chapter 36, Exercise B.3

Proof presented using Fitch notation:

1	$\forall x(J(x) \rightarrow K(x))$	PR
2	$\exists x \forall y L(x, y)$	PR
3	$\forall x J(x)$	PR
4	$\forall y L(a, y)$	AS
5	$L(a, a)$	$\forall E, 4$
6	$J(a)$	$\forall E, 3$
7	$(J(a) \rightarrow K(a))$	$\forall E, 1$
8	$K(a)$	$\rightarrow E, 7, 6$
9	$(K(a) \wedge L(a, a))$	$\wedge I, 8, 5$
10	$\exists x (K(x) \wedge L(x, x))$	$\exists I, 9$
11	$\exists x (K(x) \wedge L(x, x))$	$\exists E, 2, 4-10$

Daeducer commands needed to generate the proof:

```

1  premise forall x (J(x) -> K(x))
2  premise exists x forall y L(x, y)
3  premise forall x J(x)
4  assumption forall y L(a, y)
5  forall_elim 4, a
6  forall_elim 3, a
7  forall_elim 1, a
8  imp_elim 7, 6
9  and_intro 8, 5
10 exists_intro 9, (K(x) ^ L(x, x))
11 discharge
12 exists_elim 2, 4, 10
13 qed

```

Chapter 36, Exercise D.1

Proof presented using Fitch notation:

1	$\exists x H(x)$	PR
2	$\forall x (G(x) \rightarrow F(x))$	PR
3	$\exists x (H(x) \rightarrow G(x))$	PR
4	$(H(a) \rightarrow G(a))$	AS
5	$H(a)$	AS
6	$G(a)$	$\rightarrow$ E, 4, 5
7	$(G(a) \rightarrow F(a))$	$\forall$ E, 2
8	$F(a)$	$\rightarrow$ E, 7, 6
9	$(H(a) \rightarrow F(a))$	$\rightarrow$ I, 5, 8
10	$\exists x (H(x) \rightarrow F(x))$	$\exists$ I, 9
11	$\exists x (H(x) \rightarrow F(x))$	$\exists$ E, 3, 4–10

Daeducer commands needed to generate the proof:

```

1 premise exists x H(x)
2 premise forall x (G(x) -> F(x))
3 premise exists x (H(x) -> G(x))
4 assumption (H(a) -> G(a))
5 assumption H(a)
6 imp_elim 4, 5
7 forall_elim 2, a
8 imp_elim 7, 6
9 discharge
10 imp_intro 5, 8
11 exists_intro 10, (H(x) -> F(x))
12 discharge
13 exists_elim 3, 4, 11
14 qed

```

Chapter 36, Exercise D.2

Proof presented using Fitch notation:

1	$\exists x H(x)$	PR
2	$\forall x (G(x) \rightarrow \neg F(x))$	PR
3	$\exists x (H(x) \rightarrow G(x))$	PR
4	$(H(c) \rightarrow G(c))$	AS
5	$(G(c) \rightarrow \neg F(c))$	$\forall E$, 2
6	$H(c)$	AS
7	$G(c)$	$\rightarrow E$, 4, 6
8	$\neg F(c)$	$\rightarrow E$, 5, 7
9	$(H(c) \rightarrow \neg F(c))$	$\rightarrow I$, 6, 8
10	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I$, 9
11	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E$, 3, 4–10

Daeducer commands needed to generate the proof:

```

1  premise exists x H(x)
2  premise forall x (G(x) -> !F(x))
3  premise exists x (H(x) -> G(x))
4  assumption (H(c) -> G(c))
5  forall_elim 2, c
6  assumption H(c)
7  imp_elim 4, 6
8  imp_elim 5, 7
9  discharge
10 imp_intro 6, 8
11 exists_intro 10, (H(x) -> !F(x))
12 discharge
13 exists_elim 3, 4, 11
14 qed

```

Chapter 36, Exercise D.3

Proof presented using Fitch notation:

1	$\exists x H(x)$	PR
2	$\forall x (F(x) \rightarrow \neg G(x))$	PR
3	$\forall x (H(x) \rightarrow G(x))$	PR
4	$H(c)$	AS
5	$(F(c) \rightarrow \neg G(c))$	$\forall E, 2$
6	$(H(c) \rightarrow G(c))$	$\forall E, 3$
7	$H(c)$	AS
8	$G(c)$	$\rightarrow E, 6, 7$
9	$F(c)$	AS
10	$\neg G(c)$	$\rightarrow E, 5, 9$
11	$\perp$	$\neg E, 10, 8$
12	$\neg F(c)$	$\neg I, 9-11$
13	$(H(c) \rightarrow \neg F(c))$	$\rightarrow I, 7, 12$
14	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I, 13$
15	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E, 1, 4-14$

Daeducer commands needed to generate the proof:

```

1 premise exists x H(x)
2 premise forall x (F(x) -> !G(x))
3 premise forall x (H(x) -> G(x))
4 assumption H(c)
5 forall_elim 2, c
6 forall_elim 3, c
7 assumption H(c)
8 imp_elim 6, 7
9 assumption F(c)
10 imp_elim 5, 9
11 not_elim 10, 8
12 discharge
13 not_intro 9, 11
14 discharge
15 imp_intro 7, 13
16 exists_intro 15, (H(x) -> !F(x))
17 discharge
18 exists_elim 1, 4, 16
19 qed

```

Chapter 36, Exercise D.4

Proof presented using Fitch notation:

1	$\exists x H(x)$	PR
2	$\forall x (F(x) \rightarrow G(x))$	PR
3	$\forall x (H(x) \rightarrow \neg G(x))$	PR
4	$H(a)$	AS
5	$(F(a) \rightarrow G(a))$	$\forall E, 2$
6	$(H(a) \rightarrow \neg G(a))$	$\forall E, 3$
7	$H(a)$	AS
8	$\neg G(a)$	$\rightarrow E, 6, 4$
9	$\neg F(a)$	MT, 5, 8
10	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I, 7, 9$
11	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I, 10$
12	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E, 1, 4-11$

Daeducer commands needed to generate the proof:

```

1 premise exists x H(x)
2 premise forall x (F(x) -> G(x))
3 premise forall x (H(x) -> !G(x))
4 assumption H(a)
5 forall_elim 2, a
6 forall_elim 3, a
7 assumption H(a)
8 imp_elim 6, 4
9 modus_tollens 5, 8
10 discharge
11 imp_intro 7, 9
12 exists_intro 11, (H(x) -> !F(x))
13 discharge
14 exists_elim 1, 4, 12
15 qed

```

Chapter 36, Exercise D.5

Proof presented using Fitch notation:

1	$\exists x G(x)$	PR
2	$\forall x (G(x) \rightarrow \neg F(x))$	PR
3	$\forall x (G(x) \rightarrow H(x))$	PR
4	$G(a)$	AS
5	$(G(a) \rightarrow \neg F(a))$	$\forall E$, 2
6	$\neg F(a)$	$\rightarrow E$, 5, 4
7	$H(a)$	AS
8	$\neg F(a)$	R, 6
9	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I$, 7, 8
10	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I$, 9
11	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E$, 1, 4–10

Daeducer commands needed to generate the proof:

```

1  premise exists x G(x)
2  premise forall x (G(x) -> !F(x))
3  premise forall x (G(x) -> H(x))
4  assumption G(a)
5  forall_elim 2, a
6  imp_elim 5, 4
7  assumption H(a)
8  reiteration 6
9  discharge
10 imp_intro 7, 8
11 exists_intro 10, (H(x) -> !F(x))
12 discharge
13 exists_elim 1, 4, 11
14 qed

```

Chapter 36, Exercise D.6

Proof presented using Fitch notation:

1	$\exists x G(x)$	PR
2	$\forall x (G(x) \rightarrow F(x))$	PR
3	$\forall x (G(x) \rightarrow H(x))$	PR
4	$G(c)$	AS
5	$(G(c) \rightarrow F(c))$	$\forall E, 2$
6	$F(c)$	$\rightarrow E, 5, 4$
7	$H(c)$	AS
8	$F(c)$	R, 6
9	$(H(c) \rightarrow F(c))$	$\rightarrow I, 7, 8$
10	$\exists x (H(x) \rightarrow F(x))$	$\exists I, 9$
11	$\exists x (H(x) \rightarrow F(x))$	$\exists E, 1, 4-10$

Daeducer commands needed to generate the proof:

```

1  premise exists x G(x)
2  premise forall x (G(x) -> F(x))
3  premise forall x (G(x) -> H(x))
4  assumption G(c)
5  forall_elim 2, c
6  imp_elim 5, 4
7  assumption H(c)
8  reiteration 6
9  discharge
10 imp_intro 7, 8
11 exists_intro 10, (H(x) -> F(x))
12 discharge
13 exists_elim 1, 4, 11
14 qed

```

Chapter 36, Exercise D.7

Proof presented using Fitch notation:

1	$\exists x H(x)$	PR
2	$\forall x (F(x) \rightarrow G(x))$	PR
3	$\forall x (G(x) \rightarrow \neg H(x))$	PR
4	$H(c)$	AS
5	$(F(c) \rightarrow G(c))$	$\forall E, 2$
6	$(G(c) \rightarrow \neg H(c))$	$\forall E, 3$
7	$F(c)$	AS
8	$G(c)$	$\rightarrow E, 5, 7$
9	$\neg H(c)$	$\rightarrow E, 6, 8$
10	$\perp$	$\neg E, 9, 4$
11	$\neg F(c)$	$\neg I, 7-10$
12	$H(c)$	AS
13	$\neg F(c)$	R, 11
14	$(H(c) \rightarrow \neg F(c))$	$\rightarrow I, 12, 13$
15	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I, 14$
16	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E, 1, 4-15$

Daeducer commands needed to generate the proof:

```

1  premise exists x H(x)
2  premise forall x (F(x) -> G(x))
3  premise forall x (G(x) -> !H(x))
4  assumption H(c)
5  forall_elim 2, c
6  forall_elim 3, c
7  assumption F(c)
8  imp_elim 5, 7
9  imp_elim 6, 8
10 not_elim 9, 4
11 discharge
12 not_intro 7, 10
13 assumption H(c)
14 reiteration 12
15 discharge
16 imp_intro 13, 14
17 exists_intro 16, (H(x) -> !F(x))
18 discharge
19 exists_elim 1, 4, 17
20 qed

```


Chapter 36, Exercise D.8

Proof presented using Fitch notation:

1	$\exists x G(x)$	PR
2	$\forall x (F(x) \rightarrow \neg G(x))$	PR
3	$\forall x (G(x) \rightarrow H(x))$	PR
4	$G(a)$	AS
5	$(F(a) \rightarrow \neg G(a))$	$\forall E, 2$
6	$(G(a) \rightarrow H(a))$	$\forall E, 3$
7	$H(a)$	$\rightarrow E, 6, 4$
8	$F(a)$	AS
9	$\neg G(a)$	$\rightarrow E, 5, 8$
10	$\perp$	$\neg E, 9, 4$
11	$\neg F(a)$	$\neg I, 8-10$
12	$H(a)$	AS
13	$\neg F(a)$	R, 11
14	$(H(a) \rightarrow \neg F(a))$	$\rightarrow I, 12, 13$
15	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists I, 14$
16	$\exists x (H(x) \rightarrow \neg F(x))$	$\exists E, 1, 4-15$

Daeducer commands needed to generate the proof:

```

1 premise exists x G(x)
2 premise forall x (F(x) -> !G(x))
3 premise forall x (G(x) -> H(x))
4 assumption G(a)
5 forall_elim 2, a
6 forall_elim 3, a
7 imp_elim 6, 4
8 assumption F(a)
9 imp_elim 5, 8
10 not_elim 9, 4
11 discharge
12 not_intro 8, 10
13 assumption H(a)
14 reiteration 12
15 discharge
16 imp_intro 13, 14
17 exists_intro 16, (H(x) -> !F(x))
18 discharge
19 exists_elim 1, 4, 17
20 qed

```

Chapter 36, Exercise D.9

Proof presented using Fitch notation:

1	$\exists x F(x)$	PR
2	$\forall x (F(x) \rightarrow G(x))$	PR
3	$\forall x (G(x) \rightarrow H(x))$	PR
4	$F(a)$	AS
5	$H(a)$	AS
6	$F(a)$	R, 4
7	$(H(a) \rightarrow F(a))$	$\rightarrow$ I, 5, 6
8	$\exists x (H(x) \rightarrow F(x))$	$\exists$ I, 7
9	$\exists x (H(x) \rightarrow F(x))$	$\exists$ E, 1, 4–8

Daeducer commands needed to generate the proof:

```

1 premise exists x F(x)
2 premise forall x (F(x) -> G(x))
3 premise forall x (G(x) -> H(x))
4 assumption F(a)
5 assumption H(a)
6 reiteration 4
7 discharge
8 imp_intro 5, 6
9 exists_intro 8, (H(x) -> F(x))
10 discharge
11 exists_elim 1, 4, 9
12 qed

```

Chapter 36, Exercise E.1

Proof presented using Fitch notation:

1	$\forall x F(x)$	AS
2	$F(y)$	$\forall$ E, 1
3	$(F(y) \wedge F(y))$	$\wedge$ I, 2, 2
4	$\forall y (F(y) \wedge F(y))$	$\forall$ I, 3
5	$(\forall x F(x) \rightarrow \forall y (F(y) \wedge F(y)))$	$\rightarrow$ I, 1, 4

Daeducer commands needed to generate the proof:

```

1 assumption forall x F(x)
2 forall_elim 1, y
3 and_intro 2, 2
4 forall_intro 3, y, y
5 discharge
6 imp_intro 1, 4
7 qed

```

Chapter 36, Exercise E.2

Proof presented using Fitch notation:

1	$\forall x(A(x) \rightarrow B(x))$	PR
2	$\exists x A(x)$	PR
3	$A(a)$	AS
4	$(A(a) \rightarrow B(a))$	$\forall E, 1$
5	$B(a)$	$\rightarrow E, 4, 3$
6	$\exists x B(x)$	$\exists I, 5$
7	$\exists x B(x)$	$\exists E, 2, 3-6$

Daeducer commands needed to generate the proof:

```

1 premise forall x (A(x) -> B(x))
2 premise exists x A(x)
3 assumption A(a)
4 forall_elim 1, a
5 imp_elim 4, 3
6 exists_intro 5, B(x)
7 discharge
8 exists_elim 2, 3, 6
9 qed

```

Chapter 36, Exercise E.3

Proof presented using Fitch notation:

1	$\forall x(M(x) \leftrightarrow N(x))$	PR
2	$(M(a) \wedge \exists x R(x, a))$	PR
3	$M(a)$	$\wedge E, 2$
4	$(M(a) \leftrightarrow N(a))$	$\forall E, 1$
5	$N(a)$	$\leftrightarrow E, 4, 3$
6	$\exists x N(x)$	$\exists I, 5$

Daeducer commands needed to generate the proof:

```

1 premise forall x (M(x) <-> N(x))
2 premise (M(a) ^ exists x R(x, a))
3 and_elim_left 2
4 forall_elim 1, a
5 bi_elim 4, 3
6 exists_intro 5, N(x)
7 qed

```

Chapter 36, Exercise E.4

Proof presented using Fitch notation:

1	$\forall x \forall y G(x, y)$	PR
2	$\forall y G(a, y)$	$\forall E, 1$
3	$G(a, a)$	$\forall E, 2$
4	$\exists x G(x, x)$	$\exists I, 3$

Daeducer commands needed to generate the proof:

```
1 premise forall x forall y G(x, y)
2 forall_elim 1, a
3 forall_elim 2, a
4 exists_intro 3, G(x, x)
5 qed
```

Chapter 36, Exercise E.5

Proof presented using Fitch notation:

1	$\forall x R(x, x)$	AS
2	$R(a, a)$	$\forall E, 1$
3	$\exists y R(a, y)$	$\exists I, 2$
4	$\exists x \exists y R(x, y)$	$\exists I, 3$
5	$(\forall x R(x, x) \rightarrow \exists x \exists y R(x, y))$	$\rightarrow I, 1, 4$

Daeducer commands needed to generate the proof:

```
1 assumption forall x R(x, x)
2 forall_elim 1, a
3 exists_intro 2, R(a, y)
4 exists_intro 3, exists y R(x, y)
5 discharge
6 imp_intro 1, 4
7 qed
```

Chapter 36, Exercise E.6

Proof presented using Fitch notation:

1		$Q(a)$	AS
2		$Q(a)$	R, 1
3		$(Q(a) \rightarrow Q(a))$	$\rightarrow$ I, 1, 2
4		$\exists x(Q(a) \rightarrow Q(x))$	$\exists$ I, 3
5		$\forall y \exists x(Q(y) \rightarrow Q(x))$	$\forall$ I, 4

Daeducer commands needed to generate the proof:

```

1  assumption Q(a)
2  reiteration 1
3  discharge
4  imp_intro 1, 2
5  exists_intro 4, (Q(a) -> Q(x))
6  forall_intro 5, a, y
7  qed

```

Chapter 36, Exercise E.7

Proof presented using Fitch notation:

1		$(N(a) \rightarrow \forall x(M(x) \leftrightarrow M(a)))$	PR
2		$M(a)$	PR
3		$\neg M(b)$	PR
4		$N(a)$	AS
5		$\forall x(M(x) \leftrightarrow M(a))$	$\rightarrow$ E, 1, 4
6		$(M(b) \leftrightarrow M(a))$	$\forall$ E, 5
7		$M(b)$	$\leftrightarrow$ E, 6, 2
8		$\perp$	$\neg$ E, 3, 7
9		$\neg N(a)$	$\neg$ I, 4-8

Daeducer commands needed to generate the proof:

```

1  premise (N(a) -> forall x (M(x) <-> M(a)))
2  premise M(a)
3  premise !M(b)
4  assumption N(a)
5  imp_elim 1, 4
6  forall_elim 5, b
7  bi_elim 6, 2
8  not_elim 3, 7
9  discharge
10 not_intro 4, 8
11 qed

```

Chapter 36, Exercise E.8

Proof presented using Fitch notation:

1	$\forall x \forall y (G(x, y) \rightarrow G(y, x))$	PR
2	$\forall y (G(a, y) \rightarrow G(y, a))$	$\forall E, 1$
3	$(G(a, b) \rightarrow G(b, a))$	$\forall E, 2$
4	$G(a, b)$	AS
5	$G(b, a)$	$\rightarrow E, 3, 4$
6	$\forall y (G(b, y) \rightarrow G(y, b))$	$\forall E, 1$
7	$(G(b, a) \rightarrow G(a, b))$	$\forall E, 6$
8	$G(b, a)$	AS
9	$G(a, b)$	$\rightarrow E, 7, 8$
10	$(G(a, b) \leftrightarrow G(b, a))$	$\leftrightarrow I, 4-5, 8-9$
11	$\forall y (G(a, y) \leftrightarrow G(y, a))$	$\forall I, 10$
12	$\forall x \forall y (G(x, y) \leftrightarrow G(y, x))$	$\forall I, 11$

Daeducer commands needed to generate the proof:

```

1 premise forall x forall y (G(x, y) -> G(y, x))
2 forall_elim 1, a
3 forall_elim 2, b
4 assumption G(a, b)
5 imp_elim 3, 4
6 discharge
7 forall_elim 1, b
8 forall_elim 7, a
9 assumption G(b, a)
10 imp_elim 8, 9
11 discharge
12 bi_intro 4, 5, 9, 10
13 forall_intro 12, b, y
14 forall_intro 13, a, x
15 qed

```

Chapter 36, Exercise E.9

Proof presented using Fitch notation:

1	$\forall x(\neg M(x) \vee L(j, x))$	PR
2	$\forall x(B(x) \rightarrow L(j, x))$	PR
3	$\forall x(M(x) \vee B(x))$	PR
4	$(M(a) \vee B(a))$	$\forall E, 3$
5	$(\neg M(a) \vee L(j, a))$	$\forall E, 1$
6	$(B(a) \rightarrow L(j, a))$	$\forall E, 2$
7	$M(a)$	AS
8	$\neg M(a)$	AS
9	$\perp$	$\neg E, 8, 7$
10	$L(j, a)$	X, 9
11	$L(j, a)$	AS
12	$L(j, a)$	R, 11
13	$L(j, a)$	$\vee E, 5, 8-10, 11-12$
14	$B(a)$	AS
15	$L(j, a)$	$\rightarrow E, 6, 14$
16	$L(j, a)$	$\vee E, 4, 7-13, 14-15$
17	$\forall x L(j, x)$	$\forall I, 16$

Daeducer commands needed to generate the proof:

```

1 premise forall x (!M(x) v L(j, x))
2 premise forall x (B(x) -> L(j, x))
3 premise forall x (M(x) v B(x))
4 forall_elim 3, a
5 forall_elim 1, a
6 forall_elim 2, a
7 assumption M(a)
8 assumption !M(a)
9 not_elim 8, 7
10 explosion 9, L(j, a)
11 discharge
12 assumption L(j, a)
13 reiteration 12
14 discharge
15 or_elim 5, 8, 10, 12, 13
16 discharge
17 assumption B(a)
18 imp_elim 6, 17
19 discharge
20 or_elim 4, 7, 15, 17, 18
21 forall_intro 20, a, x
22 qed

```

ch36-exF

Proof presented using Fitch notation:

1	$\exists x \forall y \forall z ((L(x, y) \rightarrow L(z, y)) \leftrightarrow L(y, z))$	PR
2	$\forall y \forall z ((L(a, y) \rightarrow L(z, y)) \leftrightarrow L(y, z))$	AS
3	$\forall z ((L(a, a) \rightarrow L(z, a)) \leftrightarrow L(a, z))$	$\forall E, 2$
4	$((L(a, a) \rightarrow L(a, a)) \leftrightarrow L(a, a))$	$\forall E, 3$
5	$L(a, a)$	AS
6	$L(a, a)$	R, 5
7	$(L(a, a) \rightarrow L(a, a))$	$\rightarrow I, 5, 6$
8	$L(a, a)$	$\leftrightarrow E, 4, 7$
9	$\exists x L(x, x)$	$\exists I, 8$
10	$\exists x L(x, x)$	$\exists E, 1, 2-9$

Daeducer commands needed to generate the proof:

```

1 premise exists x forall y forall z ((L(x, y) -> L(z, y)) <-> L(y, z))
2 assumption forall y forall z ((L(a, y) -> L(z, y)) <-> L(y, z))
3 forall_elim 2, a
4 forall_elim 3, a
5 assumption L(a, a)
6 reiteration 5
7 discharge
8 imp_intro 5, 6
9 bi_elim 4, 8
10 exists_intro 9, L(x, x)
11 discharge
12 exists_elim 1, 2, 10
13 qed

```

Chapter 36, Exercise G.1.a

Proof presented using Fitch notation:

1	$\forall x (A(x) \rightarrow \neg B(x))$	PR
2	$(A(x) \rightarrow \neg B(x))$	$\forall E, 1$
3	$(A(x) \wedge B(x))$	AS
4	$A(x)$	$\wedge E, 3$
5	$B(x)$	$\wedge E, 3$
6	$\neg B(x)$	$\rightarrow E, 2, 4$
7	$\perp$	$\neg E, 6, 5$
8	$\neg(A(x) \wedge B(x))$	$\neg I, 3-7$
9	$\forall x \neg(A(x) \wedge B(x))$	$\forall I, 8$
10	$\neg \exists x (A(x) \wedge B(x))$	CQ, 9

Daeducer commands needed to generate the proof:


```

1 premise forall x (A(x) -> !B(x))
2 forall_elim 1, x
3 assumption (A(x) ^ B(x))
4 and_elim_left 3
5 and_elim_right 3
6 imp_elim 2, 4
7 not_elim 6, 5
8 discharge
9 not_intro 3, 7
10 forall_intro 9, x, x
11 conv_forall_not 10
12 qed

```

Chapter 36, Exercise G.1.b

Proof presented using Fitch notation:

1	$\neg \exists x(A(x) \wedge B(x))$	PR
2	$A(a)$	AS
3	$B(a)$	AS
4	$(A(a) \wedge B(a))$	$\wedge I$, 2, 3
5	$\exists x(A(x) \wedge B(x))$	$\exists I$, 4
6	$\perp$	$\neg E$, 1, 5
7	$\neg B(a)$	$\neg I$, 3-6
8	$(A(a) \rightarrow \neg B(a))$	$\rightarrow I$, 2, 7
9	$\forall x(A(x) \rightarrow \neg B(x))$	$\forall I$, 8

Daeducer commands needed to generate the proof:

```

1 premise !exists x (A(x) ^ B(x))
2 assumption A(a)
3 assumption B(a)
4 and_intro 2, 3
5 exists_intro 4, (A(x) ^ B(x))
6 not_elim 1, 5
7 discharge
8 not_intro 3, 6
9 discharge
10 imp_intro 2, 8
11 forall_intro 10, a, x
12 qed

```

Chapter 36, Exercise G.2.a

Proof presented using Fitch notation:

1	$\forall x(\neg A(x) \rightarrow B(d))$	PR
2	$\neg(\forall x A(x) \vee B(d))$	AS
3	$(\neg \forall x A(x) \wedge \neg B(d))$	DeM, 2
4	$\neg \forall x A(x)$	$\wedge E$, 3
5	$\neg B(d)$	$\wedge E$, 3
6	$\exists x \neg A(x)$	CQ, 4
7	$\neg A(y)$	AS
8	$(\neg A(y) \rightarrow B(d))$	$\forall E$, 1
9	$B(d)$	$\rightarrow E$, 8, 7
10	$B(d)$	$\exists E$, 6, 7–9
11	$\perp$	$\neg E$, 5, 10
12	$(\forall x A(x) \vee B(d))$	IP, 2–11

Daeducer commands needed to generate the proof:

```

1 premise forall x (!A(x) -> B(d))
2 assumption !(forall x A(x) v B(d))
3 demorgan_not_or 2
4 and_elim_left 3
5 and_elim_right 3
6 conv_not_forall 4
7 assumption !A(y)
8 forall_elim 1, y
9 imp_elim 8, 7
10 discharge
11 exists_elim 6, 7, 9
12 not_elim 5, 11
13 discharge
14 indirect 2, 12
15 qed

```

Chapter 36, Exercise G.2.b

Proof presented using Fitch notation:

1	$(\forall x A(x) \vee B(d))$	PR
2	$\neg A(y)$	AS
3	$\forall x A(x)$	AS
4	$A(y)$	$\forall E$, 3
5	$\perp$	$\neg E$, 2, 4
6	$B(d)$	X, 5
7	$B(d)$	AS
8	$B(d)$	R, 7
9	$B(d)$	$\vee E$, 1, 3–6, 7–8
10	$(\neg A(y) \rightarrow B(d))$	$\rightarrow I$, 2, 9
11	$\forall x (\neg A(x) \rightarrow B(d))$	$\forall I$, 10

Daeducer commands needed to generate the proof:

```

1  premise (forall x A(x) ∨ B(d))
2  assumption !A(y)
3  assumption forall x A(x)
4  forall_elim 3, y
5  not_elim 2, 4
6  explosion 5, B(d)
7  discharge
8  assumption B(d)
9  reiteration 8
10 discharge
11 or_elim 1, 3, 6, 8, 9
12 discharge
13 imp_intro 2, 11
14 forall_intro 13, y, x
15 qed

```

Chapter 36, Exercise G.3.a

Proof presented using Fitch notation:

1	$(\exists x P(x) \rightarrow Q(c))$	PR
2	$P(a)$	AS
3	$\exists x P(x)$	$\exists$ I, 2
4	$Q(c)$	$\rightarrow$ E, 1, 3
5	$(P(a) \rightarrow Q(c))$	$\rightarrow$ I, 2, 4
6	$\forall x (P(x) \rightarrow Q(c))$	$\forall$ I, 5

Daeducer commands needed to generate the proof:

```

1 premise (exists x P(x) -> Q(c))
2 assumption P(a)
3 exists_intro 2, P(x)
4 imp_elim 1, 3
5 discharge
6 imp_intro 2, 4
7 forall_intro 6, a, x
8 qed

```

Chapter 36, Exercise G.3.b

Proof presented using Fitch notation:

1	$\forall x (P(x) \rightarrow Q(c))$	PR
2	$\exists x P(x)$	AS
3	$P(a)$	AS
4	$(P(a) \rightarrow Q(c))$	$\forall$ E, 1
5	$Q(c)$	$\rightarrow$ E, 4, 3
6	$Q(c)$	$\exists$ E, 2, 3-5
7	$(\exists x P(x) \rightarrow Q(c))$	$\rightarrow$ I, 2, 6

Daeducer commands needed to generate the proof:

```

1 premise forall x (P(x) -> Q(c))
2 assumption exists x P(x)
3 assumption P(a)
4 forall_elim 1, a
5 imp_elim 4, 3
6 discharge
7 exists_elim 2, 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```

Chapter 36, Exercise H.2

Proof presented using Fitch notation:

1	$\forall x \forall y \forall z B(x, y, z)$	PR
2	$\forall y \forall z B(x, y, z)$	$\forall E, 1$
3	$\forall z B(x, x, z)$	$\forall E, 2$
4	$B(x, x, x)$	$\forall E, 3$
5	$\forall x B(x, x, x)$	$\forall I, 4$

Daeducer commands needed to generate the proof:

```
1 premise forall x forall y forall z B(x, y, z)
2 forall_elim 1, x
3 forall_elim 2, x
4 forall_elim 3, x
5 forall_intro 4, x, x
6 qed
```

Chapter 36, Exercise H.3.b

Proof presented using Fitch notation:

1	$\forall y \forall x D(x, y)$	PR
2	$\forall x D(x, b)$	$\forall E, 1$
3	$D(a, b)$	$\forall E, 2$
4	$\forall y D(a, y)$	$\forall I, 3$
5	$\forall x \forall y D(x, y)$	$\forall I, 4$

Daeducer commands needed to generate the proof:

```
1 premise forall y forall x D(x, y)
2 forall_elim 1, b
3 forall_elim 2, a
4 forall_intro 3, b, y
5 forall_intro 4, a, x
6 qed
```

Chapter 36, Exercise H.4

Proof presented using Fitch notation:

1	$\exists x \forall y D(x, y)$	PR
2	$\forall y D(a, y)$	AS
3	$D(a, z)$	$\forall E, 2$
4	$\exists x D(x, z)$	$\exists I, 3$
5	$\forall y \exists x D(x, y)$	$\forall I, 4$
6	$\forall y \exists x D(x, y)$	$\exists E, 1, 2-5$

Daeducer commands needed to generate the proof:

```

1 premise exists x forall y D(x, y)
2 assumption forall y D(a, y)
3 forall_elim 2, z
4 exists_intro 3, D(x, z)
5 forall_intro 4, z, y
6 discharge
7 exists_elim 1, 2, 5
8 qed

```

Chapter 36, Exercise H.5

Proof presented using Fitch notation:

1	$\forall x (R(c, a) \leftrightarrow R(x, a))$	PR
2	$R(c, a)$	AS
3	$(R(c, a) \leftrightarrow R(y, a))$	$\forall E, 1$
4	$R(y, a)$	$\leftrightarrow E, 3, 2$
5	$\forall x R(x, a)$	$\forall I, 4$
6	$\forall x R(x, a)$	AS
7	$R(c, a)$	$\forall E, 6$
8	$(R(c, a) \leftrightarrow \forall x R(x, a))$	$\leftrightarrow I, 2-5, 6-7$

Daeducer commands needed to generate the proof:

```

1 premise forall x (R(c, a) <-> R(x, a))
2 assumption R(c, a)
3 forall_elim 1, y
4 bi_elim 3, 2
5 forall_intro 4, y, x
6 discharge
7 assumption forall x R(x, a)
8 forall_elim 7, c
9 discharge
10 bi_intro 2, 5, 7, 8
11 qed

```

Chapter 36, Exercise I.1

Proof presented using Fitch notation:

1	$\exists y \forall x R(x, y)$	PR
2	$\forall x R(x, c)$	AS
3	$R(z, c)$	$\forall E, 2$
4	$\exists y R(z, y)$	$\exists I, 3$
5	$\forall x \exists y R(x, y)$	$\forall I, 4$
6	$\forall x \exists y R(x, y)$	$\exists E, 1, 2-5$

Daeducer commands needed to generate the proof:

```

1 premise exists y forall x R(x, y)
2 assumption forall x R(x, c)
3 forall_elim 2, z
4 exists_intro 3, R(z, y)
5 forall_intro 4, z, x
6 discharge
7 exists_elim 1, 2, 5
8 qed

```

Chapter 36, Exercise I.4

Proof presented using Fitch notation:

1	$\forall x (S(x) \rightarrow T(a))$	PR
2	$S(d)$	PR
3	$(S(d) \rightarrow T(a))$	$\forall E, 1$
4	$T(a)$	$\rightarrow E, 3, 2$

Daeducer commands needed to generate the proof:

```

1 premise forall x (S(x) -> T(a))
2 premise S(d)
3 forall_elim 1, d
4 imp_elim 3, 2
5 qed

```

Chapter 36, Exercise I.5

Proof presented using Fitch notation:

1	$\forall x(A(x) \rightarrow B(x))$	PR
2	$\forall x(B(x) \rightarrow C(x))$	PR
3	$(B(a) \rightarrow C(a))$	$\forall E, 2$
4	$A(a)$	AS
5	$(A(a) \rightarrow B(a))$	$\forall E, 1$
6	$B(a)$	$\rightarrow E, 5, 4$
7	$C(a)$	$\rightarrow E, 3, 6$
8	$(A(a) \rightarrow C(a))$	$\rightarrow I, 4, 7$
9	$\forall x(A(x) \rightarrow C(x))$	$\forall I, 8$

Daeducer commands needed to generate the proof:

```

1 premise forall x (A(x) -> B(x))
2 premise forall x (B(x) -> C(x))
3 forall_elim 2, a
4 assumption A(a)
5 forall_elim 1, a
6 imp_elim 5, 4
7 imp_elim 3, 6
8 discharge
9 imp_intro 4, 7
10 forall_intro 9, a, x
11 qed

```

Chapter 36, Exercise I.7

Proof presented using Fitch notation:

1	$\forall x \forall y (R(x, y) \vee R(y, x))$	PR
2	$\forall y (R(j, y) \vee R(y, j))$	$\forall E, 1$
3	$(R(j, j) \vee R(j, j))$	$\forall E, 2$
4	$R(j, j)$	AS
5	$R(j, j)$	R, 4
6	$R(j, j)$	AS
7	$R(j, j)$	R, 6
8	$R(j, j)$	$\vee E, 3, 4-5, 6-7$

Daeducer commands needed to generate the proof:

```

1 premise forall x forall y (R(x, y) v R(y, x))
2 forall_elim 1, j
3 forall_elim 2, j
4 assumption R(j, j)
5 reiteration 4
6 discharge
7 assumption R(j, j)
8 reiteration 7

```



```

9  discharge
10 or_elim 3, 4, 5, 7, 8
11 qed

```

Chapter 36, Exercise I.10

Proof presented using Fitch notation:

1	$(\exists x M(x) \rightarrow \exists x N(x))$	PR
2	$\neg \exists x N(x)$	PR
3	$\neg \exists x M(x)$	MT, 1, 2
4	$\forall x \neg M(x)$	CQ, 3

Daeducer commands needed to generate the proof:

```

1  premise (exists x M(x) -> exists x N(x))
2  premise !exists x N(x)
3  modus_tollens 1, 2
4  conv_not_exists 3
5  qed

```

Chapter 37, Example p.318

Proof presented using Fitch notation:

1	$\forall x (A(x) \rightarrow B)$	PR
2	$\exists x A(x)$	AS
3	$A(a)$	AS
4	$\forall x (A(x) \rightarrow B)$	R, 1
5	$(A(a) \rightarrow B)$	$\forall E$, 4
6	B	$\rightarrow E$, 5, 3
7	B	$\exists E$, 2, 3-6
8	$(\exists x A(x) \rightarrow B)$	$\rightarrow I$, 2, 7

Daeducer commands needed to generate the proof:

```

1  premise forall x (A(x) -> B)
2  assumption exists x A(x)
3  assumption A(a)
4  reiteration 1
5  forall_elim 4, a
6  imp_elim 5, 3
7  discharge
8  exists_elim 2, 3, 6
9  discharge
10 imp_intro 2, 8
11 qed

```

Chapter 37, Example p.320

Proof presented using Fitch notation:

1	$(\exists x A(x) \rightarrow B)$	PR
2	$A(a)$	AS
3	$\exists x A(x)$	$\exists$ I, 2
4	B	$\rightarrow$ E, 1, 3
5	$(A(a) \rightarrow B)$	$\rightarrow$ I, 2, 4
6	$\forall x(A(x) \rightarrow B)$	$\forall$ I, 5

Daeducer commands needed to generate the proof:

```
1 premise (exists x A(x) -> B)
2 assumption A(a)
3 exists_intros 2, A(x)
4 imp_elim 1, 3
5 discharge
6 imp_intro 2, 4
7 forall_intro 6, a, x
8 qed
```

Chapter 37, Exercise A.1

Proof presented using Fitch notation:

1	$(A \rightarrow \forall x B(x))$	PR
2	A	AS
3	$\forall x B(x)$	$\rightarrow$ E, 1, 2
4	$B(x)$	$\forall$ E, 3
5	$(A \rightarrow B(x))$	$\rightarrow$ I, 2, 4
6	$\forall x(A \rightarrow B(x))$	$\forall$ I, 5

Daeducer commands needed to generate the proof:

```
1 premise (A -> forall x B(x))
2 assumption A
3 imp_elim 1, 2
4 forall_elim 3, x
5 discharge
6 imp_intro 2, 4
7 forall_intro 6, x, x
8 qed
```

Chapter 37, Exercise A.2

Proof presented using Fitch notation:

1	$\exists x(A \rightarrow B(x))$	PR
2	$(A \rightarrow B(c))$	AS
3	A	AS
4	$B(c)$	$\rightarrow$ E, 2, 3
5	$\exists xB(x)$	$\exists$ I, 4
6	$(A \rightarrow \exists xB(x))$	$\rightarrow$ I, 3, 5
7	$(A \rightarrow \exists xB(x))$	$\exists$ E, 1, 2-6

Daeducer commands needed to generate the proof:

```

1 premise exists x (A -> B(x))
2 assumption (A -> B(c))
3 assumption A
4 imp_elim 2, 3
5 exists_intro 4, B(x)
6 discharge
7 imp_intro 3, 5
8 discharge
9 exists_elim 1, 2, 7
10 qed

```

Chapter 37, Exercise A.3

Proof presented using Fitch notation:

1	$\forall x(A(x) \wedge B(x))$	AS
2	$(A(a) \wedge B(a))$	$\forall E$, 1
3	$A(a)$	$\wedge E$, 2
4	$\forall xA(x)$	$\forall I$, 3
5	$B(a)$	$\wedge E$, 2
6	$\forall xB(x)$	$\forall I$, 5
7	$(\forall xA(x) \wedge \forall xB(x))$	$\wedge I$, 4, 6
8	$(\forall x(A(x) \wedge B(x)) \rightarrow (\forall xA(x) \wedge \forall xB(x)))$	$\rightarrow I$, 1, 7
9	$(\forall xA(x) \wedge \forall xB(x))$	AS
10	$\forall xA(x)$	$\wedge E$, 9
11	$\forall xB(x)$	$\wedge E$, 9
12	$A(a)$	$\forall E$, 10
13	$B(a)$	$\forall E$, 11
14	$(A(a) \wedge B(a))$	$\wedge I$, 12, 13
15	$\forall x(A(x) \wedge B(x))$	$\forall I$, 14
16	$((\forall xA(x) \wedge \forall xB(x)) \rightarrow \forall x(A(x) \wedge B(x)))$	$\rightarrow I$, 9, 15
17	$(\forall x(A(x) \wedge B(x)) \leftrightarrow (\forall xA(x) \wedge \forall xB(x)))$	$\leftrightarrow I$, 1–7, 9–15

Daeducer commands needed to generate the proof:

```

1  assumption forall x (A(x) ^ B(x))
2  forall_elim 1, a
3  and_elim_left 2
4  forall_intro 3, a, x
5  and_elim_right 2
6  forall_intro 5, a, x
7  and_intro 4, 6
8  discharge
9  imp_intro 1, 7
10 assumption (forall x A(x) ^ forall x B(x))
11 and_elim_left 10
12 and_elim_right 10
13 forall_elim 11, a
14 forall_elim 12, a
15 and_intro 13, 14
16 forall_intro 15, a, x
17 discharge
18 imp_intro 10, 16
19 bi_intro 1, 7, 10, 16
20 qed

```

Chapter 37, Exercise A.4

Proof presented using Fitch notation:

1		$\exists x(A(x) \vee B(x))$	AS
2		$(A(c) \vee B(c))$	AS
3		$A(c)$	AS
4		$\exists xA(x)$	$\exists$ I, 3
5		$(\exists xA(x) \vee \exists xB(x))$	$\vee$ E, 4
6		$B(c)$	AS
7		$\exists xB(x)$	$\exists$ I, 6
8		$(\exists xA(x) \vee \exists xB(x))$	$\vee$ E, 7
9		$(\exists xA(x) \vee \exists xB(x))$	$\vee$ E, 2, 3–5, 6–8
10		$(\exists xA(x) \vee \exists xB(x))$	$\exists$ E, 1, 2–9
11		$(\exists x(A(x) \vee B(x)) \rightarrow (\exists xA(x) \vee \exists xB(x)))$	$\rightarrow$ I, 1, 10
12		$(\exists xA(x) \vee \exists xB(x))$	AS
13		$\exists xA(x)$	AS
14		$A(d)$	AS
15		$(A(d) \vee B(d))$	$\vee$ E, 14
16		$\exists x(A(x) \vee B(x))$	$\exists$ I, 15
17		$\exists x(A(x) \vee B(x))$	$\exists$ E, 13, 14–16
18		$\exists xB(x)$	AS
19		$B(d)$	AS
20		$(A(d) \vee B(d))$	$\vee$ E, 19
21		$\exists x(A(x) \vee B(x))$	$\exists$ I, 20
22		$\exists x(A(x) \vee B(x))$	$\exists$ E, 18, 19–21
23		$\exists x(A(x) \vee B(x))$	$\vee$ E, 12, 13–17, 18–22
24		$((\exists xA(x) \vee \exists xB(x)) \rightarrow \exists x(A(x) \vee B(x)))$	$\rightarrow$ I, 12, 23
25		$(\exists x(A(x) \vee B(x)) \leftrightarrow (\exists xA(x) \vee \exists xB(x)))$	$\leftrightarrow$ I, 1–10, 12–23

Daeducer commands needed to generate the proof:

```

1  assumption exists x (A(x) ∨ B(x))
2  assumption (A(c) ∨ B(c))
3  assumption A(c)
4  exists_intro 3, A(x)
5  or_intro_left 4, exists x B(x)
6  discharge
7  assumption B(c)
8  exists_intro 7, B(x)
9  or_intro_right 8, exists x A(x)
10 discharge
11 or_elim 2, 3, 5, 7, 9
12 discharge
13 exists_elim 1, 2, 11
14 discharge

```

```

15 imp_intro 1, 13
16 assumption (exists x A(x) ∨ exists x B(x))
17 assumption exists x A(x)
18 assumption A(d)
19 or_intro_left 18, B(d)
20 exists_intro 19, (A(x) ∨ B(x))
21 discharge
22 exists_elim 17, 18, 20
23 discharge
24 assumption exists x B(x)
25 assumption B(d)
26 or_intro_right 25, A(d)
27 exists_intro 26, (A(x) ∨ B(x))
28 discharge
29 exists_elim 24, 25, 27
30 discharge
31 or_elim 16, 17, 22, 24, 29
32 discharge
33 imp_intro 16, 31
34 bi_intro 1, 13, 16, 31
35 qed

```

Chapter 37, Exercise A.5

Proof presented using Fitch notation:

1	$(A \vee \forall x B(x))$	PR
2	A	AS
3	$(A \vee B(y))$	$\vee E, 2$
4	$\forall x(A \vee B(x))$	$\forall I, 3$
5	$\forall x B(x)$	AS
6	$B(y)$	$\forall E, 5$
7	$(A \vee B(y))$	$\vee E, 6$
8	$\forall x(A \vee B(x))$	$\forall I, 7$
9	$\forall x(A \vee B(x))$	$\vee E, 1, 2-4, 5-8$

Daeducer commands needed to generate the proof:

```

1 premise (A ∨ forall x B(x))
2 assumption A
3 or_intro_left 2, B(y)
4 forall_intro 3, y, x
5 discharge
6 assumption forall x B(x)
7 forall_elim 6, y
8 or_intro_right 7, A
9 forall_intro 8, y, x
10 discharge
11 or_elim 1, 2, 4, 6, 9
12 qed

```

Chapter 37, Exercise A.6

Proof presented using Fitch notation:

1	$\forall x(A(x) \rightarrow B)$	PR
2	$\exists x A(x)$	AS
3	$A(c)$	AS
4	$(A(c) \rightarrow B)$	$\forall E, 1$
5	B	$\rightarrow E, 4, 3$
6	B	$\exists E, 2, 3-5$
7	$(\exists x A(x) \rightarrow B)$	$\rightarrow I, 2, 6$

Daeducer commands needed to generate the proof:

```

1 premise forall x (A(x) -> B)
2 assumption exists x A(x)
3 assumption A(c)
4 forall_elim 1, c
5 imp_elim 4, 3
6 discharge
7 exists_elim 2, 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```

Chapter 37, Exercise A.7

Proof presented using Fitch notation:

1	$\exists x(A(x) \rightarrow B)$	PR
2	$\forall x A(x)$	AS
3	$(A(c) \rightarrow B)$	AS
4	$A(c)$	$\forall E, 2$
5	B	$\rightarrow E, 3, 4$
6	B	$\exists E, 1, 3-5$
7	$(\forall x A(x) \rightarrow B)$	$\rightarrow I, 2, 6$

Daeducer commands needed to generate the proof:

```

1 premise exists x (A(x) -> B)
2 assumption forall x A(x)
3 assumption (A(c) -> B)
4 forall_elim 2, c
5 imp_elim 3, 4
6 discharge
7 exists_elim 1, 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```

Chapter 37, Exercise A.8

Proof presented using Fitch notation:

1		$A(x)$	AS
2		$\exists y A(y)$	$\exists I, 1$
3		$(A(x) \rightarrow \exists y A(y))$	$\rightarrow I, 1, 2$
4		$\forall x (A(x) \rightarrow \exists y A(y))$	$\forall I, 3$

Daeducer commands needed to generate the proof:

```

1  assumption A(x)
2  exists_intro 1, A(y)
3  discharge
4  imp_intro 1, 2
5  forall_intro 4, x, x
6  qed

```

Chapter 37, Exercise B.1

Proof presented using Fitch notation:

1		$\forall x R(x, x)$	PR
2		$R(a, a)$	$\forall E, 1$
3		$\exists y R(a, y)$	$\exists I, 2$
4		$\forall x \exists y R(x, y)$	$\forall I, 3$

Daeducer commands needed to generate the proof:

```

1  premise forall x R(x, x)
2  forall_elim 1, a
3  exists_intro 2, R(a, y)
4  forall_intro 3, a, x
5  qed

```


Chapter 37, Exercise B.2

Proof presented using Fitch notation:

1	$\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$	PR
2	$\forall y \forall z ((R(a, y) \wedge R(y, z)) \rightarrow R(a, z))$	$\forall E, 1$
3	$\forall z ((R(a, b) \wedge R(b, z)) \rightarrow R(a, z))$	$\forall E, 2$
4	$((R(a, b) \wedge R(b, c)) \rightarrow R(a, c))$	$\forall E, 3$
5	$R(a, b)$	AS
6	$R(b, c)$	AS
7	$(R(a, b) \wedge R(b, c))$	$\wedge I, 5, 6$
8	$R(a, c)$	$\rightarrow E, 4, 7$
9	$(R(b, c) \rightarrow R(a, c))$	$\rightarrow I, 6, 8$
10	$\forall z (R(b, z) \rightarrow R(a, z))$	$\forall I, 9$
11	$(R(a, b) \rightarrow \forall z (R(b, z) \rightarrow R(a, z)))$	$\rightarrow I, 5, 10$
12	$\forall y (R(a, y) \rightarrow \forall z (R(y, z) \rightarrow R(a, z)))$	$\forall I, 11$
13	$\forall x \forall y (R(x, y) \rightarrow \forall z (R(y, z) \rightarrow R(x, z)))$	$\forall I, 12$

Daeducer commands needed to generate the proof:

```

1 premise forall x forall y forall z ((R(x, y) ^ R(y, z)) -> R(x, z))
2 forall_elim 1, a
3 forall_elim 2, b
4 forall_elim 3, c
5 assumption R(a, b)
6 assumption R(b, c)
7 and_intro 5, 6
8 imp_elim 4, 7
9 discharge
10 imp_intro 6, 8
11 forall_intro 10, c, z
12 discharge
13 imp_intro 5, 11
14 forall_intro 13, b, y
15 forall_intro 14, a, x
16 qed

```

Chapter 37, Exercise B.3

Proof presented using Fitch notation:

1	$\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$	PR
2	$\forall x \forall y (R(x, y) \rightarrow R(y, x))$	PR
3	$\forall y \forall z ((R(b, y) \wedge R(y, z)) \rightarrow R(b, z))$	$\forall E, 1$
4	$\forall z ((R(b, a) \wedge R(a, z)) \rightarrow R(b, z))$	$\forall E, 3$
5	$((R(b, a) \wedge R(a, c)) \rightarrow R(b, c))$	$\forall E, 4$
6	$\forall y (R(a, y) \rightarrow R(y, a))$	$\forall E, 2$
7	$(R(a, b) \rightarrow R(b, a))$	$\forall E, 6$
8	$(R(a, b) \wedge R(a, c))$	AS
9	$R(a, b)$	$\wedge E, 8$
10	$R(a, c)$	$\wedge E, 8$
11	$R(b, a)$	$\rightarrow E, 7, 9$
12	$(R(b, a) \wedge R(a, c))$	$\wedge I, 11, 10$
13	$R(b, c)$	$\rightarrow E, 5, 12$
14	$((R(a, b) \wedge R(a, c)) \rightarrow R(b, c))$	$\rightarrow I, 8, 13$
15	$\forall z ((R(a, b) \wedge R(a, z)) \rightarrow R(b, z))$	$\forall I, 14$
16	$\forall y \forall z ((R(a, y) \wedge R(a, z)) \rightarrow R(y, z))$	$\forall I, 15$
17	$\forall x \forall y \forall z ((R(x, y) \wedge R(y, z)) \rightarrow R(x, z))$	$\forall I, 16$

Daeducer commands needed to generate the proof:

```

1 premise forall x forall y forall z ((R(x, y) ^ R(y, z)) -> R(x, z))
2 premise forall x forall y (R(x, y) -> R(y, x))
3 forall_elim 1, b
4 forall_elim 3, a
5 forall_elim 4, c
6 forall_elim 2, a
7 forall_elim 6, b
8 assumption (R(a, b) ^ R(a, c))
9 and_elim_left 8
10 and_elim_right 8
11 imp_elim 7, 9
12 and_intro 11, 10
13 imp_elim 5, 12
14 discharge
15 imp_intro 8, 13
16 forall_intro 15, c, z
17 forall_intro 16, b, y
18 forall_intro 17, a, x
19 qed

```

Chapter 37, Exercise B.4

Proof presented using Fitch notation:

1	$\forall x \forall y (R(x, y) \rightarrow R(y, x))$	PR
2	$(R(a, b) \wedge R(a, c))$	AS
3	$R(a, b)$	$\wedge E, 2$
4	$R(a, c)$	$\wedge E, 2$
5	$\forall y (R(a, y) \rightarrow R(y, a))$	$\forall E, 1$
6	$(R(a, b) \rightarrow R(b, a))$	$\forall E, 5$
7	$R(b, a)$	$\rightarrow E, 6, 3$
8	$\forall y (R(a, y) \rightarrow R(y, a))$	$\forall E, 1$
9	$(R(a, c) \rightarrow R(c, a))$	$\forall E, 8$
10	$R(c, a)$	$\rightarrow E, 9, 4$
11	$(R(b, a) \wedge R(c, a))$	$\wedge I, 7, 10$
12	$\exists u (R(b, u) \wedge R(c, u))$	$\exists I, 11$
13	$((R(a, b) \wedge R(a, c)) \rightarrow \exists u (R(b, u) \wedge R(c, u)))$	$\rightarrow I, 2, 12$
14	$\forall z ((R(a, b) \wedge R(a, z)) \rightarrow \exists u (R(b, u) \wedge R(z, u)))$	$\forall I, 13$
15	$\forall y \forall z ((R(a, y) \wedge R(a, z)) \rightarrow \exists u (R(y, u) \wedge R(z, u)))$	$\forall I, 14$
16	$\forall x \forall y \forall z ((R(x, y) \wedge R(x, z)) \rightarrow \exists u (R(y, u) \wedge R(z, u)))$	$\forall I, 15$

Daeducer commands needed to generate the proof:

```

1  premise forall x forall y (R(x, y) -> R(y, x))
2  assumption (R(a, b) ^ R(a, c))
3  and_elim_left 2
4  and_elim_right 2
5  forall_elim 1, a
6  forall_elim 5, b
7  imp_elim 6, 3
8  forall_elim 1, a
9  forall_elim 8, c
10 imp_elim 9, 4
11 and_intro 7, 10
12 exists_intro 11, (R(b, u) ^ R(c, u))
13 discharge
14 imp_intro 2, 12
15 forall_intro 14, c, z
16 forall_intro 15, b, y
17 forall_intro 16, a, x
18 qed

```

Chapter 37, Exercise B.5

Proof presented using Fitch notation:

1			$\exists x \forall y (A(x, y) \leftrightarrow \neg A(y, y))$	AS	
2			$\forall y (A(a, y) \leftrightarrow \neg A(y, y))$	AS	
3			$(A(a, a) \leftrightarrow \neg A(a, a))$	$\forall E, 2$	
4				$A(a, a)$	
5				$\neg A(a, a)$	$\leftrightarrow E, 3, 4$
6				$\perp$	$\neg E, 5, 4$
7			$\neg A(a, a)$	$\neg I, 4-6$	
8			$A(a, a)$	$\leftrightarrow E, 3, 7$	
9			$\perp$	$\neg E, 7, 8$	
10			$\perp$	$\exists E, 1, 2-9$	
11			$\neg \exists x \forall y (A(x, y) \leftrightarrow \neg A(y, y))$	$\neg I, 1-10$	

Daeducer commands needed to generate the proof:

```

1  assumption exists x forall y (A(x, y) <-> !A(y, y))
2  assumption forall y (A(a, y) <-> !A(y, y))
3  forall_elim 2, a
4  assumption A(a, a)
5  bi_elim 3, 4
6  not_elim 5, 4
7  discharge
8  not_intro 4, 6
9  bi_elim 3, 8
10 not_elim 8, 9
11 discharge
12 exists_elim 1, 2, 10
13 discharge
14 not_intro 1, 12
15 qed

```

Chapter 37, Exercise C.1

Proof presented using Fitch notation:

1	$(\forall x A(x) \rightarrow B)$	PR
2	B	AS
3	$A(y)$	AS
4	B	R, 2
5	$(A(y) \rightarrow B)$	$\rightarrow$ I, 3, 4
6	$\exists x(A(x) \rightarrow B)$	$\exists$ I, 5
7	$(B \rightarrow \exists x(A(x) \rightarrow B))$	$\rightarrow$ I, 2, 6
8	$\neg B$	AS
9	$\neg \forall x A(x)$	MT, 1, 8
10	$\exists x \neg A(x)$	CQ, 9
11	$\neg A(y)$	AS
12	$A(y)$	AS
13	$\perp$	$\neg$ E, 11, 12
14	B	X, 13
15	$(A(y) \rightarrow B)$	$\rightarrow$ I, 12, 14
16	$\exists x(A(x) \rightarrow B)$	$\exists$ I, 15
17	$\exists x(A(x) \rightarrow B)$	$\exists$ E, 10, 11–16
18	$(\neg B \rightarrow \exists x(A(x) \rightarrow B))$	$\rightarrow$ I, 8, 17
19	$\exists x(A(x) \rightarrow B)$	LEM, 7, 18

Daeducer commands needed to generate the proof:

```

1 premise (forall x A(x) -> B)
2 assumption B
3 assumption A(y)
4 reiteration 2
5 discharge
6 imp_intro 3, 4
7 exists_intro 6, (A(x) -> B)
8 discharge
9 imp_intro 2, 7
10 assumption !B
11 modus_tollens 1, 10
12 conv_not_forall 11
13 assumption !A(y)
14 assumption A(y)
15 not_elim 13, 14
16 explosion 15, B
17 discharge
18 imp_intro 14, 16
19 exists_intro 18, (A(x) -> B)
20 discharge
21 exists_elim 12, 13, 19
22 discharge
23 imp_intro 10, 21
24 excluded_middle 9, 23
25 qed

```

Chapter 37, Exercise C.2

Proof presented using Fitch notation:

1	$(A \rightarrow \exists x B(x))$	PR
2	A	AS
3	$\exists x B(x)$	$\rightarrow$ E, 1, 2
4	$B(c)$	AS
5	A	AS
6	$B(c)$	R, 4
7	$(A \rightarrow B(c))$	$\rightarrow$ I, 5, 6
8	$\exists x(A \rightarrow B(x))$	$\exists$ I, 7
9	$\exists x(A \rightarrow B(x))$	$\exists$ E, 3, 4–8
10	$(A \rightarrow \exists x(A \rightarrow B(x)))$	$\rightarrow$ I, 2, 9
11	$\neg A$	AS
12	A	AS
13	$\perp$	$\neg$ E, 11, 12
14	$B(c)$	X, 13
15	$(A \rightarrow B(c))$	$\rightarrow$ I, 12, 14
16	$\exists x(A \rightarrow B(x))$	$\exists$ I, 15
17	$(\neg A \rightarrow \exists x(A \rightarrow B(x)))$	$\rightarrow$ I, 11, 16
18	$\exists x(A \rightarrow B(x))$	LEM, 10, 17

Daeducer commands needed to generate the proof:

```

1 premise (A -> exists x B(x))
2 assumption A
3 imp_elim 1, 2
4 assumption B(c)
5 assumption A
6 reiteration 4
7 discharge
8 imp_intro 5, 6
9 exists_intro 8, (A -> B(x))
10 discharge
11 exists_elim 3, 4, 9
12 discharge
13 imp_intro 2, 11
14 assumption !A
15 assumption A
16 not_elim 14, 15
17 explosion 16, B(c)
18 discharge
19 imp_intro 15, 17
20 exists_intro 19, (A -> B(x))
21 discharge
22 imp_intro 14, 20
23 excluded_middle 13, 22
24 qed

```

Chapter 37, Exercise C.3

Proof presented using Fitch notation:

1	$\forall x(A \vee B(x))$	PR
2	A	AS
3	$(A \vee \forall xB(x))$	$\vee E$, 2
4	$(A \rightarrow (A \vee \forall xB(x)))$	$\rightarrow I$, 2, 3
5	$\neg A$	AS
6	$(A \vee B(c))$	$\vee E$, 1
7	$B(c)$	DS, 6, 5
8	$\forall xB(x)$	$\forall I$, 7
9	$(A \vee \forall xB(x))$	$\vee E$, 8
10	$(\neg A \rightarrow (A \vee \forall xB(x)))$	$\rightarrow I$, 5, 9
11	$(A \vee \forall xB(x))$	LEM, 4, 10

Daeducer commands needed to generate the proof:

```

1  premise forall x (A v B(x))
2  assumption A
3  or_intro_left 2, forall x B(x)
4  discharge
5  imp_intro 2, 3
6  assumption !A
7  forall_elim 1, c
8  or_syllogism_left 7, 6
9  forall_intro 8, c, x
10 or_intro_right 9, A
11 discharge
12 imp_intro 6, 10
13 excluded_middle 5, 12
14 qed

```

Chapter 37, Exercise C.4

Proof presented using Fitch notation:

1	$\forall y A(y)$	AS
2	$A(a)$	AS
3	$\forall y A(y)$	R, 1
4	$(A(a) \rightarrow \forall y A(y))$	$\rightarrow$ I, 2, 3
5	$\exists x (A(x) \rightarrow \forall y A(y))$	$\exists$ I, 4
6	$(\forall y A(y) \rightarrow \exists x (A(x) \rightarrow \forall y A(y)))$	$\rightarrow$ I, 1, 5
7	$\neg \forall y A(y)$	AS
8	$\exists y \neg A(y)$	CQ, 7
9	$\neg A(b)$	AS
10	$A(b)$	AS
11	$\perp$	$\neg$ E, 9, 10
12	$\forall y A(y)$	X, 11
13	$(A(b) \rightarrow \forall y A(y))$	$\rightarrow$ I, 10, 12
14	$\exists x (A(x) \rightarrow \forall y A(y))$	$\exists$ I, 13
15	$\exists x (A(x) \rightarrow \forall y A(y))$	$\exists$ E, 8, 9–14
16	$(\neg \forall y A(y) \rightarrow \exists x (A(x) \rightarrow \forall y A(y)))$	$\rightarrow$ I, 7, 15
17	$\exists x (A(x) \rightarrow \forall y A(y))$	LEM, 6, 16

Daeducer commands needed to generate the proof:

```

1  assumption forall y A(y)
2  assumption A(a)
3  reiteration 1
4  discharge
5  imp_intro 2, 3
6  exists_intro 5, (A(x) -> forall y A(y))
7  discharge
8  imp_intro 1, 6
9  assumption !forall y A(y)
10 conv_not_forall 9
11 assumption !A(b)
12 assumption A(b)
13 not_elim 11, 12
14 explosion 13, forall y A(y)
15 discharge
16 imp_intro 12, 14
17 exists_intro 16, (A(x) -> forall y A(y))
18 discharge
19 exists_elim 10, 11, 17
20 discharge
21 imp_intro 9, 19
22 excluded_middle 8, 21
23 qed

```


Chapter 37, Exercise C.5

Proof presented using Fitch notation:

1		$\exists y A(y)$	AS
2		$A(a)$	AS
3		$\exists y A(y)$	AS
4		$A(a)$	R, 2
5		$(\exists y A(y) \rightarrow A(a))$	$\rightarrow$ I, 3, 4
6		$\exists x (\exists y A(y) \rightarrow A(x))$	$\exists$ I, 5
7		$\exists x (\exists y A(y) \rightarrow A(x))$	$\exists$ E, 1, 2-6
8		$(\exists y A(y) \rightarrow \exists x (\exists y A(y) \rightarrow A(x)))$	$\rightarrow$ I, 1, 7
9		$\neg \exists y A(y)$	AS
10		$\exists y A(y)$	AS
11		$\perp$	$\neg$ E, 9, 10
12		$A(b)$	X, 11
13		$(\exists y A(y) \rightarrow A(b))$	$\rightarrow$ I, 10, 12
14		$\exists x (\exists y A(y) \rightarrow A(x))$	$\exists$ I, 13
15		$(\neg \exists y A(y) \rightarrow \exists x (\exists y A(y) \rightarrow A(x)))$	$\rightarrow$ I, 9, 14
16		$\exists x (\exists y A(y) \rightarrow A(x))$	LEM, 8, 15

Daeducer commands needed to generate the proof:

```

1  assumption exists y A(y)
2  assumption A(a)
3  assumption exists y A(y)
4  reiteration 2
5  discharge
6  imp_intro 3, 4
7  exists_intro 6, (exists y A(y) -> A(x))
8  discharge
9  exists_elim 1, 2, 7
10 discharge
11 imp_intro 1, 9
12 assumption !exists y A(y)
13 assumption exists y A(y)
14 not_elim 12, 13
15 explosion 14, A(b)
16 discharge
17 imp_intro 13, 15
18 exists_intro 17, (exists y A(y) -> A(x))
19 discharge
20 imp_intro 12, 18
21 excluded_middle 11, 20
22 qed

```

Chapter 38, Exercise B.1.a

Proof presented using Fitch notation:

1	$\forall x(A(x) \rightarrow \neg B(x))$	PR
2	$\exists x(A(x) \wedge B(x))$	AS
3	$(A(c) \wedge B(c))$	AS
4	$A(c)$	$\wedge E$, 3
5	$B(c)$	$\wedge E$, 3
6	$(A(c) \rightarrow \neg B(c))$	$\forall E$, 1
7	$\neg B(c)$	$\rightarrow E$, 6, 4
8	$\perp$	$\neg E$, 7, 5
9	$\perp$	$\exists E$, 2, 3–8
10	$\neg \exists x(A(x) \wedge B(x))$	$\neg I$, 2–9

Daeducer commands needed to generate the proof:

```

1 premise forall x (A(x) -> !B(x))
2 assumption exists x (A(x) ^ B(x))
3 assumption (A(c) ^ B(c))
4 and_elim_left 3
5 and_elim_right 3
6 forall_elim 1, c
7 imp_elim 6, 4
8 not_elim 7, 5
9 discharge
10 exists_elim 2, 3, 8
11 discharge
12 not_intro 2, 10
13 qed

```

Chapter 38, Exercise B.1.b

Proof presented using Fitch notation:

1	$\neg \exists x(A(x) \wedge B(x))$	PR
2	$\forall x \neg(A(x) \wedge B(x))$	CQ, 1
3	$A(y)$	AS
4	$B(y)$	AS
5	$(A(y) \wedge B(y))$	$\wedge I$, 3, 4
6	$\neg(A(y) \wedge B(y))$	$\forall E$, 2
7	$\perp$	$\neg E$, 6, 5
8	$\neg B(y)$	$\neg I$, 4–7
9	$(A(y) \rightarrow \neg B(y))$	$\rightarrow I$, 3, 8
10	$\forall x(A(x) \rightarrow \neg B(x))$	$\forall I$, 9

Daeducer commands needed to generate the proof:

```

1 premise !exists x (A(x) ~ B(x))
2 conv_not_exists 1
3 assumption A(y)
4 assumption B(y)
5 and_intro 3, 4
6 forall_elim 2, y
7 not_elim 6, 5
8 discharge
9 not_intro 4, 7
10 discharge
11 imp_intro 3, 9
12 forall_intro 11, y, x
13 qed

```

Chapter 38, Exercise B.2.a

Proof presented using Fitch notation:

1	$\forall x(\neg A(x) \rightarrow B(d))$	PR
2	$B(d)$	AS
3	$(\forall x A(x) \vee B(d))$	$\vee E$, 2
4	$(B(d) \rightarrow (\forall x A(x) \vee B(d)))$	$\rightarrow I$, 2, 3
5	$\neg B(d)$	AS
6	$(\neg A(a) \rightarrow B(d))$	$\vee E$, 1
7	$\neg \neg A(a)$	MT, 6, 5
8	$A(a)$	DNE, 7
9	$\forall x A(x)$	$\forall I$, 8
10	$(\forall x A(x) \vee B(d))$	$\vee E$, 9
11	$(\neg B(d) \rightarrow (\forall x A(x) \vee B(d)))$	$\rightarrow I$, 5, 10
12	$(\forall x A(x) \vee B(d))$	LEM, 4, 11

Daeducer commands needed to generate the proof:

```

1 premise forall x (!A(x) -> B(d))
2 assumption B(d)
3 or_intro_right 2, forall x A(x)
4 discharge
5 imp_intro 2, 3
6 assumption !B(d)
7 forall_elim 1, a
8 modus_tollens 7, 6
9 double_not_elim 8
10 forall_intro 9, a, x
11 or_intro_left 10, B(d)
12 discharge
13 imp_intro 6, 11
14 excluded_middle 5, 13
15 qed

```

Chapter 38, Exercise B.2.b

Proof presented using Fitch notation:

1	$(\forall x A(x) \vee B(d))$	PR
2	$\neg A(a)$	AS
3	$\forall x A(x)$	AS
4	$A(a)$	$\forall E$, 3
5	$\perp$	$\neg E$, 2, 4
6	$B(d)$	X, 5
7	$B(d)$	AS
8	$B(d)$	R, 7
9	$B(d)$	$\vee E$, 1, 3–6, 7–8
10	$(\neg A(a) \rightarrow B(d))$	$\rightarrow I$, 2, 9
11	$\forall x (\neg A(x) \rightarrow B(d))$	$\forall I$, 10

Daeducer commands needed to generate the proof:

```

1  premise (forall x A(x) ∨ B(d))
2  assumption !A(a)
3  assumption forall x A(x)
4  forall_elim 3, a
5  not_elim 2, 4
6  explosion 5, B(d)
7  discharge
8  assumption B(d)
9  reiteration 8
10 discharge
11 or_elim 1, 3, 6, 8, 9
12 discharge
13 imp_intro 2, 11
14 forall_intro 13, a, x
15 qed

```

Chapter 38, Exercise C.1.a

Proof presented using Fitch notation:

1	$\forall x(F(x) \wedge G(a))$	PR
2	$(F(y) \wedge G(a))$	$\forall E, 1$
3	$F(y)$	$\wedge E, 2$
4	$\forall xF(x)$	$\forall I, 3$
5	$G(a)$	$\wedge E, 2$
6	$(\forall xF(x) \wedge G(a))$	$\wedge I, 4, 5$

Daeducer commands needed to generate the proof:

```
1 premise forall x (F(x) ^ G(a))
2 forall_elim 1, y
3 and_elim_left 2
4 forall_intro 3, y, x
5 and_elim_right 2
6 and_intro 4, 5
7 qed
```

Chapter 38, Exercise C.1.b

Proof presented using Fitch notation:

1	$(\forall xF(x) \wedge G(a))$	PR
2	$\forall xF(x)$	$\wedge E, 1$
3	$G(a)$	$\wedge E, 1$
4	$F(x)$	$\forall E, 2$
5	$(F(x) \wedge G(a))$	$\wedge I, 4, 3$
6	$\forall x(F(x) \wedge G(a))$	$\forall I, 5$

Daeducer commands needed to generate the proof:

```
1 premise (forall x F(x) ^ G(a))
2 and_elim_left 1
3 and_elim_right 1
4 forall_elim 2, x
5 and_intro 4, 3
6 forall_intro 5, x, x
7 qed
```

Chapter 38, Exercise C.2.a

Proof presented using Fitch notation:

1	$\exists x(F(x) \vee G(a))$	PR
2	$(F(c) \vee G(a))$	AS
3	$F(c)$	AS
4	$\exists xF(x)$	$\exists I$, 3
5	$(\exists xF(x) \vee G(a))$	$\vee E$, 4
6	$G(a)$	AS
7	$(\exists xF(x) \vee G(a))$	$\vee E$, 6
8	$(\exists xF(x) \vee G(a))$	$\vee E$, 2, 3–5, 6–7
9	$(\exists xF(x) \vee G(a))$	$\exists E$, 1, 2–8

Daeducer commands needed to generate the proof:

```

1 premise exists x (F(x) ∨ G(a))
2 assumption (F(c) ∨ G(a))
3 assumption F(c)
4 exists_intro 3, F(x)
5 or_intro_left 4, G(a)
6 discharge
7 assumption G(a)
8 or_intro_right 7, exists x F(x)
9 discharge
10 or_elim 2, 3, 5, 7, 8
11 discharge
12 exists_elim 1, 2, 10
13 qed

```

Chapter 38, Exercise C.2.b

Proof presented using Fitch notation:

1	$(\exists xF(x) \vee G(a))$	PR
2	$\exists xF(x)$	AS
3	$F(y)$	AS
4	$(F(y) \vee G(a))$	$\vee E$, 3
5	$\exists x(F(x) \vee G(a))$	$\exists I$, 4
6	$\exists x(F(x) \vee G(a))$	$\exists E$, 2, 3–5
7	$G(a)$	AS
8	$(F(a) \vee G(a))$	$\vee E$, 7
9	$\exists x(F(x) \vee G(a))$	$\exists I$, 8
10	$\exists x(F(x) \vee G(a))$	$\vee E$, 1, 2–6, 7–9

Daeducer commands needed to generate the proof:

```

1 premise (exists x F(x) ∨ G(a))

```

```

2  assumption exists x F(x)
3  assumption F(y)
4  or_intro_left 3, G(a)
5  exists_intro 4, (F(x) ∨ G(a))
6  discharge
7  exists_elim 2, 3, 5
8  discharge
9  assumption G(a)
10 or_intro_right 9, F(a)
11 exists_intro 10, (F(x) ∨ G(a))
12 discharge
13 or_elim 1, 2, 7, 9, 11
14 qed

```

Chapter 38, Exercise C.3.a

Proof presented using Fitch notation:

1	$\forall x(G(a) \rightarrow F(x))$	PR
2	$G(a)$	AS
3	$(G(a) \rightarrow F(x))$	$\forall E, 1$
4	$F(x)$	$\rightarrow E, 3, 2$
5	$\forall x F(x)$	$\forall I, 4$
6	$(G(a) \rightarrow \forall x F(x))$	$\rightarrow I, 2, 5$

Daeducer commands needed to generate the proof:

```

1  premise forall x (G(a) -> F(x))
2  assumption G(a)
3  forall_elim 1, x
4  imp_elim 3, 2
5  forall_intro 4, x, x
6  discharge
7  imp_intro 2, 5
8  qed

```

Chapter 38, Exercise C.3.b

Proof presented using Fitch notation:

1		$(G(a) \rightarrow \forall x F(x))$	PR
2			
3			
4			
5		$(G(a) \rightarrow F(y))$	$\rightarrow I, 2, 4$
6		$\forall x(G(a) \rightarrow F(x))$	$\forall I, 5$

Daeducer commands needed to generate the proof:

```

1 premise (G(a) -> forall x F(x))
2 assumption G(a)
3 imp_elim 1, 2
4 forall_elim 3, y
5 discharge
6 imp_intro 2, 4
7 forall_intro 6, y, x
8 qed

```

Chapter 38, Exercise C.4.a

Proof presented using Fitch notation:

1		$\forall x(F(x) \rightarrow G(a))$	PR
2			
3			
4			
5			
6		$G(a)$	$\exists E, 2, 3-5$
7		$(\exists x F(x) \rightarrow G(a))$	$\rightarrow I, 2, 6$

Daeducer commands needed to generate the proof:

```

1 premise forall x (F(x) -> G(a))
2 assumption exists x F(x)
3 assumption F(c)
4 forall_elim 1, c
5 imp_elim 4, 3
6 discharge
7 exists_elim 2, 3, 5
8 discharge
9 imp_intro 2, 7
10 qed

```


Chapter 38, Exercise C.4.b

Proof presented using Fitch notation:

1	$(\exists x F(x) \rightarrow G(a))$	PR
2	$F(y)$	AS
3	$\exists x F(x)$	$\exists I, 2$
4	$G(a)$	$\rightarrow E, 1, 3$
5	$(F(y) \rightarrow G(a))$	$\rightarrow I, 2, 4$
6	$\forall x (F(x) \rightarrow G(a))$	$\forall I, 5$

Daeducer commands needed to generate the proof:

```

1 premise (exists x F(x) -> G(a))
2 assumption F(y)
3 exists_intro 2, F(x)
4 imp_elim 1, 3
5 discharge
6 imp_intro 2, 4
7 forall_intro 6, y, x
8 qed

```

Chapter 38, Exercise C.5.a

Proof presented using Fitch notation:

1	$\exists x (G(a) \rightarrow F(x))$	PR
2	$G(a)$	AS
3	$(G(a) \rightarrow F(y))$	AS
4	$G(a)$	R, 2
5	$F(y)$	$\rightarrow E, 3, 4$
6	$\exists x F(x)$	$\exists I, 5$
7	$\exists x F(x)$	$\exists E, 1, 3-6$
8	$(G(a) \rightarrow \exists x F(x))$	$\rightarrow I, 2, 7$

Daeducer commands needed to generate the proof:

```

1 premise exists x (G(a) -> F(x))
2 assumption G(a)
3 assumption (G(a) -> F(y))
4 reiteration 2
5 imp_elim 3, 4
6 exists_intro 5, F(x)
7 discharge
8 exists_elim 1, 3, 6
9 discharge
10 imp_intro 2, 8
11 qed

```

Chapter 38, Exercise C.5.b

Proof presented using Fitch notation:

1	$(G(a) \rightarrow \exists x F(x))$	PR
2	$G(a)$	AS
3	$\exists x F(x)$	$\rightarrow$ E, 1, 2
4	$F(b)$	AS
5	$G(a)$	AS
6	$F(b)$	R, 4
7	$(G(a) \rightarrow F(b))$	$\rightarrow$ I, 5, 6
8	$\exists x (G(a) \rightarrow F(x))$	$\exists$ I, 7
9	$\exists x (G(a) \rightarrow F(x))$	$\exists$ E, 3, 4-8
10	$(G(a) \rightarrow \exists x (G(a) \rightarrow F(x)))$	$\rightarrow$ I, 2, 9
11	$\neg G(a)$	AS
12	$G(a)$	AS
13	$\perp$	$\neg$ E, 11, 12
14	$F(a)$	X, 13
15	$(G(a) \rightarrow F(a))$	$\rightarrow$ I, 12, 14
16	$\exists x (G(a) \rightarrow F(x))$	$\exists$ I, 15
17	$(\neg G(a) \rightarrow \exists x (G(a) \rightarrow F(x)))$	$\rightarrow$ I, 11, 16
18	$\exists x (G(a) \rightarrow F(x))$	LEM, 10, 17

Daeducer commands needed to generate the proof:

```

1 premise (G(a) -> exists x F(x))
2 assumption G(a)
3 imp_elim 1, 2
4 assumption F(b)
5 assumption G(a)
6 reiteration 4
7 discharge
8 imp_intro 5, 6
9 exists_intro 8, (G(a) -> F(x))
10 discharge
11 exists_elim 3, 4, 9
12 discharge
13 imp_intro 2, 11
14 assumption !G(a)
15 assumption G(a)
16 not_elim 14, 15
17 explosion 16, F(a)
18 discharge
19 imp_intro 15, 17
20 exists_intro 19, (G(a) -> F(x))
21 discharge
22 imp_intro 14, 20
23 excluded_middle 13, 22
24 qed

```

Chapter 38, Exercise C.6.a

Proof presented using Fitch notation:

1	$\exists x(F(x) \rightarrow G(a))$	PR
2	$\forall xF(x)$	AS
3	$(F(c) \rightarrow G(a))$	AS
4	$\forall xF(x)$	R, 2
5	$F(c)$	$\forall E$, 4
6	$G(a)$	$\rightarrow E$, 3, 5
7	$G(a)$	$\exists E$, 1, 3-6
8	$(\forall xF(x) \rightarrow G(a))$	$\rightarrow I$, 2, 7

Daeducer commands needed to generate the proof:

```

1 premise exists x (F(x) -> G(a))
2 assumption forall x F(x)
3 assumption (F(c) -> G(a))
4 reiteration 2
5 forall_elim 4, c
6 imp_elim 3, 5
7 discharge
8 exists_elim 1, 3, 6
9 discharge
10 imp_intro 2, 8
11 qed

```

Chapter 38, Exercise C.6.b

Proof presented using Fitch notation:

1	$(\forall x F(x) \rightarrow G(a))$	PR
2	$G(a)$	AS
3	$F(a)$	AS
4	$G(a)$	R, 2
5	$(F(a) \rightarrow G(a))$	$\rightarrow$ I, 3, 4
6	$\exists x(F(x) \rightarrow G(a))$	$\exists$ I, 5
7	$(G(a) \rightarrow \exists x(F(x) \rightarrow G(a)))$	$\rightarrow$ I, 2, 6
8	$\neg G(a)$	AS
9	$\neg \forall x F(x)$	MT, 1, 8
10	$\exists x \neg F(x)$	CQ, 9
11	$\neg F(c)$	AS
12	$F(c)$	AS
13	$\perp$	$\neg$ E, 11, 12
14	$G(a)$	X, 13
15	$(F(c) \rightarrow G(a))$	$\rightarrow$ I, 12, 14
16	$\exists x(F(x) \rightarrow G(a))$	$\exists$ I, 15
17	$\exists x(F(x) \rightarrow G(a))$	$\exists$ E, 10, 11–16
18	$(\neg G(a) \rightarrow \exists x(F(x) \rightarrow G(a)))$	$\rightarrow$ I, 8, 17
19	$\exists x(F(x) \rightarrow G(a))$	LEM, 7, 18

Daeducer commands needed to generate the proof:

```

1 premise (forall x F(x) -> G(a))
2 assumption G(a)
3 assumption F(a)
4 reiteration 2
5 discharge
6 imp_intro 3, 4
7 exists_intro 6, (F(x) -> G(a))
8 discharge
9 imp_intro 2, 7
10 assumption !G(a)
11 modus_tollens 1, 10
12 conv_not_forall 11
13 assumption !F(c)
14 assumption F(c)
15 not_elim 13, 14
16 explosion 15, G(a)
17 discharge
18 imp_intro 14, 16
19 exists_intro 18, (F(x) -> G(a))
20 discharge
21 exists_elim 12, 13, 19
22 discharge
23 imp_intro 10, 21
24 excluded_middle 9, 23
25 qed

```

Chapter 39, Example p.328

Proof presented using Fitch notation:

1		$a = b$	AS
2		$a = a$	=I
3		$b = a$	=E, 1, 2
4		$(a = b \rightarrow b = a)$	$\rightarrow$ I, 1, 3
5		$\forall y(a = y \rightarrow y = a)$	$\forall$ I, 4
6		$\forall x\forall y(x = y \rightarrow y = x)$	$\forall$ I, 5

Daeducer commands needed to generate the proof:

```

1  assumption a = b
2  id_intro a
3  id_elim 1, 2, b = a
4  discharge
5  imp_intro 1, 3
6  forall_intro 5, b, y
7  forall_intro 6, a, x
8  qed

```

Chapter 39, Example p.328

Proof presented using Fitch notation:

1		$(x = y \wedge y = z)$	AS
2		$x = y$	$\wedge$ E, 1
3		$y = z$	$\wedge$ E, 1
4		$x = x$	=I
5		$y = x$	=E, 2, 4
6		$x = z$	=E, 5, 3
7		$((x = y \wedge y = z) \rightarrow x = z)$	$\rightarrow$ I, 1, 6
8		$\forall z((x = y \wedge y = z) \rightarrow x = z)$	$\forall$ I, 7
9		$\forall y\forall z((x = y \wedge y = z) \rightarrow x = z)$	$\forall$ I, 8
10		$\forall x\forall y\forall z((x = y \wedge y = z) \rightarrow x = z)$	$\forall$ I, 9

Daeducer commands needed to generate the proof:

```

1  assumption (x = y ^ y = z)
2  and_elim_left 1
3  and_elim_right 1
4  id_intro x
5  id_elim 2, 4, y = x
6  id_elim 5, 3, x = z
7  discharge
8  imp_intro 1, 6
9  forall_intro 8, z, z
10 forall_intro 9, y, y
11 forall_intro 10, x, x
12 qed

```

Chapter 39, Exercise A.1

Proof presented using Fitch notation:

1		$(P(a) \vee Q(b))$	PR
2		$(Q(b) \rightarrow b = c)$	PR
3		$\neg P(a)$	PR
4		$Q(b)$	DS, 1, 3
5		$b = c$	$\rightarrow$ E, 2, 4
6		$Q(c)$	$=$ E, 5, 4

Daeducer commands needed to generate the proof:

```

1 premise (P(a) v Q(b))
2 premise (Q(b) -> b = c)
3 premise !P(a)
4 or_syllogism_left 1, 3
5 imp_elim 2, 4
6 id_elim 5, 4, Q(c)
7 qed

```

Chapter 39, Exercise A.2

Proof presented using Fitch notation:

1		$(m = n \vee n = o)$	PR
2		$A(n)$	PR
3		$m = n$	AS
4		$A(n)$	R, 2
5		$(A(n) \vee A(o))$	$\vee$ E, 4
6		$n = o$	AS
7		$A(n)$	R, 2
8		$A(o)$	$=$ E, 6, 7
9		$(A(n) \vee A(o))$	$\vee$ E, 8
10		$(A(n) \vee A(o))$	$\vee$ E, 1, 3–5, 6–9

Daeducer commands needed to generate the proof:

```

1 premise (m = n v n = o)
2 premise A(n)
3 assumption m = n
4 reiteration 2
5 or_intro_left 4, A(o)
6 discharge
7 assumption n = o
8 reiteration 2
9 id_elim 7, 8, A(o)
10 or_intro_right 9, A(n)
11 discharge
12 or_elim 1, 3, 5, 7, 10
13 qed

```

Chapter 39, Exercise A.3

Proof presented using Fitch notation:

1	$\forall x x = m$	PR
2	$R(m, a)$	PR
3	$a = m$	$\forall E, 1$
4	$R(m, m)$	$=E, 3, 2$
5	$\exists x R(x, x)$	$\exists I, 4$

Daeducer commands needed to generate the proof:

```
1 premise forall x x = m
2 premise R(m, a)
3 forall_elim 1, a
4 id_elim 3, 2, R(m, m)
5 exists_intro 4, R(x, x)
6 qed
```

Chapter 39, Exercise A.4

Proof presented using Fitch notation:

1	$\forall x \forall y (R(x, y) \rightarrow x = y)$	PR
2	$R(a, b)$	AS
3	$\forall y (R(a, y) \rightarrow a = y)$	$\forall E, 1$
4	$(R(a, b) \rightarrow a = b)$	$\forall E, 3$
5	$a = b$	$\rightarrow E, 4, 2$
6	$a = a$	$=I$
7	$b = a$	$=E, 5, 6$
8	$R(b, b)$	$=E, 5, 2$
9	$R(b, a)$	$=E, 7, 8$
10	$(R(a, b) \rightarrow R(b, a))$	$\rightarrow I, 2, 9$

Daeducer commands needed to generate the proof:

```
1 premise forall x forall y (R(x, y) -> x = y)
2 assumption R(a, b)
3 forall_elim 1, a
4 forall_elim 3, b
5 imp_elim 4, 2
6 id_intro a
7 id_elim 5, 6, b = a
8 id_elim 5, 2, R(b, b)
9 id_elim 7, 8, R(b, a)
10 discharge
11 imp_intro 2, 9
12 qed
```

Chapter 39, Exercise A.5

Proof presented using Fitch notation:

1	$\neg\exists x\neg x = m$	PR
2	$\forall x\neg\neg x = m$	CQ, 1
3	$\neg\neg a = m$	$\forall E$, 2
4	$a = m$	DNE, 3
5	$\neg\neg b = m$	$\forall E$, 2
6	$b = m$	DNE, 5
7	$P(a)$	AS
8	$P(m)$	$=E$, 4, 7
9	$P(b)$	$=E$, 6, 8
10	$(P(a) \rightarrow P(b))$	$\rightarrow I$, 7, 9
11	$\forall y(P(a) \rightarrow P(y))$	$\forall I$, 10
12	$\forall x\forall y(P(x) \rightarrow P(y))$	$\forall I$, 11

Daeducer commands needed to generate the proof:

```

1 premise !exists x !x = m
2 conv_not_exists 1
3 forall_elim 2, a
4 double_not_elim 3
5 forall_elim 2, b
6 double_not_elim 5
7 assumption P(a)
8 id_elim 4, 7, P(m)
9 id_elim 6, 8, P(b)
10 discharge
11 imp_intro 7, 9
12 forall_intro 11, b, y
13 forall_intro 12, a, x
14 qed

```


Chapter 39, Exercise A.6

Proof presented using Fitch notation:

1	$\exists x J(x)$	PR
2	$\exists x \neg J(x)$	PR
3	$J(a)$	AS
4	$\neg J(b)$	AS
5	$a = b$	AS
6	$J(a)$	R, 3
7	$J(b)$	=E, 5, 6
8	$\neg J(b)$	R, 4
9	$\perp$	$\neg$ E, 8, 7
10	$\neg a = b$	$\neg$ I, 5–9
11	$\exists y \neg a = y$	$\exists$ I, 10
12	$\exists x \exists y \neg x = y$	$\exists$ I, 11
13	$\exists x \exists y \neg x = y$	$\exists$ E, 2, 4–12
14	$\exists x \exists y \neg x = y$	$\exists$ E, 1, 3–13

Daeducer commands needed to generate the proof:

```

1 premise exists x J(x)
2 premise exists x !J(x)
3 assumption J(a)
4 assumption !J(b)
5 assumption a = b
6 reiteration 3
7 id_elim 5, 6, J(b)
8 reiteration 4
9 not_elim 8, 7
10 discharge
11 not_intro 5, 9
12 exists_intro 11, !a = y
13 exists_intro 12, exists y !x = y
14 discharge
15 exists_elim 2, 4, 13
16 discharge
17 exists_elim 1, 3, 15
18 qed

```

Chapter 39, Exercise A.7

Proof presented using Fitch notation:

1	$\forall x(x = n \leftrightarrow M(x))$	PR
2	$\forall x(O(x) \vee \neg M(x))$	PR
3	$(O(n) \vee \neg M(n))$	$\forall E, 2$
4	$(n = n \leftrightarrow M(n))$	$\forall E, 1$
5	$n = n$	$=I$
6	$M(n)$	$\leftrightarrow E, 4, 5$
7	$O(n)$	AS
8	$O(n)$	R, 7
9	$\neg M(n)$	AS
10	$\perp$	$\neg E, 9, 6$
11	$O(n)$	X, 10
12	$O(n)$	$\vee E, 3, 7-8, 9-11$

Daeducer commands needed to generate the proof:

```

1 premise forall x (x = n <-> M(x))
2 premise forall x (O(x) v !M(x))
3 forall_elim 2, n
4 forall_elim 1, n
5 id_intro n
6 bi_elim 4, 5
7 assumption O(n)
8 reiteration 7
9 discharge
10 assumption !M(n)
11 not_elim 10, 6
12 explosion 11, O(n)
13 discharge
14 or_elim 3, 7, 8, 10, 12
15 qed

```

Chapter 39, Exercise A.8

Proof presented using Fitch notation:

1	$\exists x D(x)$	PR
2	$\forall x(x = p \leftrightarrow D(x))$	PR
3	$D(a)$	AS
4	$(a = p \leftrightarrow D(a))$	$\forall E, 2$
5	$a = p$	$\leftrightarrow E, 4, 3$
6	$D(p)$	$=E, 5, 3$
7	$D(p)$	$\exists E, 1, 3-6$

Daeducer commands needed to generate the proof:

```

1 premise exists x D(x)
2 premise forall x (x = p <-> D(x))
3 assumption D(a)
4 forall_elim 2, a
5 bi_elim 4, 3
6 id_elim 5, 3, D(p)
7 discharge
8 exists_elim 1, 3, 6
9 qed

```

Chapter 39, Exercise A.9

Proof presented using Fitch notation:

1	$\exists x((K(x) \wedge \forall y(K(y) \rightarrow x = y)) \wedge B(x))$	PR
2	$K(d)$	PR
3	$((K(c) \wedge \forall y(K(y) \rightarrow c = y)) \wedge B(c))$	AS
4	$(K(c) \wedge \forall y(K(y) \rightarrow c = y))$	$\wedge E, 3$
5	$B(c)$	$\wedge E, 3$
6	$K(c)$	$\wedge E, 4$
7	$\forall y(K(y) \rightarrow c = y)$	$\wedge E, 4$
8	$(K(d) \rightarrow c = d)$	$\forall E, 7$
9	$c = d$	$\rightarrow E, 8, 2$
10	$B(d)$	$=E, 9, 5$
11	$B(d)$	$\exists E, 1, 3-10$

Daeducer commands needed to generate the proof:

```

1 premise exists x ((K(x) ^ forall y (K(y) -> x = y)) ^ B(x))
2 premise K(d)
3 assumption ((K(c) ^ forall y (K(y) -> c = y)) ^ B(c))
4 and_elim_left 3
5 and_elim_right 3
6 and_elim_left 4
7 and_elim_right 4

```

```

8 forall_elim 7, d
9 imp_elim 8, 2
10 id_elim 9, 5, B(d)
11 discharge
12 exists_elim 1, 3, 10
13 qed

```

Chapter 39, Exercise A.10

Proof presented using Fitch notation:

1		$P(a)$	AS
2		$y = a$	AS
3		$P(y)$	=E, 2, 1
4		$(P(y) \vee \neg y = a)$	$\vee$ E, 3
5		$(y = a \rightarrow (P(y) \vee \neg y = a))$	$\rightarrow$ I, 2, 4
6		$\neg y = a$	AS
7		$(P(y) \vee \neg y = a)$	$\vee$ E, 6
8		$(\neg y = a \rightarrow (P(y) \vee \neg y = a))$	$\rightarrow$ I, 6, 7
9		$(P(y) \vee \neg y = a)$	LEM, 5, 8
10		$\forall x(P(x) \vee \neg x = a)$	$\forall$ I, 9
11		$(P(a) \rightarrow \forall x(P(x) \vee \neg x = a))$	$\rightarrow$ I, 1, 10

Daeducer commands needed to generate the proof:

```

1  assumption P(a)
2  assumption y = a
3  id_elim 2, 1, P(y)
4  or_intro_left 3, !y = a
5  discharge
6  imp_intro 2, 4
7  assumption !y = a
8  or_intro_right 7, P(y)
9  discharge
10 imp_intro 7, 8
11 excluded_middle 6, 10
12 forall_intro 11, y, x
13 discharge
14 imp_intro 1, 12
15 qed

```

Chapter 39, Exercise B.1.a

Proof presented using Fitch notation:

1	$\exists x((F(x) \wedge \forall y(F(y) \rightarrow x = y)) \wedge x = n)$	PR
2	$((F(a) \wedge \forall y(F(y) \rightarrow a = y)) \wedge a = n)$	AS
3	$(F(a) \wedge \forall y(F(y) \rightarrow a = y))$	$\wedge E, 2$
4	$a = n$	$\wedge E, 2$
5	$F(a)$	$\wedge E, 3$
6	$\forall y(F(y) \rightarrow a = y)$	$\wedge E, 3$
7	$F(n)$	$=E, 4, 5$
8	$(F(w) \rightarrow a = w)$	$\forall E, 6$
9	$F(w)$	AS
10	$a = w$	$\rightarrow E, 8, 9$
11	$n = w$	$=E, 4, 10$
12	$(F(w) \rightarrow n = w)$	$\rightarrow I, 9, 11$
13	$\forall y(F(y) \rightarrow n = y)$	$\forall I, 12$
14	$(F(n) \wedge \forall y(F(y) \rightarrow n = y))$	$\wedge I, 7, 13$
15	$(F(n) \wedge \forall y(F(y) \rightarrow n = y))$	$\exists E, 1, 2-14$

Daeducer commands needed to generate the proof:

```

1 premise exists x ((F(x) ^ forall y (F(y) -> x = y)) ^ x = n)
2 assumption ((F(a) ^ forall y (F(y) -> a = y)) ^ a = n)
3 and_elim_left 2
4 and_elim_right 2
5 and_elim_left 3
6 and_elim_right 3
7 id_elim 4, 5, F(n)
8 forall_elim 6, w
9 assumption F(w)
10 imp_elim 8, 9
11 id_elim 4, 10, n = w
12 discharge
13 imp_intro 9, 11
14 forall_intro 13, w, y
15 and_intro 7, 14
16 discharge
17 exists_elim 1, 2, 15
18 qed

```

Chapter 39, Exercise B.1.b

Proof presented using Fitch notation:

1	$(F(n) \wedge \forall y(F(y) \rightarrow n = y))$	PR
2	$n = n$	=I
3	$((F(n) \wedge \forall y(F(y) \rightarrow n = y)) \wedge n = n)$	$\wedge$ I, 1, 2
4	$\exists x((F(x) \wedge \forall y(F(y) \rightarrow x = y)) \wedge x = n)$	$\exists$ I, 3

Daeducer commands needed to generate the proof:

```
1 premise (F(n) ^ forall y (F(y) -> n = y))
2 id_intro n
3 and_intro 1, 2
4 exists_intro 3, ((F(x) ^ forall y (F(y) -> x = y)) ^ x = n)
5 qed
```

Chapter 39, Exercise C.1.a

Proof presented using Fitch notation:

1	$(\exists x F(x) \wedge \forall x \forall y ((F(x) \wedge F(y)) \rightarrow x = y))$	PR
2	$\exists x F(x)$	$\wedge$ E, 1
3	$\forall x \forall y ((F(x) \wedge F(y)) \rightarrow x = y)$	$\wedge$ E, 1
4	$F(a)$	AS
5	$F(z)$	AS
6	$\forall y ((F(a) \wedge F(y)) \rightarrow a = y)$	$\forall$ E, 3
7	$((F(a) \wedge F(z)) \rightarrow a = z)$	$\forall$ E, 6
8	$(F(a) \wedge F(z))$	$\wedge$ I, 4, 5
9	$a = z$	$\rightarrow$ E, 7, 8
10	$(F(z) \rightarrow a = z)$	$\rightarrow$ I, 5, 9
11	$\forall y (F(y) \rightarrow a = y)$	$\forall$ I, 10
12	$(F(a) \wedge \forall y (F(y) \rightarrow a = y))$	$\wedge$ I, 4, 11
13	$\exists x (F(x) \wedge \forall y (F(y) \rightarrow x = y))$	$\exists$ I, 12
14	$\exists x (F(x) \wedge \forall y (F(y) \rightarrow x = y))$	$\exists$ E, 2, 4–13

Daeducer commands needed to generate the proof:

```
1 premise (exists x F(x) ^ forall x forall y ((F(x) ^ F(y)) -> x = y))
2 and_elim_left 1
3 and_elim_right 1
4 assumption F(a)
5 assumption F(z)
6 forall_elim 3, a
7 forall_elim 6, z
8 and_intro 4, 5
9 imp_elim 7, 8
10 discharge
11 imp_intro 5, 9
```

```

12 forall_intro 11, z, y
13 and_intro 4, 12
14 exists_intro 13, (F(x) ^ forall y (F(y) -> x = y))
15 discharge
16 exists_elim 2, 4, 14
17 qed

```

Chapter 39, Exercise C.1.b

Proof presented using Fitch notation:

1	$\exists x(F(x) \wedge \forall y(F(y) \rightarrow x = y))$	PR
2	$(F(a) \wedge \forall y(F(y) \rightarrow a = y))$	AS
3	$F(a)$	$\wedge E$, 2
4	$\forall y(F(y) \rightarrow a = y)$	$\wedge E$, 2
5	$F(z)$	AS
6	$(F(z) \rightarrow a = z)$	$\forall E$, 4
7	$a = z$	$\rightarrow E$, 6, 5
8	$a = z$	AS
9	$F(z)$	$=E$, 8, 3
10	$(F(z) \leftrightarrow a = z)$	$\leftrightarrow I$, 5–7, 8–9
11	$\forall y(F(y) \leftrightarrow a = y)$	$\forall I$, 10
12	$\exists x \forall y(F(y) \leftrightarrow x = y)$	$\exists I$, 11
13	$\exists x \forall y(F(y) \leftrightarrow x = y)$	$\exists E$, 1, 2–12

Daeducer commands needed to generate the proof:

```

1 premise exists x (F(x) ^ forall y (F(y) -> x = y))
2 assumption (F(a) ^ forall y (F(y) -> a = y))
3 and_elim_left 2
4 and_elim_right 2
5 assumption F(z)
6 forall_elim 4, z
7 imp_elim 6, 5
8 discharge
9 assumption a = z
10 id_elim 9, 3, F(z)
11 discharge
12 bi_intro 5, 7, 9, 10
13 forall_intro 12, z, y
14 exists_intro 13, forall y (F(y) <-> x = y)
15 discharge
16 exists_elim 1, 2, 14
17 qed

```

Chapter 39, Exercise C.1.c

Proof presented using Fitch notation:

1	$\exists x \forall y (F(y) \leftrightarrow x = y)$	PR
2	$\forall y (F(y) \leftrightarrow c = y)$	AS
3	$(F(c) \leftrightarrow c = c)$	$\forall E, 2$
4	$c = c$	=I
5	$F(c)$	$\leftrightarrow E, 3, 4$
6	$\exists x F(x)$	$\exists I, 5$
7	$(F(a) \wedge F(b))$	AS
8	$F(a)$	$\wedge E, 7$
9	$F(b)$	$\wedge E, 7$
10	$(F(a) \leftrightarrow c = a)$	$\forall E, 2$
11	$c = a$	$\leftrightarrow E, 10, 8$
12	$(F(b) \leftrightarrow c = b)$	$\forall E, 2$
13	$c = b$	$\leftrightarrow E, 12, 9$
14	$a = b$	=E, 11, 13
15	$((F(a) \wedge F(b)) \rightarrow a = b)$	$\rightarrow I, 7, 14$
16	$\forall y ((F(a) \wedge F(y)) \rightarrow a = y)$	$\forall I, 15$
17	$\forall x \forall y ((F(x) \wedge F(y)) \rightarrow x = y)$	$\forall I, 16$
18	$(\exists x F(x) \wedge \forall x \forall y ((F(x) \wedge F(y)) \rightarrow x = y))$	$\wedge I, 6, 17$
19	$(\exists x F(x) \wedge \forall x \forall y ((F(x) \wedge F(y)) \rightarrow x = y))$	$\exists E, 1, 2-18$

Daeducer commands needed to generate the proof:

```

1 premise exists x forall y (F(y) <-> x = y)
2 assumption forall y (F(y) <-> c = y)
3 forall_elim 2, c
4 id_intro c
5 bi_elim 3, 4
6 exists_intro 5, F(x)
7 assumption (F(a) ^ F(b))
8 and_elim_left 7
9 and_elim_right 7
10 forall_elim 2, a
11 bi_elim 10, 8
12 forall_elim 2, b
13 bi_elim 12, 9
14 id_elim 11, 13, a = b
15 discharge
16 imp_intro 7, 14
17 forall_intro 16, b, y
18 forall_intro 17, a, x
19 and_intro 6, 18
20 discharge
21 exists_elim 1, 2, 19
22 qed

```


Chapter 39, Exercise D.1

Proof presented using Fitch notation:

1	$\exists x(F(x) \wedge \forall y(F(y) \rightarrow x = y))$	PR
2	$\exists x(G(x) \wedge \forall y(G(y) \rightarrow x = y))$	PR
3	$\forall x\neg(F(x) \wedge G(x))$	PR
4	$(F(a) \wedge \forall y(F(y) \rightarrow a = y))$	AS
5	$F(a)$	$\wedge E, 4$
6	$\forall y(F(y) \rightarrow a = y)$	$\wedge E, 4$
7	$(G(b) \wedge \forall y(G(y) \rightarrow b = y))$	AS
8	$G(b)$	$\wedge E, 7$
9	$\forall y(G(y) \rightarrow b = y)$	$\wedge E, 7$
10	$a = b$	AS
11	$\forall x\neg(F(x) \wedge G(x))$	R, 3
12	$\neg(F(a) \wedge G(a))$	$\forall E, 11$
13	$G(a)$	$=E, 10, 8$
14	$(F(a) \wedge G(a))$	$\wedge I, 5, 13$
15	$\perp$	$\neg E, 12, 14$
16	$\neg a = b$	$\neg I, 10-15$
17	$(F(a) \vee G(a))$	$\vee E, 5$
18	$(F(b) \vee G(b))$	$\vee E, 8$
19	$(F(z) \vee G(z))$	AS
20	$F(z)$	AS
21	$(F(z) \rightarrow a = z)$	$\forall E, 6$
22	$a = z$	$\rightarrow E, 21, 20$
23	$a = a$	$=I$
24	$z = a$	$=E, 22, 23$
25	$(z = a \vee z = b)$	$\vee E, 24$
26	$G(z)$	AS
27	$(G(z) \rightarrow b = z)$	$\forall E, 9$
28	$b = z$	$\rightarrow E, 27, 26$
29	$b = b$	$=I$
30	$z = b$	$=E, 28, 29$
31	$(z = a \vee z = b)$	$\vee E, 30$
32	$(z = a \vee z = b)$	$\vee E, 19, 20-25, 26-31$

33	$((F(z) \vee G(z)) \rightarrow (z = a \vee z = b))$	$\rightarrow I, 19, 32$
34	$\forall z((F(z) \vee G(z)) \rightarrow (z = a \vee z = b))$	$\forall I, 33$
35	$((F(b) \vee G(b)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = a \vee z = b)))$	$\wedge I, 18, 34$
36	$((F(a) \vee G(a)) \wedge ((F(b) \vee G(b)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = a \vee z = b))))$	$\wedge I, 17, 35$
37	$(\neg a = b \wedge ((F(a) \vee G(a)) \wedge ((F(b) \vee G(b)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = a \vee z = b))))$	$\wedge I, 16, 36$
38	$\exists y(\neg a = y \wedge ((F(a) \vee G(a)) \wedge ((F(y) \vee G(y)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = a \vee z = y))))$	$\exists I, 37$
39	$\exists x \exists y(\neg x = y \wedge ((F(x) \vee G(x)) \wedge ((F(y) \vee G(y)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = x \vee z = y))))$	$\exists I, 38$
40	$\exists x \exists y(\neg x = y \wedge ((F(x) \vee G(x)) \wedge ((F(y) \vee G(y)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = x \vee z = y))))$	$\exists E, 2, 7-39$
41	$\exists x \exists y(\neg x = y \wedge ((F(x) \vee G(x)) \wedge ((F(y) \vee G(y)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = x \vee z = y))))$	$\exists E, 1, 4-40$
42	$\exists x \exists y(\neg x = y \wedge ((F(x) \vee G(x)) \wedge ((F(y) \vee G(y)) \wedge \forall z((F(z) \vee G(z)) \rightarrow (z = x \vee z = y))))$	$\exists E, 1, 4-40$

Daeducer commands needed to generate the proof:

```

1 premise exists x (F(x) ^ forall y (F(y) -> x = y))
2 premise exists x (G(x) ^ forall y (G(y) -> x = y))
3 premise forall x !(F(x) ^ G(x))
4 assumption (F(a) ^ forall y (F(y) -> a = y))
5 and_elim_left 4
6 and_elim_right 4
7 assumption (G(b) ^ forall y (G(y) -> b = y))
8 and_elim_left 7
9 and_elim_right 7
10 assumption a = b
11 reiteration 3
12 forall_elim 11, a
13 id_elim 10, 8, G(a)
14 and_intro 5, 13
15 not_elim 12, 14
16 discharge
17 not_intro 10, 15
18 or_intro_left 5, G(a)
19 or_intro_right 8, F(b)
20 assumption (F(z) v G(z))
21 assumption F(z)
22 forall_elim 6, z
23 imp_elim 22, 21
24 id_intro a
25 id_elim 23, 24, z = a
26 or_intro_left 25, z = b
27 discharge
28 assumption G(z)
29 forall_elim 9, z
30 imp_elim 29, 28
31 id_intro b
32 id_elim 30, 31, z = b

```

```

33 or_intro_right 32, z = a
34 discharge
35 or_elim 20, 21, 26, 28, 33
36 discharge
37 imp_intro 20, 35
38 forall_intro 37, z, z
39 and_intro 19, 38
40 and_intro 18, 39
41 and_intro 17, 40
42 exists_intro 41, (!a = y ^ ((F(a) ∨ G(a)) ^ ((F(y) ∨ G(y)) ^ forall z ((F(z) ∨ G(z)) -> (z = a
    ∨ z = y))))))
43 exists_intro 42, exists y (!x = y ^ ((F(x) ∨ G(x)) ^ ((F(y) ∨ G(y)) ^ forall z ((F(z) ∨ G(z))
    -> (z = x ∨ z = y))))))
44 discharge
45 exists_elim 2, 7, 43
46 discharge
47 exists_elim 1, 4, 45
48 exists_elim 1, 4, 45
49 qed

```

Chapter 40, Example p.330

Proof presented using Fitch notation:

1	$\forall x \neg A(x)$	PR
2	$\exists x A(x)$	AS
3	$A(c)$	AS
4	$\neg A(c)$	$\forall E, 1$
5	$\perp$	$\neg E, 4, 3$
6	$\perp$	$\exists E, 2, 3-5$
7	$\neg \exists x A(x)$	$\neg I, 2-6$

Daeducer commands needed to generate the proof:

```

1 premise forall x !A(x)
2 assumption exists x A(x)
3 assumption A(c)
4 forall_elim 1, c
5 not_elim 4, 3
6 discharge
7 exists_elim 2, 3, 5
8 discharge
9 not_intro 2, 7
10 qed

```

Chapter 40, Example p.331

Proof presented using Fitch notation:

1	$\exists x \neg A(x)$	PR
2	$\neg A(a)$	AS
3	$\forall x A(x)$	AS
4	$A(a)$	$\forall E, 3$
5	$\perp$	$\neg E, 2, 4$
6	$\neg \forall x A(x)$	$\neg I, 3-5$
7	$\neg \forall x A(x)$	$\exists E, 1, 2-6$
8	$\neg \forall x A(x)$	$\exists E, 1, 2-6$

Daeducer commands needed to generate the proof:

```

1 premise exists x !A(x)
2 assumption !A(a)
3 assumption forall x A(x)
4 forall_elim 3, a
5 not_elim 2, 4
6 discharge
7 not_intro 3, 5
8 discharge
9 exists_elim 1, 2, 7
10 exists_elim 1, 2, 7
11 qed

```

Chapter 40, Exercise A.1

Proof presented using Fitch notation:

1	$\neg \exists x A(x)$	PR
2	$A(y)$	AS
3	$\exists x A(x)$	$\exists I, 2$
4	$\perp$	$\neg E, 1, 3$
5	$\neg A(y)$	$\neg I, 2-4$
6	$\forall x \neg A(x)$	$\forall I, 5$

Daeducer commands needed to generate the proof:

```

1 premise !exists x A(x)
2 assumption A(y)
3 exists_intro 2, A(x)
4 not_elim 1, 3
5 discharge
6 not_intro 2, 4
7 forall_intro 6, y, x
8 qed

```

Chapter 40, Exercise A.2

Proof presented using Fitch notation:

1	$\neg\forall xA(x)$	PR
2	$\neg\exists x\neg A(x)$	AS
3	$\neg A(c)$	AS
4	$\exists x\neg A(x)$	$\exists I$, 3
5	$\perp$	$\neg E$, 2, 4
6	$A(c)$	IP, 3–5
7	$\forall xA(x)$	$\forall I$, 6
8	$\perp$	$\neg E$, 1, 7
9	$\exists x\neg A(x)$	IP, 2–8
10	$\exists x\neg A(x)$	IP, 2–8

Daeducer commands needed to generate the proof:

```

1  premise !forall x A(x)
2  assumption !exists x !A(x)
3  assumption !A(c)
4  exists_intro 3, !A(x)
5  not_elim 2, 4
6  discharge
7  indirect 3, 5
8  forall_intro 7, c, x
9  not_elim 1, 8
10 discharge
11 indirect 2, 9
12 indirect 2, 9
13 qed

```